F18CE

Multivariable Calculus

Lecturers:
Nasreddine Megrez (Dubai)
Beatrice Pelloni (Edinburgh)
Joshua Tan (Malaysia)



Department of Mathematics
Heriot-Watt University

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Contents

1 Functions of several variables - definition, limits and continuity **1**
- 1.1 Definitions . . . 1
- 1.2 Limits of functions of two variables . . . 4
- 1.3 Continuity of functions of two variables . . . 7

2 Differentiation in several variables **12**
- 2.1 Partial derivatives . . . 12
- 2.2 Graphs. Tangent Planes and normal lines. . . . 15
- 2.3 Rigorous definition of differentiability of functions of two variables . . . 22
- 2.4 Higher order partial derivatives . . . 27
- 2.5 The chain rule for functions of several variables. . . . 30
- 2.6 Transforming equations. . . . 33
- 2.7 Gradient and Directional derivatives . . . 34

3 Implicit functions and the Implicit Function Theorem **36**
- 3.1 Solution functions . . . 36
- 3.2 Implicit differentiation. . . . 37
- 3.3 The Implicit Function Theorem . . . 37
  - 3.3.1 Many variables case . . . 38

4 Taylor's expansion of functions of two variables **41**
- 4.1 Recap of Taylor's expansion of functions of one variable . . . 41
- 4.2 Taylor's expansion of functions of two variables . . . 41

5 Stationary Points, maxima and minima **45**
- 5.1 Second derivative test for maxima and minima – one variable case . . . 47
- 5.2 Second derivative test for two variables . . . 48
- 5.3 Constrained extrema - Lagrange multipliers . . . 51
- 5.4 More examples . . . 54

6 The Riemann Integral **57**
- 6.1 Riemann sums . . . 57
- 6.2 The Riemann integral of f . . . 60
- 6.3 Properties of the integral . . . 65
- 6.4 Some classes of integrable functions . . . 66

7 The Fundamental Theorem of Calculus **68**

8 Double integrals **70**
- 8.1 Double integrals: general definition and properties . . . 70
- 8.2 How to calculate double integrals . . . 72
- 8.3 Interchanging the integration order . . . 77
- 8.4 Calculating areas using double integrals . . . 79
- 8.5 Change of variables in double integrals - the Jacobian . . . 80
- 8.6 Polar coordinates – continued . . . 83
- 8.7 Other substitutions . . . 85
- 8.8 Calculating volumes using double integrals . . . 86

9 Triple integrals **90**
9.1 Change of variables in triple integrals 94
9.1.1 Cylindrical polar coordinates (r, θ, z) 95
9.1.2 Spherical polar coordinates (r, θ, ϕ) 97
9.1.3 Jacobian 98
9.1.4 Latitude and longitude 98

Part 1 - Differentiation

1 Functions of several variables - definition, limits and continuity

1.1 Definitions

A function $f(x)$ of one variable is a rule taking a single number x and producing a new number $f(x)$, e.g. $f(x) = x^2$ or $f(x) = \sin(x)$. Similarly, a function of two variables $f(x, y)$ is a rule taking two numbers x, y and producing a new number $f(x, y)$, e.g.

$$f(x, y) = x^2 + \sin(y)$$
$$f(x, y) = x^2 y^3.$$

We do not have to stop at two variables, e.g. $f(x, y, z) = x^2 y^3 z^5$ is a function of three variables. More generally,

$$f(x_1, x_2, \ldots, x_n) = x_1 + x_2 + \ldots + x_n \tag{1.1}$$

is a function of n variables. Usually, for two and three variables, we will use the notation x, y or x, y, z, respectively, while for more variables we will use subscripts such as in x_i.

So: a real-valued function of n variables $f(x_1, ..., x_n)$ starts with a point in $\mathbb{R}^n$ and sends it to a real value in $\mathbb{R}$. We write

$$f : \mathbb{R}^n \to \mathbb{R} \tag{1.2}$$

to say that f is a function of n variables taking values in the real numbers.

Reminder:
- $\mathbb{R}$ – the set of all real numbers
- $\mathbb{Z}$ – the set of all integers
- $\mathbb{N}$ – the set of all natural numbers $(1, 2, 3, \ldots)$
- $\mathbb{Q}$ – the set of all rational numbers

Sometimes a function does not make sense for arbitrary values of the variables and can only be defined on a restricted set called its *domain*.

For example:

- $$f : D_f \subset \mathbb{R}^2 \to \mathbb{R}, \quad f(x, y) = \frac{x}{y}$$

  is only defined if $y \neq 0$. Hence its domain D_f is

  $$D_f = \{(x, y) : y \neq 0\} = \mathbb{R} \times \mathbb{R} \setminus \{0\}.$$

- $$f : D_f \subset \mathbb{R}^2 \to \mathbb{R}, \quad f(x, y) = \sqrt{xy}$$

  is only defined for $xy > 0$. Hence its domain D_f is

  $$D_f = \{(x, y) : xy > 0\} = (\mathbb{R}^+ \times \mathbb{R}^+) \cup (\mathbb{R}^- \times \mathbb{R}^-).$$

-

Reminder: Domains of some elementary functions:
- $f(x) = x^\mu$

1. For $\mu \geq 0$, and $\mu \in \mathbb{Z}$ or $\mu = \frac{m}{n} \in \mathbb{Q}$ (where $m, n \in \mathbb{Z}$, and where $\mathbb{Q}$ denotes the rational numbers) with n odd,
$$D_f = \mathbb{R}^1 \,,$$
e.g. $f(x) = x^{2/3}$.
2. for $\mu < 0$, and $\mu \in \mathbb{Z}$ or $\mu = \frac{m}{n} \in \mathbb{Q}$ with n odd,
$$D_f = \{x \in \mathbb{R}^1 \;\mid\; x \neq 0\} \,,$$
e.g. $f(x) = x^{-3/5}$.
3. for $\mu > 0$, and when μ is irrational or for $\mu = \frac{m}{n} \in \mathbb{Q}$ with n even,
$$D_f = \{x \;\mid\; x \in \mathbb{R} \,,\, x \geq 0\} \,,$$
e.g. $f(x) = x^{1/2}$ or $f(x) = x^{\sqrt{2}}$.
4. for $\mu < 0$ when μ is irrational or $\mu = \frac{m}{n} \in \mathbb{Q}$ with n even,
$$D_f = \{x \;\mid\; x \in \mathbb{R} \,,\, x > 0\} \,,$$
e.g. $f(x) = x^{-\pi}$ or $f(x) = x^{-1/4}$.

- $f(x) = \log_a(x)$ is defined for $x > 0$.

We write

$$f \,:\, D_f \to \mathbb{R}$$

or

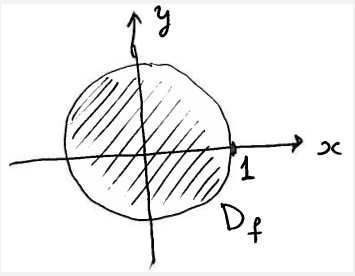
$$f \,:\, D \to \mathbb{R}$$

to say that f takes numbers from the domain $D_f \subset \mathbb{R}^n$ and assigns to them numbers in $\mathbb{R}$.

For a composite function constructed using elementary functions we can define its "natural domain" to consist of a maximal set of points at which all elementary functions involved are well defined, i.e. get arguments from their domains.

Example 1.1

$f(x, y) = \sqrt{1 - x^2 - y^2}$ only makes sense when $x^2 + y^2 \leq 1$. Then, the natural domain of this function is $D_f = \{(x, y) \in \mathbb{R}^2 | x^2 + y^2 \leqslant 1\}$:



Example 1.2

Find the natural domain for

$$f(x, y) = \ln[(1 - x)(1 - y)] \,.$$

Solution. We need

$$(1-x)(1-y) > 0\,,$$

so either

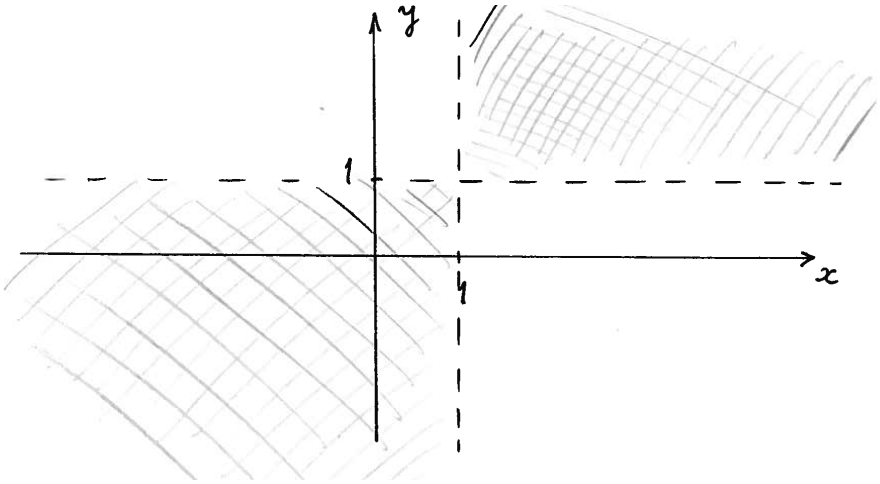
$$(1-x) > 0\,,\ (1-y) > 0$$

or

$$(1-x) < 0\,,\ (1-y) < 0\,.$$

Therefore,

$$D_f = \{(x,y) \in \mathbb{R}^2 \ \mid\ x < 1\,,\ y < 1 \text{ or } x > 1\,,\ y > 1\}\,.$$



In the illustration, dashed lines are used to indicate that we do not include the lines themselves. Thus, D_f consists of two quarter planes not including the boundaries. ■

Note that in the above two examples functions of one variable were used to construct a function of two (or more) variables. Sometimes, in such constructions, we do not specify all of the involved functions explicitly.

Example 1.3

For example, let $f : \mathbb{R}^1 \to \mathbb{R}^1$ be a function whose domain is the set of all real numbers. Then

$$g(x,y) = \sqrt{y} + f(\sqrt{x} - 1)$$

defines a function of two variables with domain

$$D_g = \{(x,y) \in \mathbb{R}^2 \ \mid\ x \geq 0\,,\ y \geq 0\} \subset \mathbb{R}^2\,.$$

Another frequent construction is when given a function of many variables one constructs out of it a function of fewer variables by restricting the original variables.

Example 1.4

Let

$$f(x,y) = \frac{x^4}{y^2} - 2xy \tag{1.3}$$

be a function of two variables defined on $\mathbb{R}^2$. We define a new function of one variable $g : \mathbb{R}^1 \to \mathbb{R}^1$ by the rule

$$g(x) = f\left(x, \frac{x}{2}\right) .$$

To obtain the explicit formula we have to substitute $y = x/2$ into (1.3). We obtain

$$g(x) = 3x^2 .$$

Example 1.5

With the relation between f and g as in example 1.3, determine the explicit forms of f and g given that

$$g(x,1) = x .$$

Solution.

$$\begin{aligned} & g(x,1) = 1 + f(\sqrt{x} - 1) = x \\ \Rightarrow \quad & f(\sqrt{x} - 1) = x - 1 \\ \Rightarrow \quad & g(x,y) = \sqrt{y} + x - 1 . \end{aligned}$$

To determine f explicitly, let $w = \sqrt{x} - 1$. Then $\sqrt{x} = 1 + w$, and for $x \geq 0$

$$x = (1+w)^2 .$$

Hence $f(w) = (1+w)^2 - 1 = w^2 + 2w$. This is the explicit form of f. ■

1.2 Limits of functions of two variables

The ϵ-δ definition of a limit for a function of one variable extends to functions of two variables in the following way.

Definition 1.6

Let f by a function of two variables with domain $D_f \subset \mathbb{R}^2$. We say that f has a limit at $(a,b) \in \mathbb{R}^2$ (not necessarily from the domain D_f) with the value L and write

$$\lim_{(x,y)\to(a,b)} f(x,y) = L$$

if for every $\epsilon > 0$ there exists $\delta > 0$ such that whenever $(x,y) \in D_f$, $(x,y) \neq (a,b)$ and

$$|x - a| < \delta , \quad |y - b| < \delta$$

we have $|f(x,y) - L| < \epsilon$.

Equivalently, by using the Euclidean norm in the plane, which means that we compute the distance from (x, y) to (a, b) as

$$dist([(x, y), (a, b)] = \sqrt{(x-a)^2 + (y-b)^2} := \|(x, y) - (a, b)\|$$

(which generalises to any dimension) this definition can be stated as follows.

Definition 1.7

Let f by a function of two variables with domain $D_f \subset \mathbb{R}^2$. We say that f has a limit at $(a, b) \in \mathbb{R}^2$ (not necessarily from the domain D_f) with the value L and write

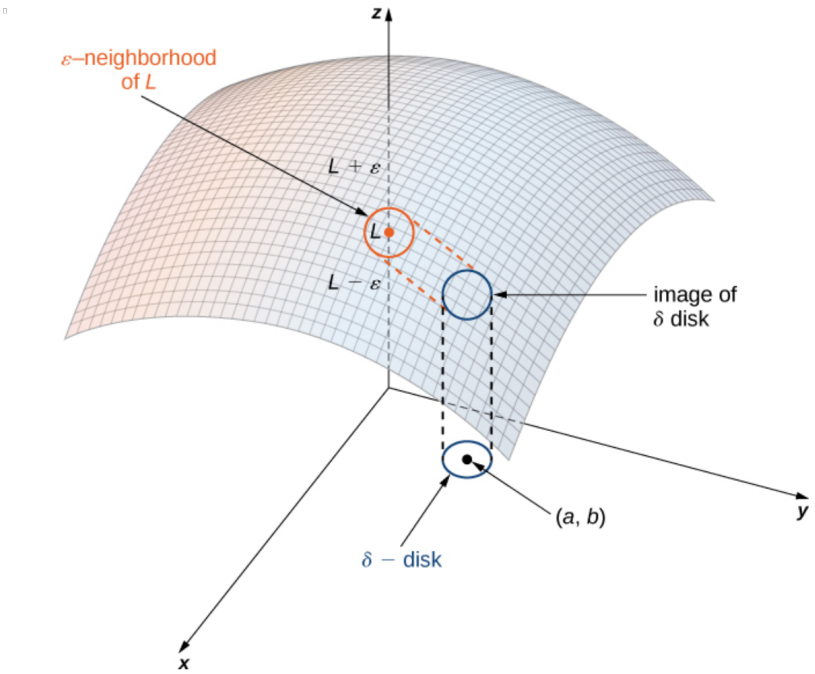
$$\lim_{(x,y)\to(a,b)} f(x, y) = L$$

if for for every $\epsilon > 0$ there exists $\delta > 0$ such that whenever $(x, y) \in D_f$, $(x, y) \neq (a, b)$ and

$$\|(x, y) - (a, b)\| < \delta$$

we have $|f(x, y) - L| < \epsilon$.

This definition, described in the figure below, tells us that no matter how the limit point is approached from within a disc around the point (a, b), the limit must exist and be always the same. In the figure, the neighbourhood of L is the set of all real values at distance less than ϵ from L, hence $\{\alpha \in \mathbb{R} : |\alpha - L| < \epsilon\}$.



We can give an equivalent definition of a limit of function of two variables in terms of sequences:

Theorem 1.8

Let $f(x,y)$ be a function of two variables with domain D_f . The following two statements are equivalent

(i)
$$\lim_{(x,y)\to(a,b)} f(x,y) = L$$

(ii) For every pair of sequences (x_n, y_n) such that $\forall n\,,\ \ (x_n,y_n) \in D_f$, $(x_n,y_n) \neq (a,b)$ and
$$\lim_{n\to\infty} x_n = a\,, \qquad \lim_{n\to\infty} y_n = b$$
we have
$$\lim_{n\to\infty} f(x_n,y_n) = L$$

The proof of this theorem is similar to the proof of the analogous theorems for one variable and we will omit it.

Limits of multivariable functions have the same linearity and composition properties as in the one variable case, with an analogous proof.

Theorem 1.9

Let $f(x), y,\ g(x,y)$ be a function of two variables with common domain D. Assume $F = \lim_{(x,y)\to(a,b)} f(x,y)$, $G = \lim_{(x,y)\to(a,b)} g(x,y)$

(i)
$$\lim_{(x,y)\to(a,b)} (af(x,y) + bg(x,y)) = aF + bG, \quad \forall a,b \in \mathbb{R}.$$

(ii)
$$\lim_{(x,y)\to(a,b)} (f(x,y)g(x,y)) = FG.$$

(iii)
$$\lim_{(x,y)\to(a,b)} \frac{f(x,y)}{g(x,y)} = \frac{F}{G}, \quad for\ G \neq 0.$$

(iv) If $X : \mathbb{R}^2 \to \mathbb{R}^2$ is such that $\lim_{(x,y)\to(x_0,y_0)} X(x,y) = (a,b)$, then
$$\lim_{(x,y)\to(x_0,y_0)} (f(X(x,y)) = F.$$

In general, the limit exists if it is the same *no matter how the limit point is approached*, namely no matter what sequence of points x_n and y_n is chosen. If you reach two different limits by selecting different sequences, the two-dimensional limit does not exist.

Example 1.10

Consider a function

$$g(x,y) = \frac{x^2 - y^2}{x^2 + y^2}$$

defined for $(x,y) \neq (0,0)$. Show that the following limit does not exist

$$\lim_{(x,y)\to(0,0)} g(x,y)\,.$$

Solution. To show that the limit does not exist it suffices to present two sequences of points from the domain which both tend to the origin but for which the limits of values (if exist) are different. Choosing first $(x_n, y_n) = (1/n, 1/n) \to (0,0)$ we find

$$\lim_{n\to\infty} f(\frac{1}{n}, \frac{1}{n}) = \lim_{n\to\infty} \frac{\frac{1}{n^2} - \frac{1}{n^2}}{\frac{1}{n^2} + \frac{1}{n^2}} = \lim_{n\to\infty} 0 = 0\,.$$

Choosing next $(x_n, y_n) = (\frac{1}{n}, 0) \to (0,0)$ we obtain

$$\lim_{n\to\infty} f(\frac{1}{n}, 0) = \lim_{n\to\infty} \frac{\frac{1}{n^2} - 0^2}{\frac{1}{n^2} + 0^2} = \lim_{n\to\infty} 1 = 1\,.$$

Since these are two different numbers the limit at hand does not exist. ■

We end this subsection by remarking that Definition 1.6 and Theorem 1.8 generalise to three and more variables in a straightforward manner.

1.3 Continuity of functions of two variables

Recall the definition of *continuity* for functions of one variable:

A function $f(x)$ is said to be *continuous at a point* $x = a$ if

$$\lim_{x\to a} f(x) = f(a)\,.$$

It is assumed in the above definition that $a \in D_f$. The limit $\lim_{x\to a} f(x)$ must exist and must be equal to the value $f(a)$.

A function continuous at any point in its domain is called *continuous*.

Recall that all the following are continuous functions of one variable in their domain of definition:

- Polynomials.
- Rational functions.
- Power functions (including n-th roots).
- Trigonometric functions and inverse trigonometric functions.
- Exponential functions.
- Logarithmic functions.

Definition 1.11 (Continuity at a point for a function of two variables)

Similarly to the above, a function of two variables $f(x, y)$ with domain $D_f \subset \mathbb{R}^2$ is called continuous at a point $(a, b) \in D_f$ if

$$\lim_{(x,y)\to(a,b)} f(x, y) = f(a, b)\,.$$

In view of Theorem 1.8, this is equivalent to saying that for any sequences $x_n \to a$ and $y_n \to b$ such that $(x_n, y_n) \in D_f$ for all n, the limit $\lim_{n\to\infty} f(x_n, y_n)$ exists and

$$\lim_{n\to\infty} f(x_n, y_n) = f(a, b)\,.$$

The above definitions generalise to 3 and more variables in a straightforward manner. Using vector notation with $x \in \mathbb{R}^n$ it can be stated as follows. Given a function f of n-variables it is said to be continuous at a point $a \in \mathbb{R}^n$ from its domain if

$$\lim_{x\to a} f(x) = f(a)\,.$$

A function (of any number of variables) continuous at very point in its domain is called continuous.

Example 1.12

Consider a function of two variables

$$f(x, y) := \begin{cases} \frac{xy}{x^2+y^2} & \text{for } (x, y) \neq (0, 0) \\ 0 & \text{for } (x, y) = (0, 0) \end{cases}$$

defined on the entire $\mathbb{R}^2$. Show that this function is not continuous at $(x, y) = (0, 0)$.

Solution. Let $x_n = \frac{1}{n}$, $y_n = 0$ and $n \geq 1$ such that (x_n, y_n) approaches the point $(0, 0)$ along the x-axis as $n \to \infty$. Then

$$\lim_{n\to\infty} \frac{x_n y_n}{x_n^2 + y_n^2} = \lim_{n\to\infty} 0 = 0\,.$$

Now let $x_n = \frac{1}{n}$ as before and $y_n = \frac{1}{n}$, again with $n > 1$. Then

$$\lim_{n\to\infty} \frac{x_n y_n}{x_n^2 + y_n^2} = \lim_{n\to\infty} \frac{\frac{1}{n}\cdot\frac{1}{n}}{\frac{1}{n^2} + \frac{1}{n^2}} = \frac{1}{2}\,.$$

Thus, depending on how we approach the point $(0, 0)$, we get different limits. In particular for the last sequence the limit $\lim_{n\to\infty} f(x_n, y_n) = \frac{1}{2} \neq f(0, 0) = 0$, and we conclude that f is not continuous at the point $(0, 0)$. ■

It is not hard to show that the above function is continuous at every point $(x, y) \in \mathbb{R}^2$ except for $(x, y) = (0, 0)$, which is thus an isolated discontinuity. In general, discontinuities can be concentrated on curves for functions of two variables, on surfaces for functions of three variables and so on.

We now modify the above example only slightly but obtain a different result.

Example 1.13

Consider a function of two variables

$$f(x,y) := \begin{cases} \frac{x^2 y}{x^2+y^2} & \text{for } (x,y) \neq (0,0) \\ 0 & \text{for } (x,y) = (0,0) \end{cases}$$

defined on the entire $\mathbb{R}^2$. Show that this function is continuous at $(x,y) = (0,0)$.

Solution. This function is continuous at every point $(x,y) \in \mathbb{R}^2$ except for $(x,y) = (0,0)$. So we need to understand what happens at (0,0), no matter how we approach.

To allow us to study any possible approach, we switch to polar coordinates, namely set

$$x = r\cos\theta, \quad y = r\sin\theta.$$

Points close to $(0,0)$ are points with a small value of r, irrespective of the value of the angle θ. HEnce the given limit becomes a limit as $r \to 0$. We have

$$f(r,\theta) = \frac{r^2\cos^2\theta r\sin\theta}{r^2(\cos^2\theta + \sin^2\theta)} = \frac{r^3\cos^2\theta\sin\theta}{r^2} = r\cos^2\theta\sin\theta.$$

Since $-1 \leq \cos^2\theta\sin\theta \leq 1$, we find

$$\lim_{r\to 0} f(r,\theta) = 0 \quad \forall\theta.$$

This implies the continuity of the function also at the origin.

While this is not a fully rigorous argument, it should help the intuition of what it mean for a functions of two variables to be continuous at a point in the plane. ■

Example 1.14

The function

$$h(x,y) := \begin{cases} \frac{x^2+y^2}{x^2-y^2} & \text{for } x^2 \neq y^2 \\ 1 & \text{for } x^2 = y^2 \end{cases}$$

defined on the entire $\mathbb{R}^2$, has a discontinuity along a pair of lines $\pm x = y$. (This means that for each point on this pair of lines function h is not continuous at it.) The function of three variables

$$g(x,y,z) := \begin{cases} \frac{1}{x^2+y^2+z^2-1} & \text{for } x^2+y^2+z^2 \neq 1 \\ 0 & \text{for } x^2+y^2+z^2 = 1 \end{cases}$$

defined on $\mathbb{R}^3$, is discontinuous on the unit sphere centred at the origin:

$$x^2 + y^2 + z^2 = 1\,.$$

Remark. Note that to prove that a function is not continuous at a given point it is convenient to use the sequential definition of a limit of a function while to show that the function is continuous one typically uses the ϵ-δ definition. In general there is no straightforward

(mechanical) way to determine whether the given function is or is not continuous at a given point and one needs to investigate the problem both ways (i.e. if you tried to show that the function is not continuous using sequences but failed perhaps it is not. Then you change strategy and try to show that it is continuous using the ϵ and δ definition).

To prove continuity of composite functions constructed using other functions which we know are continuous (e.g. elementary functions) we can use the following theorem

Theorem 1.15

Let $f, g : \mathbb{R}^n \to \mathbb{R}$ be two functions of n-variables. If f and g are each continuous at $a = (a_1, a_2, \dots a_n) \in \mathbb{R}^n$ then the functions

$$\alpha f + \beta g\,, \quad f \cdot g$$

are continuous at a. Here $\alpha, \beta \in \mathbb{R}$ are arbitrary constants. Also, if $g(a) \neq 0$ then the function $\frac{f}{g}$ is continuous at a. Moreover, given two functions

$$f : \mathbb{R}^n \to \mathbb{R} \qquad h : \mathbb{R} \to \mathbb{R}$$

we can consider a composite function $h : \mathbb{R}^n \to \mathbb{R}$ given by

$$H(\mathbf{x}) = h(f(\mathbf{x})) \quad \mathbf{x} \in \mathbb{R}^n.$$

If f is continuous at a and h is continuous at $f(a)$ then H is continuous at a.

The proof of this theorem relies on the basic properties of limits given in Theorem 1.9.

Remark. As a consequence of the above theorem, polynomials in any number of variables are continuous at any point. Also, rational functions are continuous at any point where they are defined, i.e. excluding the points where the denominator vanishes but the numerator does not.

Example 1.16

Find the limit

$$\lim_{(x,y)\to(2,3)} \frac{x + \sin y}{x^2y^2 + 1}$$

Solution We have

$$\lim_{(x,y)\to(2,3)} (x + \sin y) = \lim_{(x,y)\to(2,3)} x + \lim_{(x,y)\to(2,3)} \sin y = 2 + \sin(\lim_{(x,y)\to(2,3)}) = 2 + \sin 3$$

and

$$\lim_{(x,y)\to(2,3)} x^2y^2{+}1 = \lim_{(x,y)\to(2,3)} x^2 \lim_{(x,y)\to(2,3)} y^2{+}1 = [\lim_{(x,y)\to(2,3)} x]^2[\lim_{(x,y)\to(2,3)} y]^2{+}1 = 2^2 3^2{+}1 = 37 \neq 0.$$

Hence

$$\lim_{(x,y)\to(2,3)} \frac{x + \sin y}{x^2y^2 + 1} = \frac{2 + \sin 3}{37}.$$

■

Sometimes we may have points which are not in the domain but at which the limit of the function exists. Such point can be added to the domain by assigning the value of the

function to be given by the value of the limit. This is called *extension by continuity* as we extend the domain by adding a new point in such a way that the function is continuous at that point.

Example 1.17

Determine whether the function

$$f(x, y) = x^y$$

defined for $x > 0$ and $y > 0$, can be made continuous at $(0, 0)$. (In other words determine whether the domain of f can be extended by continuity to include $(0, 0)$.)

Solution The value of the function at $(0, 0)$ is not defined. Can we modify the definition of f such that the function becomes continuous? Note that

$$f(0, y) = 0^y = 0$$

for any $y > 0$, thus picking $(x_n, y_n) = (0, \frac{1}{n}) \to (0, 0)$ we get

$$\lim_{n\to\infty} f(x_n, y_n) = \lim_{n\to\infty} 0^{\frac{1}{n}} = \lim_{n\to\infty} 0 = 0\,.$$

On the other hand, for $(x_n, y_n) = (\frac{1}{n}, 0) \to (0, 0)$,

$$\lim_{n\to\infty} f(x_n, y_n) = \lim_{n\to\infty} \left(\frac{1}{n}\right)^0 = \lim_{n\to\infty} 1 = 1\,.$$

We can choose to assign $f(0, 0)$ to be 0 or 1 (or any other number), but f *cannot be made continuous.* ■

2 Differentiation in several variables

2.1 Partial derivatives

We start by the first, natural extension of the concept of derivative to functions of several variables.

Definition 2.1 Partial derivatives

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ (a function of two variables) and let $(x_0, y_0) \in \mathbb{R}^2$. The *partial derivatives* of f at (x_0, y_0) denoted

$$\frac{\partial f}{\partial x}(x_0, y_0)\,, \qquad \frac{\partial f}{\partial y}(x_0, y_0)$$

are defined as

$$\begin{aligned} \frac{\partial f}{\partial x}(x_0, y_0) &= \lim_{\Delta x \to 0} \frac{f(x_0 + \Delta x, y_0) - f(x_0, y_0)}{\Delta x} = \lim_{x \to x_0} \frac{f(x, y_0) - f(x_0, y_0)}{x - x_0}\,, \\ \frac{\partial f}{\partial y}(x_0, y_0) &= \lim_{\Delta y \to 0} \frac{f(x_0, y_0 + \Delta y) - f(x_0, y_0)}{\Delta y} = \lim_{y \to y_0} \frac{f(x_0, y) - f(x_0, y_0)}{y - y_0}\,. \end{aligned} \tag{2.1}$$

This definition means that to define $\frac{\partial f}{\partial x}$ we consider $f(x, y)$ as a function of x only, keeping $y = y_0$ fixed, and differentiate with respect to x at the point x_0 as a function of one variable. Similarly, to define $\frac{\partial f}{\partial y}$ one fixes x and differentiates with respect to y at y_0.

More precisely, if we define $B(x) = f(x, y_0)$ and $C(y) = f(x_0, y)$, then

$$\frac{\partial f}{\partial x}(x_0, y_0) = \frac{dB}{dx}(x_0), \quad \frac{\partial f}{\partial y}(x_0, y_0) = \frac{dC}{dy}(y_0).$$

Example 2.2

For

$$f(x, y) = x^2 + y^3\,,$$

fix $y = y_0$, and consider

$$f(x, y_0) = x^2 + y_0^3$$

as a function of x only. Then in differentiating it with respect to x, we treat y_0^3 as a constant, hence

$$\frac{\partial f}{\partial x}(x_0, y_0) = 2x_0\,.$$

Similarly, fixing $x = x_0$ we consider

$$f(x_0, y) = x_0^2 + y^3$$

as a function of y only, and obtain

$$\frac{\partial f}{\partial y}(x_0, y_0) = 3y_0^2\,.$$

Often we do not care about a specific point (x_0, y_0) and we just work out $\frac{\partial f}{\partial x}(x, y)$, $\frac{\partial f}{\partial y}(x, y)$

at any point (x, y). In this case we often just write

$$\frac{\partial f}{\partial x}\,, \quad \frac{\partial f}{\partial y}\,.$$

Other alternative notations are

$$\partial_x f(x,y)\,, \quad \partial_y f(x,y)$$

or

$$f_x(x,y)\,, \quad f_y(x,y)$$

where the arguments sometimes may be omitted for brevity.

Here are some more explicit examples of functions and their partial derivatives.

Example 2.3

For

$$f(x,y) = x^3 + 3xy + y^2\,,$$

we calculate

$$f_x = \frac{\partial f}{\partial x} = \frac{\partial f}{\partial x}(x,y) = 3x^2 + 3y\,,$$
$$f_y = \frac{\partial f}{\partial y} = \frac{\partial f}{\partial y}(x,y) = 3x + 2y\,.$$

For

$$f(x,y) = \frac{x}{y}\,,$$

we calculate

$$\frac{\partial f}{\partial x} = \frac{1}{y}\,, \qquad \frac{\partial f}{\partial y} = -\frac{x}{y^2}\,.$$

Similarly, if $f : \mathbb{R}^n \to \mathbb{R}$ we define $\frac{\partial f}{\partial x_i}$ for each label $i = 1, 2, \ldots, n$.

Example 2.4

For

$$f(x_1, x_2, x_3, x_4) = x_1^2 + x_2^3 + x_3^4 + x_4^5$$

we have

$$\frac{\partial f}{\partial x_3} = 4x_3^3\,.$$

For

$$f(x,y,z) = xy\ln(z)$$

we have

$$\frac{\partial f}{\partial z} = \frac{xy}{z}\,.$$

Example 2.5

For

$$f(x, y) = x^y \,,$$

find f_x, f_y.

Solution To find f_x, we fix y. Then x^y is just an ordinary power function of x so that

$$f_x = \frac{\partial x^y}{\partial x} = y \cdot x^{y-1} \,.$$

To find f_y, we first rewrite f as

$$f(x, y) = e^{y \ln(x)} \,.$$

Treating x as a constant and using chain rule we obtain

$$f_y = \frac{\partial e^{y \ln(x)}}{\partial y} = \ln(x) \cdot e^{y \ln(x)} = \ln(x) \cdot x^y \,.$$

Sometimes functions are not given by a single formula that uses elementary functions. In this case to calculate partial derivatives one needs to use the definition 2.1 in terms of limits.

Example 2.6

A function of two variables $f(x,y)$ is defined as follows

$$f(x,y) = \begin{cases} \dfrac{2x^3 + xy + 4y^4}{\sin^2(2x) + \sin^2(3y)} & \text{for } (x,y) \neq (0,0) \\ 0 & \text{for } (x,y) = (0,0) \end{cases} \tag{2.2}$$

Calculate the values

$$\frac{\partial f}{\partial x}(0,0)\,, \quad \frac{\partial f}{\partial y}(0,0)\,.$$

Solution Because at the point $(0,0)$ the formula defining the function at the top row of its definition (2.2) is not valid we need to use the definition of partial derivatives in terms of limits 2.1. We obtain

$$\begin{aligned} \frac{\partial f}{\partial x}(0,0) &= \lim_{\Delta x \to 0} \frac{f(\Delta x, 0) - f(0,0)}{\Delta x} \\ &= \lim_{\Delta x \to 0} \frac{2(\Delta x)^3}{\sin^2(2\Delta x)\Delta x} = \lim_{\Delta x \to 0} \frac{2(\Delta x)^2}{\sin^2(2\Delta x)} = \lim_{\Delta x \to 0} \frac{1}{2} \left(\frac{2\Delta x}{\sin(2\Delta x)} \right)^2 = \frac{1}{2} \cdot 1^2 = \frac{1}{2} \end{aligned} \tag{2.3}$$

where in the last line we have used the limit

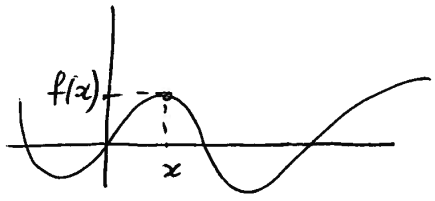
$$\lim_{x \to 0} \frac{\sin kx}{kx} = 1\,, \quad k \neq 0\,. \tag{2.4}$$

Similarly, we calculate

$$\begin{aligned} \frac{\partial f}{\partial y}(0,0) &= \lim_{\Delta y \to 0} \frac{f(0, \Delta y) - f(0,0)}{\Delta y} = \lim_{\Delta y \to 0} \frac{4(\Delta y)^4}{\sin^2(3\Delta y)\Delta y} \\ &= \lim_{\Delta y \to 0} \frac{4(\Delta y)^3}{\sin^2(3\Delta y)} = \lim_{\Delta y \to 0} \frac{4}{9} \left(\frac{3\Delta y}{\sin(3\Delta y)} \right)^2 \Delta y = \frac{4}{9} \cdot 1^2 \cdot 0 = 0\,. \end{aligned} \tag{2.5}$$

2.2 Graphs. Tangent Planes and normal lines.

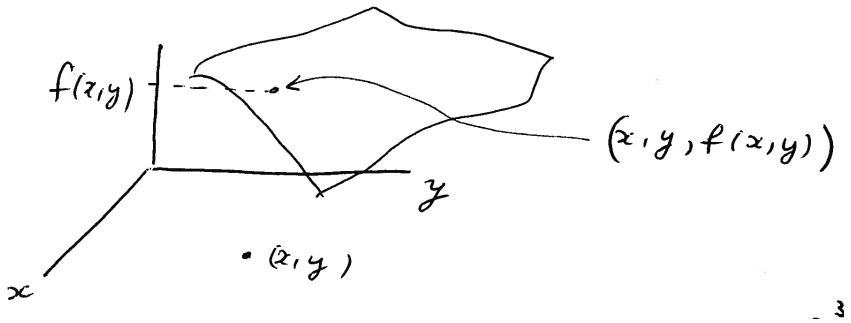
The graph of $f : \mathbb{R} \to \mathbb{R}$ is the set of points in $\mathbb{R}^2$: $\{(x, f(x) | x \in \mathbb{R}\}$, for example:



Similarly the graph of $f : \mathbb{R}^2 \to \mathbb{R}$ is the set of points in the three dimensional space

$$\{(x, y, f(x,y)) | (x,y) \in \mathbb{R}^2\} \subset \mathbb{R}^3\,,$$

for example:



Here, the graph is a two-dimensional surface in $\mathbb{R}^3$ with height $f(x, y)$ above the point (x, y). Its equation is

$$z = f(x, y).$$

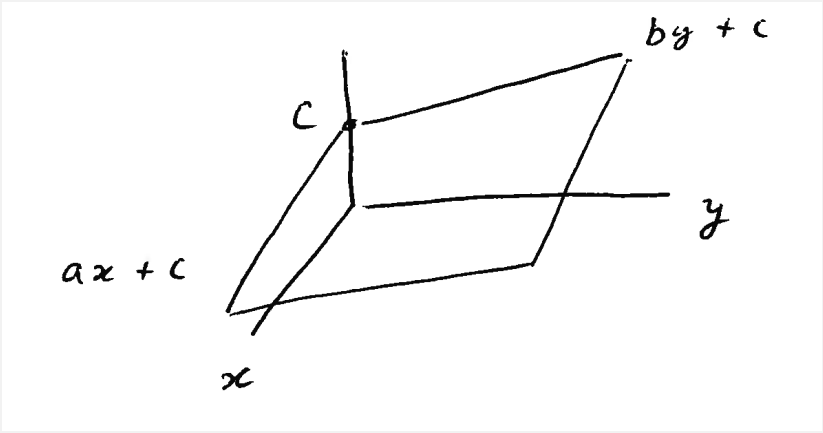
Example 2.7

For a function

$$f(x, y) = ax + by + c$$

the graph is a plane in $\mathbb{R}^3$ given by the equation

$$z = ax + by + c\,.$$



Note that the intersection of this plane with the $y = 0$ (i.e. the xz-) coordinate plane is a straight line, $z = ax + c$, with slope

$$a = \frac{\partial f}{\partial x}\,,$$

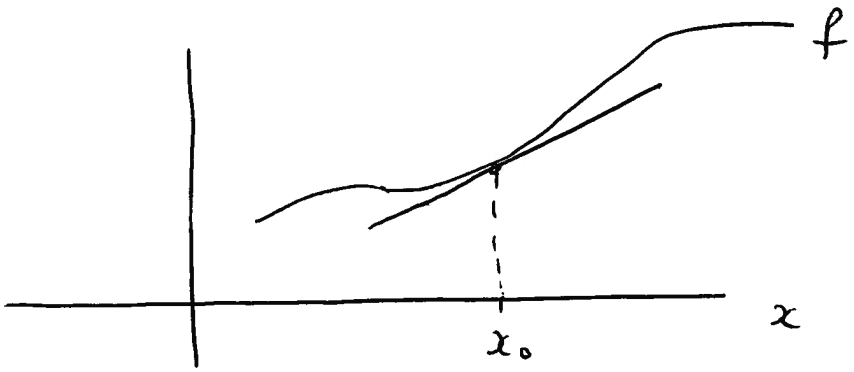
while the intersection with the $x = 0$ (i.e. the yz-) coordinate plane is a straight line, $z = by + c$, with slope

$$b = \frac{\partial f}{\partial y}\,.$$

Recall the geometric definition of a tangent line to a graph of a function of one variable. Let $f(x)$ be a function of variable defined on an interval containing a point $x = x_0$. Consider a secant line $L_{x'}$ that passes through two points on the graph of function f: $(x_0, f(x_0))$ and $(x', f(x'))$. We say that at $x = x_0$ there exists a tangent line L to the graph of $f(x)$ if the

angle θ between L and the secant line $L_{x'}$ goes to zero as $x' \to x_0$. If the function f has a derivative $f'(x_0)$ defined at $x = x_0$ then the value $f'(x_0)$ gives the slope of the tangent line at x_0, so that the equation of the tangent line is

$$y = (x - x_0) \cdot \frac{df}{dx}(x_0) + f(x_0)\,. \tag{2.6}$$



Similarly, for $f : \mathbb{R}^2 \to \mathbb{R}$, a plane T passing through a point on the graph $P = (x_0, y_0, f(x_0, y_0))$ is called a *tangent plane* at $(x, y) = (x_0, y_0)$ if for any point on the graph $P' = (x', y', f(x', y'))$ the angle between T and the vector $\overrightarrow{PP'}$ tends to zero as $P' \to P$.

Assume now that we want to find the plane tangent to the graph of such a function f. This graph is the surface of equation $z = f(x, y)$. To determine its equation, we need to specify a point on it and a vector normal (i.e. perpendicular) to it at that point. To do this, we find first two tangent vectors to the surface at the point P_0 then take their cross product - this will give us a vector perpendicular to both, hence perpendicular to the surface. This vector and the point determine the tangent plane.

If we take the intersection of the graph with the plane $y = y_0$, we obtain a curve in the (x, z) plane, given by $g(x) = f(x, y_0)$. The tangent to this curve at (x_0, z_0) has slope given by the derivative $dg/dx(x_0)$. But this derivative by definition is the partial derivative of f with respect to x, namely $\dfrac{\partial f}{\partial x}(x_0, y_0)$. Points on this tangent line have the form $(x, y_0, f(x_0, y_0) + f_x(x_0, y_0)(x - x_0))$. Hence a tangent vector, obtained by taking the derivative with respect to x of all components, is given by $(1, 0, f_x(x_0, y_0))$.

Similarly, the slope of the tangent to the curve obtained by the intersection of the graph with the plane $x = x_0$ at the point P_0 is given by $\dfrac{\partial f}{\partial y}(x_0, y_0)$, yielding the second tangent vector $(0, 1, f_y(x_0, y_0))$.

In summary, the two partial derivatives give us the directions of two vectors that are tangent to the graph of the function. These vectors are

$$(1, 0, f_x(x_0, y_0)) \quad \textit{and} \quad (0, 1, f_y(x_0, y_0).$$

We now form their cross product to find a vector normal to the graph at the point P_0:

$$(1, 0, f_x(x_0, y_0)) \times (0, 1, f_y(x_0, y_0) = \det \begin{pmatrix} i & j & k \\ 1 & 0 & f_x(x_0, y_0) \\ 0 & 1 & f_y(x_0, y_0) \end{pmatrix} = (-f_x(x_0, y_0), -f_y(x_0, y_0), 1)$$

So the plane tangent to the graph of f at P_0 is the plane through P_0 with the above as the normal vector. The equation of this plane is

$$-f_x(x_0, y_0)(x - x_0) - f_y(x_0, y_0)(y - y_0) + (z - f(x_0, y - 0)) = 0.$$

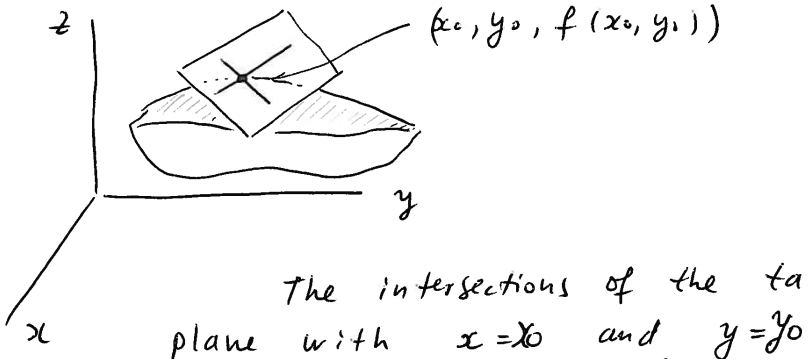
In summary, rearranging this expression we have proved the following.

Theorem 2.8

Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ and assume that the partial derivatives of f are defined at (x_0, y_0). Then the graph of f admits a tangent plane at $(x_0, y_0, f(x_0, y_0))$, which is defined by the equation

$$\boxed{z = (x - x_0) \cdot \frac{\partial f}{\partial x}(x_0, y_0) + (y - y_0) \cdot \frac{\partial f}{\partial y}(x_0, y_0) + f(x_0, y_0)\,.} \tag{2.7}$$

Note that in this equation the only variables are x, y, z, so that the equation is linear in these variables, while the rest of the quantities are numbers that depend on the chosen point (x_0, y_0).



Thus, partial derivatives allow us to find the explicit equation for a tangent plane. Indeed, the intersections of the tangent plane with $x = x_0$ and $y = y_0$ planes (parallel to the coordinate planes) are straight lines with the slopes $f_y(a, y_0)$, $f_x(x_0, y_0)$, respectively. This generalizes the geometric interpretation of the derivative of a function of one variable.

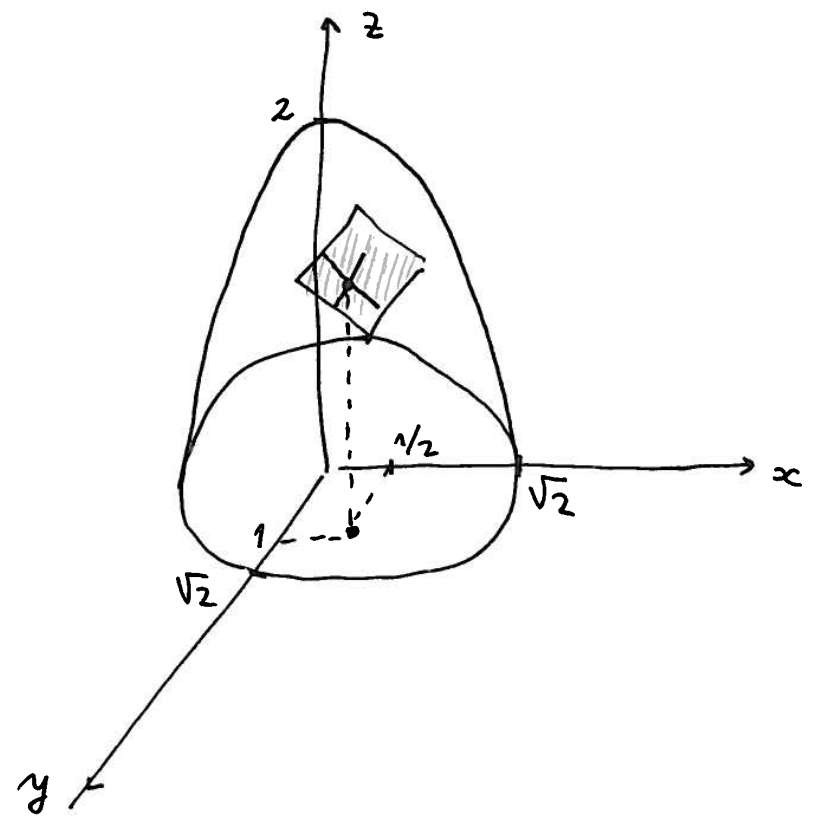
Example 2.9

Find the tangent plane to the graph of $f(x,y) = 2 - x^2 - y^2$ at the point with $x_0 = 1/2$, $y_0 = 1$.

Solution The graph of f is a surface in $\mathbb{R}^3$ with coordinates x, y, z given by the equation

$$z = 2 - x^2 - y^2 \,.$$

We work out the value $z_0 = f(x_0, y_0) = 2 - 1/4 - 1 = 3/4$.



The surface $z = 2 - x^2 - y^2$ is a paraboloid of revolution whose symmetry axis is the z-axis. The intersection with the xy-plane ($z = 0$) is the circle $x^2 + y^2 = 2$. We further find

$$f_x = -2x \,, \qquad f_y = -2y \,,$$

so that

$$f_x\left(\frac{1}{2}, 1\right) = -1 \,, \qquad f_y\left(\frac{1}{2}, 1\right) = -2 \,.$$

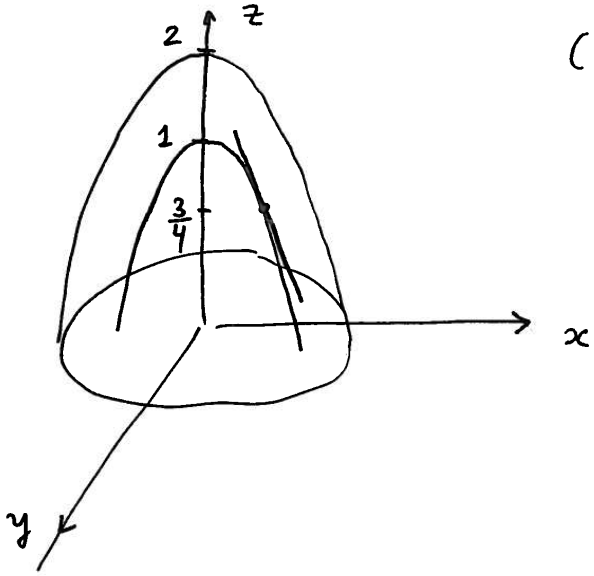
Hence the tangent plane equation is

$$z = (-1) \cdot \left(x - \frac{1}{2}\right) + (-2) \cdot (y - 1) + \frac{3}{4} = -x - 2y + \frac{13}{4} \,.$$

In this example, the intersection with the $y = 1$ plane is a parabola

$$z = 2 - x^2 - 1 = 1 - x^2 \,.$$

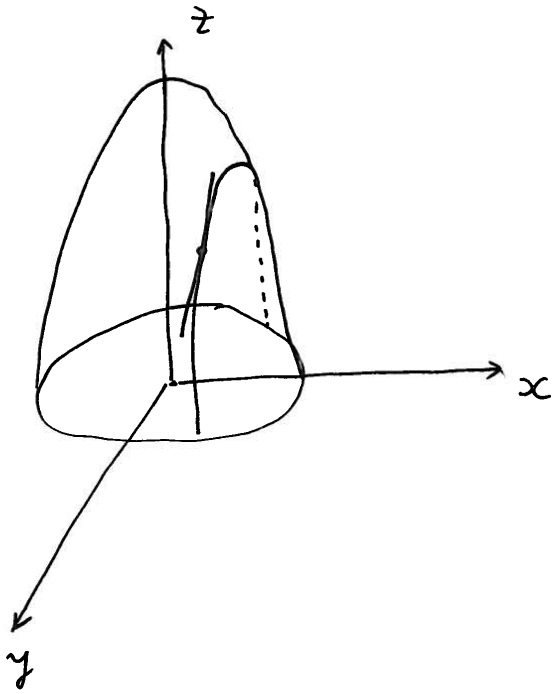
The slope of the tangent line at $x = \frac{1}{2}$ is -1, which is the value of f_x:



The intersection with the $x = \frac{1}{2}$ plane is a parabola

$$z = 2 - \frac{1}{4} - y^2 = \frac{7}{4} - y^2 \,.$$

The slope of the tangent line to this parabola at $y = 1$ is -2, which is the value of f_y:



The tangent plane

$$z = -x - 2y + \frac{13}{4}$$

contains both of the above tangent lines.

As a final example, note that if we take $(x, y) = (0, 0)$, then

$$f_x(0,0) = f_y(0,0) = 0\ ,$$

which means that at the top of the graph the tangent lines are flat and the tangent plane is parallel to the xy-plane:



From the above discussion it is clear that, thinking geometrically, partial derivatives can be thought of as derivatives in the directions along the coordinate axes. This observation leads to a natural generalization of partial derivatives to that of *directional derivatives*. To define that we first define what a gradient of a function is.

Definition 2.10 (Gradient)

Let $f : \mathbb{R}^n \to \mathbb{R}$ be a function of n-variables denoted $x_1, x_2, \ldots, x_n$. The gradient of f denoted $\mathrm{grad} f(x_1, \ldots, x_n)$ is defined to be the n-vector with components

$$\mathrm{grad} f(x_1, \ldots, x_n) = \left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \ldots, \frac{\partial f}{\partial x_n} \right) \tag{2.8}$$

The gradient of f is also denoted ∇f.

Example 2.11

Find $\mathrm{grad} f$ for $f(x, y, z) = xy^2 + zxy + e^{2z} + 7$.
Evaluating the partial derivatives we obtain

$$\mathrm{grad} f = (y^2 + zy, 2xy + zx, xy + 2e^{2z})\ .$$

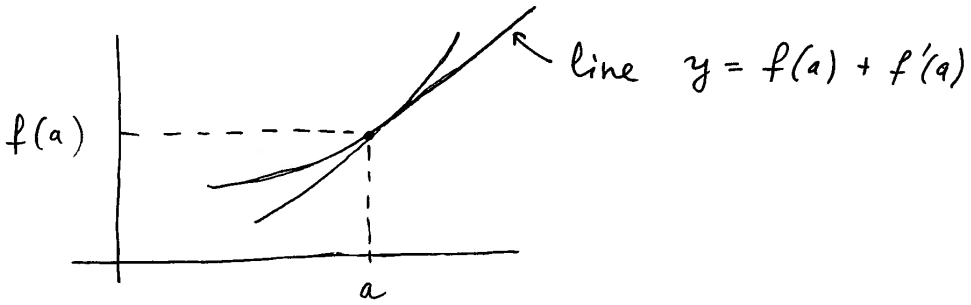
2.3 Rigorous definition of differentiability of functions of two variables

We start with the following intuitive definition: a function $f : \mathbb{R}^n \to \mathbb{R}$ is called *differentiable at* $a \in \mathbb{R}^n$ if it has a good linear approximation at a.

If a function of one variable $f = f(x)$ has a derivative f' defined at $x = a$ then we can take as a linear approximation at $x = a$ the following linear function

$$L_a(x) = f(a) + f'(a)(x - a)\,.$$

The graph of this linear function is precisely the tangent line to the graph of $f(x)$ at $x = a$.



A good approximation means that the error

$$f(x) - L_a(x)$$

is small. The smallness of the error is controlled by how close x is to a that is by how small $|x - a|$ is. For the error to be small it is reasonable to demand that $f(x) - L_a(x)$ vanishes as $x \to a$ faster than $x - a$. More precisely we demand that

$$\lim_{x\to a} \frac{f(x) - L_a(x)}{x - a} = 0\,.$$

For example if it happens that $f(x) - L_a(x) = C \cdot (x - a)^2$ for some constant C then

$$\lim_{x\to a} \frac{f(x) - L_a(x)}{x - a} = \lim_{x\to a} \frac{C(x - a)^2}{x - a} = \lim_{x\to a} C(x - a) = 0\,.$$

These considerations lead us to the following precise definition

Definition 2.12

A function of one variable $f = f(x)$ is differentiable at $x = a$ if $f'(a)$ exists and

$$\lim_{x\to a} \frac{f(x) - L_a(x)}{x - a} = 0\,. \tag{2.9}$$

We will now show that for a function of one variable differentiability at a point $x = a$ (in the sense of the above definition) follows from the existence of the derivative $\frac{df}{dx}(a)$. Substituting the explicit form of $L_a(x)$ into (2.9) we obtain

$$\begin{aligned}\lim_{x\to a} \frac{f(x) - L_a(x)}{x - a} &= \lim_{x\to a} \frac{f(x) - f'(a)(x - a) - f(a)}{x - a} \\ &= \lim_{x\to a} \left(\frac{f(x) - f(a)}{x - a} - f'(a) \right) = f'(a) - f'(a) = 0\,.\end{aligned} \tag{2.10}$$

Hence, any function of one variable that has $f'(a)$ is differentiable at a.

Consider next a function of two variables $f = f(x, y)$ for which there exists partial derivatives $f_x(a, b)$, $f_y(a, b)$ at $(x, y) = (a, b)$. As a linear approximation we can take the linear function whose graph gives the tangent plane at (a, b):

$$L_{(a,b)}(x, y) = f(a, b) + (x - a)f_x(a, b) + (y - b)f_y(a, b)\,.$$

As a measure of how close a point (x, y) is to the point (a, b) we can take the Euclidean distance between these two points

$$\sqrt{(x - a)^2 + (y - b)^2}\,.$$

This prompts the following definition

Definition 2.13 Function of two variables differentiable at a point

A function $f(x, y)$ is called *differentiable at* $(x, y) = (a, b)$ if $f_x(a, b)$, $f_y(a, b)$ exist and

$$\lim_{(x,y)\to(a,b)} \frac{f(x, y) - L_{(a,b)}(x, y)}{\sqrt{(x - a)^2 + (y - b)^2}} = 0\,.$$

This definition means that for a differentiable function the error of linear approximation vanishes faster than the distance between (x, y) and (a, b) as $(x, y) \to (a, b)$.

It can be shown that $f(x, y)$ is differentiable at (a, b) if and only if the tangent plane at $(x, y) = (a, b)$ exists (in the sense of the geometric definition we gave in section 2.2).

Definition 2.14 Differentiable function of two variables

A function of two variables $f : D_f \to \mathbb{R}$ is called *differentiable* if it is differentiable at every point in its domain D_f.

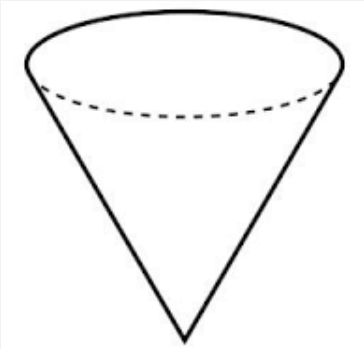
It is not hard to construct examples of functions for which one or both partial derivatives do not exist at certain exceptional points.

Example 2.15

Consider a function

$$f(x,y) = \sqrt{x^2+y^2}$$

Its graph $z = \sqrt{x^2+y^2}$ is an inverted cone with a tip at the origin which expands along the positive z direction.



It is clear geometrically that at the tip of the cone we cannot define a unique tangent plane. Let us prove formally that partial derivatives of f do not exist at the tip of the cone. If we try blindly differentiating the formula for f we obtain

$$\frac{\partial f}{\partial x} = \frac{x}{\sqrt{x^2+y^2}}$$

that does not have a value at $(x,y) = (0,0)$, we get a formal $\frac{0}{0}$ expression if we try to substitute $x = 0$, $y = 0$. Instead let us use the definition in terms of limits

$$\frac{\partial f}{\partial x}(0,0) = \lim_{h\to 0} \frac{f(h,0) - f(0,0)}{h} = \lim_{h\to 0} \frac{\sqrt{(h)^2}}{h} = \lim_{h\to 0} \frac{|h|}{h}$$

Inside the limit we have a step function for which the limit does not exist (it is different as $h \to 0^+$ from the positive side and $h \to 0^-$ from the negative side) Hence $\frac{\partial f}{\partial x}(0,0)$ does not exist. Similarly we can show that $\frac{\partial f}{\partial y}(0,0)$ also does not exist.

Example 2.16

Consider a function

$$g(x,y) = -|x|$$

Its graph looks like two half-planes: $z = x, x > 0$ and $z = -x, x < 0$, joined at the y-axis: $z = x = 0$. For all points with $x > 0$ the tangent plane is $z = x$, while for all points with $x < 0$ the tangent plane is $z = -x$. Right on the "ridge" where $x = 0$ a unique tangent plane cannot be defined. As in the previous example, using the definition, we can show that while the partial derivative $\frac{\partial g}{\partial y} = 0$ for all points including $x = 0$, $\frac{\partial g}{\partial x}$ is not defined when $x = 0$.

For a function of two variables to be differentiable at a point its partial derivatives must

exist at that point. But, unlike in the case of functions of one variable, this may not be enough. The following example gives an example of a function for which both partial derivatives exist at a certain point while the function is not differentiable at that point.

Example 2.17

Consider

$$h(x,y) = x^{1/3}y^{1/3}\,.$$

This function is defined on the entire $\mathbb{R}^2$. In particular $h(0,0) = 0$. Moreover, we find that

$$\frac{\partial h}{\partial x}(0,0) = \lim_{\Delta x\to 0}\frac{h(\Delta x,0)-h(0,0)}{\Delta x} = \lim_{\Delta x\to 0}\frac{(\Delta x)^{1/3}\cdot 0 - 0}{\Delta x} = \lim_{\Delta x\to 0} 0 = 0\,,$$

$$\frac{\partial h}{\partial y}(0,0) = \lim_{\Delta y\to 0}\frac{h(0,\Delta y)-h(0,0)}{\Delta y} = \lim_{\Delta x\to 0}\frac{0\cdot(\Delta y)^{1/3} - 0}{\Delta y} = \lim_{\Delta y\to 0} 0 = 0$$

so that both partial derivatives exist at $(0,0)$. For this function to be differentiable we need to examine the limit

$$\lim_{(x,y)\to(0,0)}\frac{h(x,y)-L_{(0,0)}(x,y)}{\sqrt{x^2+y^2}} = \lim_{(x,y)\to(0,0)}\frac{x^{1/3}y^{1/3}}{\sqrt{x^2+y^2}} \tag{2.11}$$

Using the sequential definition we can show that the above limit does not exist. Considering a sequence $(x_n,y_n) = (\frac{1}{n},\frac{1}{n}) \to (0,0)$ we get

$$\lim_{n\to\infty}\frac{(x_n)^{1/3}(y_n)^{1/3}}{\sqrt{x_n^2+y_n^2}} = \lim_{n\to\infty}\frac{(1/n)^{1/3}\cdot(1/n)^{1/3}}{\sqrt{(1/n)^2+(1/n)^2}} = \lim_{n\to\infty}\frac{n^{1/3}}{\sqrt{2}}$$

that does not exist (it goes to infinity). This suffices to conclude that the limit (2.11) does not exist and hence the function is not differentiable at $(0,0)$.

There is a simple sufficient condition for a function to be differentiable.

Theorem 2.18

If a function of two variables $f = f(x,y)$ with $(a,b) \in D_f$ has partial derivatives $f_x(x,y)$, $f_y(x,y)$ defined in some open disc centred at (a,b) and if the functions $f_x(x,y)$, $f_y(x,y)$ are continuous at $(x,y) = (a,b)$ then f is differentiable at (a,b).

This theorem implies in particular that polynomials in two variables and rational functions of two variables are differentiable functions.

Proof: Consider the tangent plane to the graph of f at $(a,b,f(a,b)$, which has equation

$$L_{(a,b)}(x,y) = f(a,b) + (x-a)f_x(a,b) + (y-b)f_y(a,b).$$

We need to show

$$\lim_{(x,y)\to(a,b)}\frac{|f(x,y)-L_{(a,b)}(x,y)|}{\sqrt{(x-a)^2+(y-b)^2}} = 0, \tag{2.12}$$

which implies that the same quantity without absolute value also has zero limit. So we need

to show that

$$\lim_{(x,y)\to(a,b)} \frac{|f(x,y)-[f(a,b)+(x-a)f_x(a,b)+(y-b)f_y(a,b)]|}{\sqrt{(x-a)^2+(y-b)^2}} = 0\,.$$

Write

$$\begin{aligned}&\frac{f(x,y)-[f(a,b)+(x-a)f_x(a,b)+(y-b)f_y(a,b)]}{\sqrt{(x-a)^2+(y-b)^2}} = \\ =&\frac{f(x,y)-f(a,y)+f(a,y)-f(a,b)-(x-a)f_x(a,b)-(y-b)f_y(a,b)]}{\sqrt{(x-a)^2+(y-b)^2}}\end{aligned}$$

By the mean value theorem in one dimension, there exist points $\tilde{a}$ in (a,x) and $\tilde{b}$ in (b,y) (we assume WLOG that $x>a$ and $y>b$) such that

$$f(x,y)-f(a,y)=f_x(\tilde{a},y)(x-a), \quad f(a,y)-f(a,b)=f_y(a,\tilde{b})(y-b).$$

Hence

$$\begin{aligned}&\frac{f(x,y)-[f(a,b)+(x-a)f_x(a,b)+(y-b)f_y(a,b)]}{\sqrt{(x-a)^2+(y-b)^2}} = \\ =&\frac{[f_x(\tilde{a},y)-f_x(a,b)](x-a)+[f_y(a,\tilde{b})-f_y(a,b)](y-b)}{\sqrt{(x-a)^2+(y-b)^2}}\end{aligned}$$

Now use the triangle inequality to deduce

$$\frac{|[f_x(\tilde{a},y)-f_x(a,b)](x-a)+[f_y(a,\tilde{b})-f_y(a,b)](y-b)|}{\sqrt{(x-a)^2+(y-b)^2}} \leq$$

$$|f_x(\tilde{a},y)-f_x(a,b)|\frac{|x-a|}{\sqrt{(x-a)^2+(y-b)^2}}+|f_y(a,\tilde{b})-f_y(a,b)||\frac{|y-b|}{\sqrt{(x-a)^2+(y-b)^2}}.$$

Finally, notice that

$$|x-a|=\sqrt{(x-a)^2}\leq\sqrt{(x-a)^2+(y-b)^2}, \quad |y-b|=\sqrt{(y-b)^2}\leq\sqrt{(x-a)^2+(y-b)^2}.$$

Summarising all these inequality, we find

$$\begin{aligned}0 \leq &\lim_{(x,y)\to(a,b)} \frac{|f(x,y)-[f(a,b)+(x-a)f_x(a,b)+(y-b)f_y(a,b)]|}{\sqrt{(x-a)^2+(y-b)^2}} \\ &\leq |f_x(\tilde{a},y)-f_x(a,b)|+|f_y(a,\tilde{b})-f_y(a,b)| \to_{(x,y)\to(a,b)} 0,\end{aligned}$$

where the limit on the right side holds because the partial derivatives are continuous. By the sandwich property, we proved that the limit in (2.12) is zero hence that the function is differentiable. ■

2.4 Higher order partial derivatives

If $f : \mathbb{R}^2 \to \mathbb{R}$, then $\frac{\partial f}{\partial x}$ is also a function of two variables. Thus we can consider its partial derivatives

$$\frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial x^2} = f_{xx}\,, \quad \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial y \partial x} = f_{xy}\,, \tag{2.13}$$

where we exhibited different notations used for these second order partial derivatives. Similarly we can consider

$$\frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial^2 f}{\partial x \partial y} = f_{yx}\,, \quad \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial^2 f}{\partial y^2} = f_{yy}\,. \tag{2.14}$$

Example 2.19

For $f(x,y) = x^3 - 4xy^2$ we calculate

$$\frac{\partial f}{\partial x} = 3x^2 - 4y^2\,, \qquad \frac{\partial f}{\partial y} = -8xy\,, \qquad \frac{\partial^2 f}{\partial x^2} = 6x\,,$$

$$\frac{\partial^2 f}{\partial y \partial x} = -8y\,, \qquad \frac{\partial^2 f}{\partial x \partial y} = -8y\,, \qquad \frac{\partial^2 f}{\partial y^2} = -8x\,.$$

Theorem 2.20 The mixed derivatives theorem

Consider $f : D \to \mathbb{R}$ for some $D \subset \mathbb{R}^2$. If f_x, f_y, f_{xy}, f_{yx} exist at any $(x,y) \in D$ and moreover the functions f_{xy}, f_{yx} are *continuous*, then

$$f_{xy}(x,y) = f_{yx}(x,y) \qquad \text{for any } (x,y) \in D\,.$$

Outline of Proof
Fix the point (x_0, y_0), and define

$$F(h,k) = \frac{1}{hk}\left[f(x_0+h, y_0+k) - f(x_0, y_0+k) - f(x_0+h, y_0) + f(x_0, y_0)\right]$$

Then a direct calculation shows that

$$\frac{\partial}{\partial y}\frac{\partial f}{\partial x}(x_0,y_0) = \lim_{k\to 0}\lim_{h\to 0} F(h,k),$$
$$\frac{\partial}{\partial x}\frac{\partial f}{\partial y}(x_0,y_0) = \lim_{h\to 0}\lim_{k\to 0} F(h,k),$$

Applying the mean value theorem four times we also find, for some numbers $\theta_1, ..\theta_4$ in $(0,1)$,

$$\begin{aligned} F(h,k) &= \frac{1}{h}\left[\frac{\partial f}{\partial y}(x_0+h, y_0+\theta_1 k) - \frac{\partial f}{\partial y}(x_0,y_0)\right] \\ &= \frac{\partial}{\partial x}\frac{\partial f}{\partial y}(x_0+\theta_2 h, y_0+\theta_1 k), \\ F(h,k) &= \frac{1}{k}\left[\frac{\partial f}{\partial x}(x_0+\theta_3 h, y_0+k) - \frac{\partial f}{\partial x}(x_0,y_0)\right] \\ &= \frac{\partial}{\partial y}\frac{\partial f}{\partial x}(x_0+\theta_3 h, y_0+\theta_4 k). \end{aligned}$$

Hence, for all h, k,

$$\frac{\partial}{\partial x}\frac{\partial f}{\partial y}(x_0+\theta_2 h, y_0+\theta_1 k) = \frac{\partial}{\partial y}\frac{\partial f}{\partial x}(x_0+\theta_3 h, y_0+\theta_4 k).$$

Taking the limit as $h, k \to 0$ and using the continuity of the mixed partial derivatives yields the result. ■

When the conditions of this theorem are violated the mixed derivatives can be different. This happens in the following example.

Example 2.21

Consider a function defined so that

$$f(x,y) = xy\frac{x^2-y^2}{x^2+y^2} \quad \text{for } (x,y) \neq (0,0)$$

and $f(0,0) = 0$. We find

$$\frac{\partial f}{\partial x} = y \cdot \left(\frac{x^2-y^2}{x^2+y^2} + \frac{4x^2y^2}{(x^2+y^2)^2}\right)$$

for $(x,y) \neq (0,0)$. To get the value $f_x(0,0)$, we use the general definition:

$$\frac{\partial f}{\partial x}(0,0) = \lim_{\Delta x \to 0}\left(\Delta x \cdot y \cdot \frac{\Delta x^2 - y^2}{(\Delta x)^2+y^2} - 0)\frac{1}{\Delta x}\right)\Big|_{y=0} = 0\,.$$

Substituting in the first expression $(x,y) = (0,y)$, we find

$$\frac{\partial f}{\partial x}(0,y) = -y\,.$$

Furthermore, by definition,

$$\frac{\partial^2 f}{\partial y \partial x}(0,0) = \lim_{\Delta y \to 0}\left(\frac{\partial f}{\partial x}(0,\Delta y) - \frac{\partial f}{\partial x}(0,0)\right)\frac{1}{\Delta y} = -1\,.$$

Similarly,

$$\frac{\partial f}{\partial y} = x\left(\frac{x^2-y^2}{x^2+y^2} - \frac{4x^2y^2}{(x^2+y^2)^2}\right)$$

for $(x,y) \neq (0,0)$. Also,

$$\frac{\partial f}{\partial y}(x,0) = x \quad \text{and} \quad \frac{\partial^2 f}{\partial x \partial y}(0,0) = 1\,.$$

Thus for this function,

$$\frac{\partial^2 f}{\partial y \partial x}(0,0) = -1 \neq \frac{\partial^2 f}{\partial x \partial y}(0,0) = 1\,.$$

In this course we will predominantly deal with functions for which all derivatives are continuous in a given domain and thus the mixed derivatives will coincide.

Remark. We can easily generalize the definition of higher derivatives to functions of any number of variables. The above theorem generalizes to higher order derivatives.

Example 2.22

Let $f : \mathbb{R} \to \mathbb{R}$ and let

$$u(x, y) = f\left(\frac{y}{x}\right) + 2xy$$

1. Show that
$$x \cdot u_x + y \cdot u_y = 4xy \tag{2.15}$$
2. Deduce that
$$x^2 \cdot u_{xx} + 2xy \cdot u_{xy} + y^2 \cdot u_{yy} = 4xy \tag{2.16}$$

Solution First we calculate using chain rule (in one variable) to obtain

$$u_x = -\frac{y}{x^2} \cdot f'\left(\frac{y}{x}\right) + 2y\,, \qquad u_y = \frac{1}{x} \cdot f'\left(\frac{y}{x}\right) + 2x\,.$$

Hence

$$x \cdot u_x + y \cdot u_y = -\frac{y}{x} \cdot f'\left(\frac{y}{x}\right) + 2xy + \frac{y}{x} \cdot f'\left(\frac{y}{x}\right) + 2xy = 4xy\,,$$

and this is what we had to show in part 1 of the problem.

To do part 2, we first differentiate formula (2.15) with respect to x. This gives

$$u_x + xu_{xx} + yu_{xy} = 4y\,.$$

Hence multiplying this by x we obtain

$$xu_x + x^2u_{xx} + xyu_{xy} = 4xy\,. \tag{2.17}$$

Differentiating (2.15) with respect to y gives

$$xu_{yx} + u_y + yu_{yy} = 4x\,.$$

Multiplying the last expression by y we obtain

$$xyu_{yx} + yu_y + y^2u_{yy} = 4xy\,.$$

Adding together the last equation and equation (2.17) we get

$$x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} + xu_x + yu_y = 8xy\,.$$

Using (2.15) in the last expression we finally obtain (2.16).

Example 2.23

For the function

$$f(x,y,z) = \frac{xyz}{(y^2+z^2)^2}\,,$$

find f_{yxzx} away from the points with $y = z = 0$.

Solution By the mixed derivatives theorem 2.20,

$$f_{yxzx} = f_{xxyz}$$

in the region at hand. We calculate

$$f_x = \frac{yz}{(y^2+z^2)^2}\,, \qquad f_{xx} = 0\,.$$

Therefore,

$$f_{yxzx} = 0\,.$$

Example 2.24

Let $f : \mathbb{R}^1 \to \mathbb{R}^1$ be a function of one variable. Consider the function $h : \mathbb{R}^2 \to \mathbb{R}^1$, defined in terms of f as

$$h(x,y) = f^3(x-y^2) = \left[f(x-y^2)\right]^3\,.$$

Determine $h_x(x,y)$, $h_y(x,y)$ and $h_{xy}(x,y)$ in terms of f, the variables x and y, and in terms of the derivatives f' and f''.

Solution

$$h_x = 3f^2(x-y^2)\cdot\frac{\partial f(x-y^2)}{\partial x} = 3f^2(x-y^2)\cdot f'(x-y^2)$$

$$h_y = 3f^2(x-y^2)\cdot\frac{\partial f(x-y^2)}{\partial y} = 3f^2(x-y^2)\cdot f'(x-y^2)\cdot(-2y)$$

$$\begin{aligned} h_{xy} &= \frac{\partial}{\partial y}\left[3f^2(x-y^2)\cdot f'(x-y^2)\right] \\ &= 3\cdot 2f(x-y^2)\cdot f'(x-y^2)\cdot(-2y)\cdot f'(x-y^2) + 3f^2(x-y^2)\cdot f''(x-y^2)\cdot(-2y)\,. \end{aligned}$$

2.5 The chain rule for functions of several variables.

Suppose f is a function of x and x is a function of u so that we have a composite function $h(u) = f(x(u))$. Then the *chain rule* in 1D (that we have already used above) says that

$$\frac{dh(u)}{du}(u) = \frac{df(x(u))}{dx}(x(u))\cdot\frac{dx(u)}{du}(u)\,. \tag{2.18}$$

It is customary to shorten the notations in this equation to

$$\frac{dh}{du} = \frac{df}{dx}\cdot\frac{dx}{du}\,,$$

or even to the form

$$\frac{df}{du} = \frac{df}{dx} \cdot \frac{dx}{du}\,.$$

Consider now a function $f : \mathbb{R}^2 \to \mathbb{R}^1$ of two variables x and y, where x and y are each functions of u and v. Let

$$h(u, v) := f(x(u, v), y(u, v))$$

denote the composite function.

Theorem 2.25

If the functions $\frac{df}{dx}$ and $\frac{df}{dy}$ are continuous, then

$$\frac{\partial h(u,v)}{\partial u} = \frac{\partial f}{\partial x}(x(u,v), y(u,v)) \cdot \frac{\partial x}{\partial u}(u,v) + \frac{\partial f}{\partial y}(x(u,v), y(u,v)) \cdot \frac{\partial y}{\partial u}(u,v)\,, \tag{2.19}$$

or, for brevity,

$$\frac{\partial h}{\partial u} = \frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial u}\,.$$

Similarly,

$$\frac{\partial h}{\partial v} = \frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial v}.$$

By some abuse of notations, we can also write

$$\begin{aligned} \frac{\partial f}{\partial u} &= \frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial u} \\ \frac{\partial f}{\partial v} &= \frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial v}\,. \end{aligned} \tag{2.20}$$

The proof uses the same ideas as the proof of the chain rule for functions of one variables, using the trick of adding and subtracting the same quantity from an expression and employing the mean value theorem. It is left as an exercise.

The theorem easily generalizes to functions of more variables, e.g. for $f(x, y, z)$ with $x = x(u, v)$, $y = y(u, v)$ and $z = z(u, v)$, we have

$$\frac{\partial f}{\partial u} = \frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial u} + \frac{\partial f}{\partial z} \cdot \frac{\partial z}{\partial u}\,.$$

Note that the number of variables can change when we form a composite function. For example, for $f(x, y)$ with $x = x(u)$ and $y = y(u)$, the composite function $h(u) = f(x(u), y(u))$ is a function of one variable, with

$$\frac{dh}{du} = \frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial u}\,.$$

As another example, consider $f(x)$ with $x = x(u, v)$, so that $h(u, v) := f(x(u, v))$ is a function of two variables, with

$$\frac{\partial h}{\partial u} = \frac{df}{dx} \cdot \frac{\partial x}{\partial u}\,, \qquad \frac{\partial h}{\partial v} = \frac{df}{dx} \cdot \frac{\partial x}{\partial v}\,.$$

Example 2.26

Let $f(x,y) = x^2 + y$, $x = u^2 + v$ and $y = v - u$. Find $\frac{\partial f}{\partial u}$
(*i*) by substitution, and
(*ii*) by using the chain rule.

Solution

$$(i) \qquad f = (u^2+v)^2 + v - u = u^4 + 2u^2v + v^2 + v - u$$
$$\Rightarrow \quad \frac{\partial f}{\partial u} = 4u^3 + 4uv - 1\,.$$

$$(ii) \qquad \frac{\partial f}{\partial u} = \frac{\partial f}{\partial x}\cdot\frac{\partial x}{\partial u} + \frac{\partial f}{\partial y}\cdot\frac{\partial y}{\partial u}$$
$$= 2x\cdot 2u + 1\cdot(-1)$$
$$= 4u(u^2+v) - 1\,.$$

Here, in the second to last line, we have inserted the explicit formula for x in terms of u and v in order to obtain an answer expressed just in terms of these two variables.

Sometimes the "native" variables are not specified for all of the constituent functions. In this case we just introduce them.

Example 2.27

Let f be a function of two variables with continuous first order partial derivatives. A function of three variables h is defined as

$$h(x,y,z) = x\cdot(f(x,e^y) + f(xyz, z^2))\,.$$

Express the partial derivatives h_x, h_y, h_z in terms of the function f and its partial derivatives. Show all arguments. **Solution** To differentiate correctly we need to denote somehow the "native" variables of f. Any notation will do as long as they are distinct from x, y, z. Let $f = f(u,v)$. Then our task is to express h_x, h_y, h_z in terms of the two types of partial derivatives of f: f_u and f_v. Applying the product rule and the chain rule we obtain

$$h_x(x,y,z) = f(x,e^y) + f(xyz,z^2) + x\cdot(f_u(x,e^y) + f_u(xyz,z^2)\cdot yz)\,.$$

Similarly, applying the chain rule we obtain

$$h_y(x,y,z) = x\cdot(f_u(xyz,z^2)\cdot xz + f_v(x,e^y)\cdot e^y)\,,$$

$$h_z(x,y,z) = x\cdot(f_u(xyz,z^2)\cdot xy + f_v(x,e^y)\cdot 2z)\,.$$

Note that the final answer should be always expressed in terms of the variables appearing on the left hand side: x, y, z.

Example 2.28

Let f be a function of x and y with $x = u + v$ and $y = u \cdot v$. Find formulae for $\frac{\partial f}{\partial v}$ and $\frac{\partial^2 f}{\partial u \partial v}$ in terms of the partial derivatives of f w.r.t. x and y.

Solution

$$\begin{aligned} & \frac{\partial x}{\partial u} = 1\,, \ \frac{\partial x}{\partial v} = 1\,, \ \frac{\partial y}{\partial u} = v\,, \ \frac{\partial y}{\partial v} = u \\ \Rightarrow \quad & \frac{\partial f}{\partial v} = \frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial v} = \frac{\partial f}{\partial x} + u \cdot \frac{\partial f}{\partial y} \\ & \frac{\partial^2 f}{\partial u \partial v} = \frac{\partial}{\partial u}\left[\frac{\partial f}{\partial x} + u \cdot \frac{\partial f}{\partial y}\right] = \frac{\partial}{\partial u}\left(\frac{\partial f}{\partial x}\right) + \frac{\partial}{\partial u}\left(u \cdot \frac{\partial f}{\partial y}\right) \\ & \qquad = \frac{\partial}{\partial u}\left(\frac{\partial f}{\partial x}\right) + u \cdot \frac{\partial}{\partial u}\left(\frac{\partial f}{\partial y}\right) + \frac{\partial f}{\partial y}\,. \end{aligned}$$

Here $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ are composite functions of u and v. So, we may use the same chain rule to evaluate the remaining partial derivatives w.r.t. u:

$$\begin{aligned} \frac{\partial}{\partial u}\left(\frac{\partial f}{\partial x}\right) &= \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) \cdot \frac{\partial x}{\partial u} + \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) \cdot \frac{\partial y}{\partial u} = \frac{\partial^2 f}{\partial x^2} + v \cdot \frac{\partial^2 f}{\partial y \partial x} \\ \frac{\partial}{\partial u}\left(\frac{\partial f}{\partial y}\right) &= \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) \cdot \frac{\partial x}{\partial u} + \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) \cdot \frac{\partial y}{\partial u} = \frac{\partial^2 f}{\partial x \partial y} + v \cdot \frac{\partial^2 f}{\partial y^2}\,. \end{aligned}$$

Therefore, using the mixed derivatives theorem, we finally obtain

$$\frac{\partial^2 f}{\partial u \partial v} = \frac{\partial^2 f}{\partial x^2} + uv \cdot \frac{\partial^2 f}{\partial y^2} + (u + v) \cdot \frac{\partial^2 f}{\partial x \partial y} + \frac{\partial f}{\partial y}\,.$$

2.6 Transforming equations.

Sometimes equations can be simplified by changing variables.

Example 2.29

Consider an equation involving the partial derivatives of some function $w = w(x, y)$:

$$w_x + w_y = 0\,.$$

Suppose that $x = u + v$ and $y = u - v$. Transform the equation so that it is expressed in terms of u and v.

Solution

$$w_x = \frac{\partial w}{\partial u}\cdot\frac{\partial u}{\partial x} + \frac{\partial w}{\partial v}\cdot\frac{\partial v}{\partial x}\,, \quad w_y = \frac{\partial w}{\partial u}\cdot\frac{\partial u}{\partial y} + \frac{\partial w}{\partial v}\cdot\frac{\partial v}{\partial y}\,.$$

We have that $u = \frac{(x+y)}{2}$ and $v = \frac{(x-y)}{2}$, so

$$\frac{\partial u}{\partial x} = \frac{1}{2}\,, \quad \frac{\partial u}{\partial y} = \frac{1}{2}\,, \quad \frac{\partial v}{\partial x} = \frac{1}{2}\,, \quad \frac{\partial v}{\partial y} = -\frac{1}{2}\,.$$

Hence

$$w_x = \frac{1}{2}w_u + \frac{1}{2}w_v\,, \quad w_y = \frac{1}{2}w_u - \frac{1}{2}w_v\,,$$

so the equation becomes

$$w_x + w_y = \frac{1}{2}\left(w_u + w_v + w_u - w_v\right) = 0\,,$$

i.e. $w_u = 0\,.$

2.7 Gradient and Directional derivatives

Consider now a vector

$$\vec{A} = (A_1, A_2, \ldots, A_n) \in \mathbb{R}^n$$

of unit length. The unit length condition means that

$$\sqrt{A_1^2 + A_2^2 + \cdots + A_n^2} = 1\,.$$

Unit length vectors define directions.

Definition 2.30

The directional derivative for f along $\vec{A}$ as

$$\partial_{\vec{A}} f(x_1, \ldots, x_n) = \lim_{t\to 0}\frac{f(x_1 + tA_1, x_2 + tA_2, \ldots, x_n + tA_n) - f(x_1, x_2, \ldots, x_n)}{t}.$$

We observe from this definition that when $\vec{A}$ is directed along one of the coordinate axes

this definition gives the partial derivative in the corresponding variable. More generally, we have

Theorem 2.31 Given a unit vector $\vec{A}$ and a function $f : \mathbb{R}^n \to \mathbb{R}$ which admits all partial derivatives, the directional derivative of f is given by

$$\partial_{\vec{A}} f(x_1, \ldots, x_n) = \nabla f \cdot \vec{A} = A_1 \partial_1 f + A_2 \partial_2 f + \ldots A_n \partial_n f \, .$$

Proof (in 2D)
Let us fix a direction $\vec{v} = (v_1, v_2)$. Then by the definition,

$$\partial_{\vec{v}} f(x, y) = \lim_{t \to 0} \frac{f(x + tv_1, y + tv_2) - f(x, y)}{t}.$$

Setting $g(t) = f(x + tv_1, y + tv_2)$, we find that the expression above is equal to

$$\lim_{t \to 0} \frac{g(t) - g(0)}{t} = \frac{dg}{dt}(0) = \frac{d}{dt}[f(x + tv_1, y + tv_2)]_{|t=0}$$

Using the chain rule, we can write

$$\frac{d}{dt}[f(x + tv_1, y + tv_2)] = f_x(x + tv_1, y + tv_2)v_1 + f_y(x + tv_1, y + tv_2)v_2$$

and so

$$\begin{aligned} \lim_{t \to 0} \frac{f(x + tv_1, y + tv_2) - f(x, y)}{t} &= [f_x(x + tv_1, y + tv_2)v_1 + f_y(x + tv_1, y + tv_2)v_2]_{|t=0} \\ &= f_x(x, y)v_1 + f_y(x, y)v_2 = \nabla f \cdot \vec{v}. \end{aligned}$$

■

This means that directional derivatives can be evaluated as linear combinations of partial derivatives.

Example 2.32

Let $f(x, y) = \frac{1}{2}x^4 + y^3$. Calculate the directional derivative along

$$\vec{A} = \frac{1}{\sqrt{5}}(1, 2) \in \mathbb{R}^2$$

at the point $(-1, 1)$ Since $f_x = 2x^3$, $f_y = 3y^2$ we have $f_x(-1, 1) = -2$, $f_y(-1, 1) = 3$ so that

$$\partial_{\vec{A}} f(-1, 1) = A_1 f_x(-1, 1) + A_2 f_y(-1, 1) = \frac{1}{\sqrt{5}} \cdot (-2) + \frac{2}{\sqrt{5}} \cdot 3 = \frac{4}{\sqrt{5}}$$

Gradient and directional derivatives are very important for applications in mathematical physics and geometry.

3 Implicit functions and the Implicit Function Theorem

3.1 Solution functions

Definition 3.1 C^1**-class function**

A function $f : \mathbb{R}^n \to \mathbb{R}$ is called a *C^1-class function* if all of its partial derivatives exist and are continuous.

Definition 3.2 Solution function

Consider an equation of the form

$$f(x, y) = 0\,. \tag{3.1}$$

A solution of this equation is simply any pair of numbers, say, (a, b), for which $f(a, b) = 0$. We will say that a function $y(x)$ is a *solution function* if $f(x, y(x)) = 0$ for all $x \in D_y$, where D_y denotes the domain of $y(x)$. Similarly, we could have a solution function $x(y)$ such that $f(x(y), y) = 0$ for all $y \in D_x$, where D_x denotes the domain of $x(y)$.

Example 3.3

Consider the equation

$$x^2 + y^2 = 1\,.$$

We can solve this equation for y as a function of x, equivalently solve $f(x, y) = 0$ for $f(x, y) = x^2 + y^2 - 1$, and we find two solution functions:

$$y = \sqrt{1 - x^2}\,, \quad y = -\sqrt{1 - x^2}\,, \quad -1 \le x \le 1\,.$$

We could also solve for x as a function of y, resulting in

$$x = \sqrt{1 - y^2}\,, \quad x = -\sqrt{1 - y^2}\,, \quad -1 \le y \le 1\,.$$

In principle, there is nothing special about which variable we solve for, but in practice it might be much easier to solve for one variable rather than the other.

Example 3.4

The equation

$$y - x - \frac{1}{2}\sin y = 0$$

can be easily solved for $x(y) = y - \frac{1}{2}\sin y$, but cannot be solved explicitly for $y(x)$.

Very often, we cannot solve explicitly for either variable.

Example 3.5

The equation

$$\ln\sqrt{x^2 + y^2} = \tan^{-1}\left(\frac{y}{x}\right)$$

cannot be solved explicitly for either variable.

In general, any equation of the form $f(x, y) = 0$ is said to define its solution functions *implicitly* (if they exist).

3.2 Implicit differentiation.

Suppose $y(x)$ is a solution function for $f(x, y) = 0$. We can calculate the derivative $\frac{dy}{dx}$ using partial derivatives and the chain rule. Differentiating $f(x, y(x)) = 0$ with respect to x, we obtain

$$\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \cdot \frac{dy}{dx} = 0\,,$$

hence

$$\boxed{\frac{dy}{dx} = -\frac{f_x}{f_y}\,.} \tag{3.2}$$

This formula is called the *implicit differentiation formula*.

Example 3.6

Let us find $\frac{dy}{dx}$ for $y = \sqrt{1 - x^2}$ from Example 3.3 (with $f(x, y) = x^2 + y^2 - 1$) by
(i) explicit differentiation, and by
(ii) using the implicit differentiation formula.

Solution.

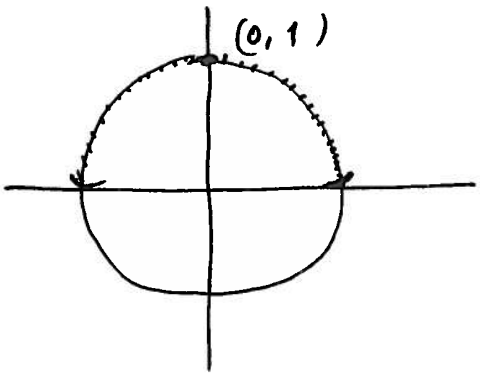
(i) $$\frac{d}{dx}\sqrt{1 - x^2} = -\frac{x}{\sqrt{1 - x^2}}$$

(ii) $$\frac{dy}{dx} = -\frac{f_x}{f_y} = -\frac{2x}{2y} = -\frac{x}{\sqrt{1 - x^2}}\,,$$

where in the last step we substituted $y = \sqrt{1 - x^2}$. ■

3.3 The Implicit Function Theorem

A general solution to $f(x, y) = 0$ may consist either of many solutions, exactly one, or no solutions. The *Implicit Function Theorem* tat wee discuss in this section is a useful theorem about the existence of solutions to such equations. The basic idea of the theorem is that if you can find a particular solution point (a, b), you can then try to find a solution function giving nearby solutions. In Example 3.3, all solutions are given by points on a unit circle:



The solution $y = \sqrt{1-x^2}$ with $-1 \leq x \leq 1$ covers the upper semi-circle – in the graphics above, the solution point $(0,1)$ is marked as an example. The other solution, $y = -\sqrt{1-x^2}$ with $-1 \leq x \leq 1$, covers the lower semi-circle. Note in particular that the lower semi-circle does not include the point $(0,1)$.

For a general equation, we cannot expect to find solutions defined everywhere (*globally*). Finding a *local* solution function near a given solution point is the best we can do in general.

Theorem 3.7 The implicit function theorem (simple case)

Suppose $f : \mathbb{R}^2 \to \mathbb{R}$ is a C^1-class function and that there exists a point $(a,b) \in \mathbb{R}^2$ such that

$$f(a,b) = 0 \quad \text{and} \quad f_y(a,b) \neq 0\,. \tag{3.3}$$

Then there exists $\delta > 0$, $\epsilon > 0$ and a neighborhood $U = \{x : |x-a| < \delta\}$ of $x = a$ such that

$$\begin{aligned} y(a) &= b, \\ f(x, y(x)) = 0, \quad \frac{dy(x)}{dx} &= -\frac{f_x(x,y(x))}{f_y(x,y(x))} \quad \text{for all } x \in U. \end{aligned} \tag{3.4}$$

Moreover, there is a unique y with $|y-b| < \epsilon$ such that $\tilde{y}(a) = b$.

The proof of this theorem, in its general form with $x \in \mathbb{R}^n$, is given in an optional handout.

Remark. In Section 3.2 we found that $f_y(a,b) \neq 0$ was a necessary condition for the C^1-class solution function $y(x)$ to exist - the above theorem states that this condition is also sufficient.

In Example 3.4, $(a,b) = (0,0)$ is a solution to $y - x - \frac{1}{2}\sin y = 0$. Since $f_y = 1 - \frac{1}{2}\cos y$ and $f_y(0) = 1 - \frac{1}{2} \neq 0$, by virtue of the implicit function theorem there exists a C^1-class solution function $y(x)$ in some neighborhood of $x = 0$.

In Example 3.5, $(x,y) = (1,0)$ is a solution point. We have

$$f_y = \frac{y}{x^2+y^2} - \frac{1}{1+\left(\frac{y}{x}\right)^2} \cdot \left(\frac{1}{x}\right)\,,$$

so $f_y(1,0) = -1 \neq 0$. Hence, according to the implicit function theorem there exists a C^1-class solution function $y(x)$ in some neighborhood of $x = 1$.

3.3.1 Many variables case

All of the above generalizes to functions of more variables and to more equations. To start of with let us generalise to the 'many' variables case where the variable x is replaced by the many variables $(x_1, x_2, \cdots, x_n)$. Then we may state the following generalisation of the Implicit Function Theorem to many variables as follows:

Theorem 3.8 The implicit function theorem (many-variables case)

Suppose that the function $f : \mathbb{R}^{n+1} \to \mathbb{R}$ has partial derivatives with respect to each or its arguments, and that there exists a point $(x_1, x_2, \cdots, x_n, y) = (a_1, a_2, \cdots, a_n, b)$ such that

$$f(a_1, a_1, \cdots, a_n, b) = 0 \quad \text{and} \quad f_y(a_1, a_1, \cdots, a_n, b) \neq 0\,. \tag{3.5}$$

Then there exists a solution function $y(x_1, x_2, \cdots, x_n)$ defined in a neighbourhood U of $(x_1, x_2, \cdots, x_n) = (a_1, a_2, \cdots, a_n)$ such that:

$$\begin{aligned} y(a_1, a_2, \cdots, a_n) &= b \\ f(x_1, x_2, \cdots, x_n, y(x_1, x_2, \cdots, x_n)) &= 0, \\ \frac{\partial y}{\partial x_i}(x_1, x_2, \cdots, x_n) &= -\frac{f_{x_i}(x_1, x_2, \cdots, a_n, y(x_1, x_2, \cdots, x_n))}{f_y(x_1, x_2, \cdots, x_n, y(x_1, x_2, \cdots, x_n))}, \\ &\text{for} \quad i \in \{1, 2, \cdots, n\}. \end{aligned} \tag{3.6}$$

Moreover, there exists no other differentiable solution function $\tilde{z}(x, y)$ with $\tilde{z}(a, b) = c$.

For example, an equation like

$$f(x_1, x_2, y) = 0 \tag{3.7}$$

defines y *implicitly* as a function of (x_1, x_2) if the conditions in the theorem with $n = 2$ are met. In this case the partial derivatives are given by

$$\frac{\partial y}{\partial x_1} = -\frac{f_{x_1}(x_1, x_2, y)}{f_y(x_1, x_2, y)}, \quad \frac{\partial y}{\partial x_2}(x_1, x_2, y) = -\frac{f_{x_2}(x_1, x_2, y)}{f_y(x_1, x_2, y)}. \tag{3.8}$$

Example 3.9

Suppose y is defined implicitly by

$$y^2 + x_2^7 + \frac{5yx_2}{x_1^2 + 1} = x_1\,.$$

in the neighbourhood of some point. Find $\frac{\partial y}{\partial x_1}$ and $\frac{\partial y}{\partial x_2}$.

Solution.

$$\begin{aligned} & f(x_1, x_2, y) = y^2 + x_2^7 + \frac{5yx_2}{x_1^2 + 1} - x_1 \\ \Rightarrow \qquad & \frac{\partial y}{\partial x_1} = -\frac{\left(-\frac{5yx_2}{(x_1^2+1)^2} \cdot (2x_1) - 1\right)}{\left(2y + \frac{5x_2}{x_1^2+1}\right)}, \quad \frac{\partial y}{\partial x_2} = -\frac{\left(7x_2^6 + \frac{5y}{x_1^2+1}\right)}{\left(2y + \frac{5x_2}{x_1^2+1}\right)}. \end{aligned}$$

■

Example 3.10

Consider the equation

$$z^2 + 2yx + zx = 0\,.$$

(*i*) Does the implicit function theorem apply, and can we solve for $z = z(x, y)$ near the points

(*a*) $x = 1\ \ y = -1,\ \ z = 1$

(*b*) $x = -2\,,\ y = -\frac{1}{4},\ \ z = 1$?

(*ii*) At the second particular solution (*b*), can one obtain a solution function for $y = y(x, z)$?

Solution. Now we need to be a little careful in using the implicit value theorem, simply because our variable names are different: we need to use the theorem with the relabelling of variables $(x_1, x_2, y) \to (x, y, z)$ in part (i) (since we trying to find $z(x, y)$), and $(x_1, x_2, y) \to (x, z, y)$ in part (ii) (since we are trying to find $y(x, z)$) . Hence, (*i*) We compute $f_z = 2z + x$ and find that

$$f_z(1, -1, 1) = 2 + 1 = 3 \neq 0\,,$$

so a solution function $z(x, y)$ exists near this solution point. On the other hand,

$$f_z(-2, -\frac{1}{4}, 1) = 2 - 2 = 0\,,$$

so the implicit function theorem is not applicable in this case and there exists no solution function $z(x, y)$ near this solution point.

(*ii*) We compute that with $f_y = 2x$,

$$f_y(-2, -\frac{1}{4}, 1) = -4 \neq 0\,,$$

so there exists a solution function $y = y(x, z)$ near that particular solution point. ■

4 Taylor's expansion of functions of two variables

4.1 Recap of Taylor's expansion of functions of one variable

Recall the key theorem regarding polynomial approximation of functions of one variable:

Theorem 4.1

For $f : \mathbb{R}^1 \to \mathbb{R}^1$ such that $f, f', \ldots, f^{(n)}$ are continuous at $x = x_0$, the following formula holds:

$$f(x) = p_{n-1}(x) + r_n(x)\,, \tag{4.1}$$

where $p_{n-1}(x)$ denotes the *Taylor polynomial of degree* $n-1$ *centered at* x_0, and

$$\boxed{R_n(x) = \frac{1}{n!}(x - x_0)^n f^{(n)}(x_*)\,,} \tag{4.2}$$

and where x_* is some point between x and x_0,

$$x_* = x_0 + c(x - x_0)\,, \quad 0 \le c \le 1\,.$$

If we know that the values of the n-th order derivative on the interval between x_0 and x are bounded by a certain constant, i.e. $|f^{(n)}(y)| \le M$, then

$$|R_n(x)| \le \frac{|x - x_0|^n}{n!} \cdot M\,. \tag{4.3}$$

4.2 Taylor's expansion of functions of two variables

Consider now a function $f : \mathbb{R}^2 \to \mathbb{R}$ such that $f, f_x, f_y, f_{xx}, f_{xy}, f_{yy}, \ldots$ up to the order n are all continuous at a given point $(x_0, y_0) \in \mathbb{R}^2$. We may then construct an n-th degree polynomial

$$P(x,y) = C_0 + B_1(x-x_0) + B_2(y-y_0) + D_1(x-x_0)^2 + D_2(x-x_0)(y-y_0) + D_3(y-y_0)^2 + \ldots \tag{4.4}$$

such that

$$\begin{aligned}
P(x_0, y_0) &= f(x_0, y_0) \\
\frac{\partial P}{\partial x}(x_0, y_0) &= \frac{\partial f}{\partial x}(x_0, y_0) \\
\frac{\partial P}{\partial y}(x_0, y_0) &= \frac{\partial f}{\partial y}(x_0, y_0) \\
\frac{\partial^2 P}{\partial x^2}(x_0, y_0) &= \frac{\partial^2 f}{\partial x^2}(x_0, y_0) \\
\frac{\partial^2 P}{\partial x \partial y}(x_0, y_0) &= \frac{\partial^2 f}{\partial x \partial y}(x_0, y_0) \\
\frac{\partial^2 P}{\partial y^2}(x_0, y_0) &= \frac{\partial^2 f}{\partial y^2}(x_0, y_0) \\
&\vdots
\end{aligned} \tag{4.5}$$

It is not difficult to check what the coefficients of this polynomial could be. Indeed:

- Imposing $P(x_0, y_0) = f(x_0, y_0)$ gives $C_0 = f(x_0, y_0)$

- Imposing $\frac{\partial P}{\partial x}(x_0, y_0) = \frac{\partial f}{\partial x}(x_0, y_0)$ gives $B_1 = f_x(x_0, y_0)$.
  Similarly, $B_2 = f_y(x_0, y_0)$.
- At second order we find $D_1 = \frac{1}{2} f_{xx}(x_0, y_0)$, $D_2 = f_{xy}(x_0, y_0)$, $D_3 = \frac{1}{2} f_{yy}(x_0, y_0)$.
- ....

where we can continue this calculation up to the derivatives of arbitrary order n. Such polynomials of two variables give us approximations of $f(x, y)$ for (x, y) near (x_0, y_0).

The polynomial $P_n[f](x_0, y_0)$ obtained in this way is called the *Taylor polynomial of order* n associated with f at (x_0, y_0). How well does it approximate $f(x, y)$?

Example 4.2

We want to find an approximate value for the number $\dfrac{(0.995)^2}{1.002}$. (The calculator on my phone gives 0.98804...)

Set $f(x, y) = \dfrac{x^2}{y}$. Then we want to approximate the value of $f(0.995, 1.002)$. Since $x = 0.995 \approx 1$ and $y = 1.002 \sim 1$, as a first step we consider a linear approximation near $(x_0, y_0) = (1, 1)$.

$$f(x,y) = f(1,1) + f_x(1,1)(x-1) + f_y(1,1)(y-1) = 1 + \frac{2x}{y}\Big|_{(1,1)}(x-1) - \frac{x^2}{y^2}\Big|_{(1,1)}(y-1),$$

hence

$$f(x, y) = 1 + 2(x - 1) - (y - 1).$$

With $x = 0.995$ and $y = 1.002$ we obtain

$$f(0.995, 1.002) \sim 1 + 2(-0.005) - (0.002) = 0.988.$$

Adding the quadratic term yields in addition the terms

$$\frac{1}{2} f_{xx}(1,1)(x-1)^2 + f_{xy}(1,1)(x-1)(y-1) + \frac{1}{2} f_{yy}(1,1)(y-1)^2$$

which is equal to

$$(x-1)^2 - 2(x-1)(y-1) + (y-1)^2 = 0.000025 + 0.00002 + 0.00004 = 0.000085$$

yielding the value 0.988085 for the approximation, one digit more accurate than the linear approximation.

The example shows how, as long as $(x - x_0)$ and $(y - y_0)$ are small, the approximation can be quite good. But how good, exactly?

We can use a generalisation of the mean value theorem (call the *n-th mean value theorem*) to give the following general estimate on the error in the approximation of $f(x, y)$ by the Taylor polynomial $P_n(x, y)$ of order n:

Theorem 4.3 Taylor's formula in two dimensions

Set $\Delta x = x - x_0$, $\Delta y = y - y_0$. Under the above assumptions on f,

$$f(x,y) = f(x_0,y_0) + \Delta x f_x(x_0,y_0) + \Delta y f_y(x_0,y_0) + \frac{1}{2!}\left(\Delta x\frac{\partial}{\partial x} + \Delta y\frac{\partial}{\partial y}\right)^2 f(x_0,y_0) + \ldots + \frac{1}{(n-1)!}\left(\Delta x\frac{\partial}{\partial x} + \Delta y\frac{\partial}{\partial y}\right)^{n-1} f(x_0,y_0) + R_n(x_0,y_0)\,. \tag{4.6}$$

The error term reads explicitly

$$R_n(x_0,y_0) = \frac{1}{n!}\left(\Delta x\frac{\partial}{\partial x} + \Delta y\frac{\partial}{\partial y}\right)^n f(x_0 + c\Delta x, y_0 + c\Delta y) \tag{4.7}$$

for some constant value $0 \le c \le 1$.

To estimate the error term R_n we need to know how large the n-th partial derivatives of f can be near (x_0, y_0).

If we know $|\frac{\partial^n}{(\partial x)^{n-k}(\partial y)^k} f(x,y)| \le M$ for all $k = 0, .., n$ and all (x, y) in a ball $B_r(x_0, y_0)$ of radius r centered at (x_0, y_0), then we have

$$|R_n(x_0,y_0)| \le \frac{M}{n!}(\Delta x + \Delta y)^n\,, \quad \forall (x,y) \in B_r(x_0,y_0).$$

Since for points in the ball we also have $|\Delta x| < r$, $|\Delta y| < r$, the size of the error is controlled by the size of the ball. For a small error, we need small r.

Note how, for the quadratic approximation we recover the formula we expect, since

$$\begin{aligned}\frac{1}{2!}\left(\Delta x\frac{\partial}{\partial x} + \Delta y\frac{\partial}{\partial y}\right)^2 f(x_0,y_0) &= \frac{1}{2}\left(\Delta x^2\frac{\partial^2}{\partial x^2} + 2\Delta x\Delta y\frac{\partial^2}{\partial x\partial y} + \Delta y^2\frac{\partial^2}{\partial y^2}\right) f(x_0,y_0)\\ &= \frac{1}{2}\left(\Delta x^2 f_{xx}(x_0,y_0) + 2\Delta x\Delta y f_{xy}(x_0,y_0) + \Delta y^2 f_{yy}(x_0,y_0)\right)\\ &= \frac{(x-x_0)^2}{2} f_{xx}(x_0,y_0) + (x-x_0)(y-y_0) f_{xy}(x_0,y_0)\\ &\quad + \frac{(y-y_0)^2}{2} f_{yy}(x_0,y_0)\,.\end{aligned}$$

Idea of Proof

Let us start from the 1D case. By the $n-th$ mean value theorem, for some $c \in [0,1]$, we have

$$f(x_0+t) - f(x_0) = tf'(x_0) + \frac{t^2}{2}f''(x_0) + \ldots + \frac{t^{n-1}}{(n-1)!}f^{(n-1)}(x_0) + \frac{t^n}{n!}f^{(n)}(x+ct).$$

For example, for $n = 2$ we find

$$f(x_0+t) = f(x_0) + tf'(x_0) + \frac{t^2}{2}f''(x_0+ct) := f(x_0) + tf'(x_0) + E_2(x_0,t).$$

The last term on the right hand side, $E_2(x_0, t)$, is precisely the error in the approximation given by $P_1(x)$, the first Taylor polynomial. For the quadratic approximation, this error is given by the term

$$E_3(x_0,t) = \frac{t^3}{6}f'''(x_0+ct), \quad \textit{some } c \in [0,1].$$

We want to use this to derive the analogous formula in two variables, given in the statement. To this end, given $f(x, y)$, define

$$G(t) = f(x_0 + t\Delta_x, y_0 + t\Delta_y), \quad where \;\; \Delta_x = x - x_0, \;\; \Delta y = y = y_0, \;\; t \in [0, 1].$$

Then G is a function of one variable such that

$$G(0) = f(x_0, y_0), \quad G(1) = f(x_0 + \Delta_x, y_0 + \Delta_y) = f(x, y).$$

The linear approximation of G is given by

$$G(t) \sim G(0) + G'(0)t, \qquad with\ error: \;\; \frac{t^2}{2}G''(ct.)$$

But, using the chain rule, we can compute that

$$G'(0) = \frac{dG}{dt}_{|0} = ... = f_x(x_0, y_0)\Delta_x + f_y(x_0, y_0)\Delta_y.$$

Similarly,

$$G''(t) = \frac{d}{dt}[f_x(x_0, y_0)\Delta_x + f_y(x_0, y_0)\Delta_y]_{|t} = ...$$

$$= f_{xx}(x_0 + t\Delta_x, y_0 + t\Delta_y)\Delta_x^2 + 2f_{xy}(x_0 + t\Delta_x, y_0 + t\Delta_y)\Delta_x\Delta_y + f_{yy}(x_0 + t\Delta_x, y_0 + t\Delta_y)\Delta_y^2.$$

By setting $t = c$, the last expression gives the error term. ■

Example 4.4

Consider the function $f(x,y) = \ln(x+y^2)$ in the vicinity of the point $(x_0,y_0) = (1,1)$:

$$\begin{aligned} f(1,1) &= \ln 2 \\ f_x(x,y) &= \frac{1}{x+y^2}\,, \quad f_x(1,1) = \frac{1}{2} \\ f_y(x,y) &= \frac{2y}{x+y^2}\,, \quad f_y(1,1) = 1\,. \end{aligned}$$

Thus, the linear approximation reads

$$f(x,y) \approx \ln 2 + \frac{1}{2}(x-1) + 1\cdot(y-1) = \frac{x}{2} + y + \left(\ln 2 - \frac{3}{2}\right)$$

and the error, for some $c \in [0,1]$, is given by

$$R_2((1,1),t) = \frac{1}{2}f_{xx}(cx+1-c)^2 + \frac{1}{2}f_{yy}(cy+1-c)^2 + f_{xy}(cx+1-c, cy+1-c).$$

The quadratic terms are

$$\begin{aligned} f_{xx}(x,y) &= -\frac{1}{(x+y^2)^2}\,, \quad f_{xx}(1,1) = -\frac{1}{4} \\ f_{yy}(x,y) &= 2\cdot\frac{x-y^2}{(x+y^2)^2}\,, \quad f_{yy}(1,1) = 0 \\ f_{xy}(x,y) &= -\frac{2y}{(x+y^2)^2}\,, \quad f_{xy}(1,1) = -\frac{1}{2}\,. \end{aligned}$$

Substituting these expression in the error R_2 gives an explciit estimate.
Using these, the quadratic approximation reads

$$\begin{aligned} f(x,y) &\approx \ln 2 + \frac{(x-1)}{2} + (y-1) + \frac{1}{2}\left[-\frac{(x-1)^2}{4} + 2\cdot(x-1)(y-1)\cdot\left(-\frac{1}{2}\right)\right] \\ &= \ln 2 + \frac{(x-1)}{2} + (y-1) - \frac{(x-1)^2}{8} - \frac{1}{2}(x-1)(y-1)\,. \end{aligned}$$

Taylor expansions tend to give reasonable approximations of a given function $f(x,y)$ *near* the expansion point (x_0,y_0), but they get worse as x, y move away from (x_0,y_0). The remainder term $R_n(x,y)$ can be used to estimate how good the approximation is.

5 Stationary Points, maxima and minima

Let $f : \mathbb{R}^3 \to \mathbb{R}$. It is said that f has a *global*

$$\begin{matrix}\text{minimum}\\ \text{maximum}\end{matrix} \quad \text{at } (x_0,y_0) \text{ if} \quad \begin{matrix} f(x_0,y_0) \le f(x,y) \\ f_(x_0,y_0) \ge f(x,y)\end{matrix}$$

for all $(x, y) \in \mathbb{R}^2$. It is said that f has a *local*

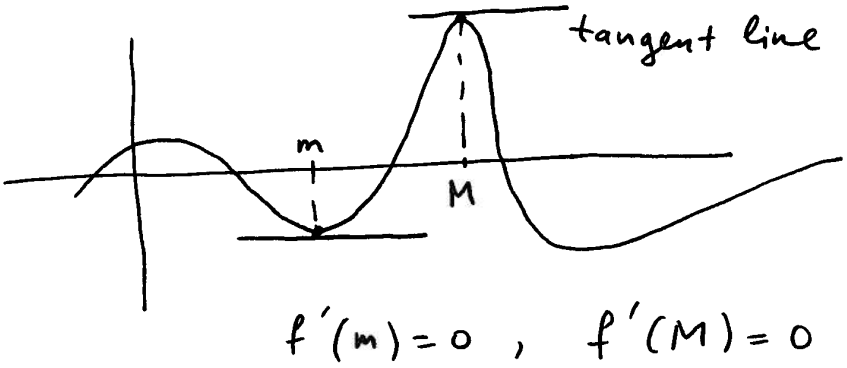
$$\begin{matrix}\text{minimum} \\ \text{maximum}\end{matrix} \text{ at } (x_0, y_0) \text{ if } \begin{matrix} f(x_0, y_0) \leq f(x, y) \\ f_(x_0, y_0) \geq f(x, y)\end{matrix}$$

for all (x, y) in a neighborhood of (x_0, y_0).

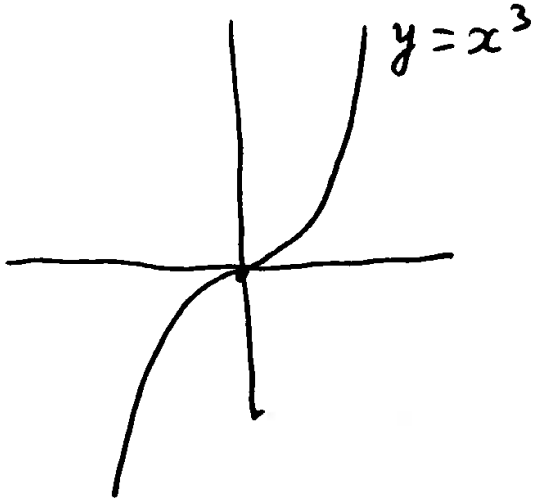
Example 5.1

Let f be the height function defined on the Scottish highlands. Then the top of every mountain is a local maximum (i.e. all nearby points are lower). The top of Ben Nevis is the global maximum.

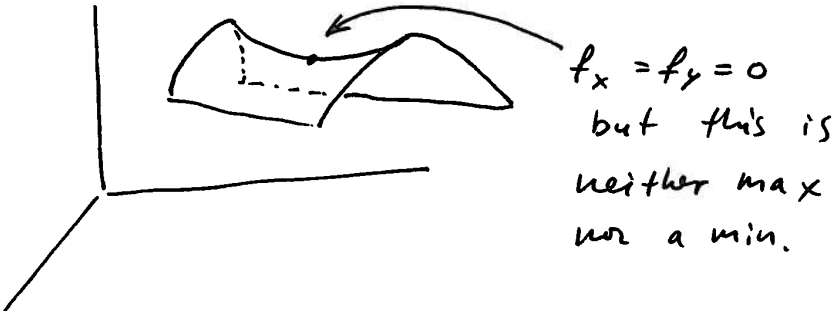
Recall that for $f : \mathbb{R}^1 \to \mathbb{R}^1$, $f' = 0$ at any local maximum or minimum.



The converse need not be true. For example, the function $f(x) = x^3$ has an inflection point at the origin, where $f_x(0) = 0$, but this is neither a local maximum nor a local minimum.



The same happens for $f : \mathbb{R}^2 \to \mathbb{R}$ at a saddle point:



Definition 5.2

If $f_x(x_0, y_0) = f_y(x_0, y_0) = 0$, then (x_0, y_0) is called a *stationary point*.

Example 5.3

1. For the function $f(x, y) = x^2 + y^2$, $(x, y) = (0, 0)$ is a stationary point which happens to be a local minimum.
2. For the function $f(x, y) = -x^2 - y^2$, the point $(x, y) = (0, 0)$ is a stationary point and is a local maximum.
3. For $f(x, y) = x^2 - y^2$, the point $(x, y) = (0, 0)$ is a stationary point and a saddle point.

5.1 Second derivative test for maxima and minima – one variable case

Suppose $f : \mathbb{R}^1 \to \mathbb{R}^1$ and $f'(a) = 0$, i.e. a is a stationary point of f. Suppose further that
(a) f has derivatives up to third order near a,
(b) $f''(a) \neq 0$,
(c) $|f'''(x)| \leq C$ for all x near a and some constant $c > 0$.
Then by virtue of the *third mean value theorem* (where we approximate f with a quadratic polynomial near $x = a$):

$$\begin{aligned} f(a + \Delta x) &= f(a) + \frac{(\Delta x)^2}{2!} f''(a) + r_3(x) \\ &= f(a) + \frac{(\Delta x)^2}{2} f''(a) \cdot \left\{ 1 + \frac{2r_3(x)}{(\Delta x)^2 f''(a)} \right\} \end{aligned} \tag{5.1}$$

We may estimate the remainder term (cf. Thoerem **??**)

$$r_3(x) = \frac{(\Delta x)^3 f'''(x_*)}{3!},$$

as

$$|r_3(x)| \leq \frac{|\Delta x|^3 \cdot c}{6},$$

hence:

$$\begin{aligned} \left| \frac{2r_3(x)}{(\Delta x)^2 f''(a)} \right| &\leq \frac{2|\Delta x|^3 \cdot c}{6(\Delta x)^2 |f''(a)|} \\ &= \frac{c}{3|f''(a)|} \cdot |\Delta x| < 1 \quad \text{if } |\Delta x| < \frac{3|f''(a)|}{c}. \end{aligned}$$

Thus, for sufficiently small Δx the term $\{\dots\}$ in (5.1) above is positive, hence if

$$\begin{aligned} f''(a) < 0 \quad &\Rightarrow \quad x = a \text{ is a local maximum} \\ f''(a) > 0 \quad &\Rightarrow \quad x = a \text{ is a local minimum}\,. \end{aligned}$$

5.2 Second derivative test for two variables

Let $f : \mathbb{R}^2 \to \mathbb{R}^1$ and $f_x(a,b) = f_y(a,b) = 0$, i.e. (a,b) is a stationary point. Taylor's formula reveals that

$$f(a+\Delta x, b+\Delta y) = f(a,b) + \frac{1}{2}Q(\Delta x, \Delta y) + R_3(x,y)\,, \tag{5.2}$$

where

$$\begin{aligned} Q(\Delta x, \Delta y) &= A\cdot(\Delta x)^2 + 2B\Delta x\Delta y + C\cdot(\Delta y)^2 \\ A = f_{xx}(a,b)\,,\ B &= f_{xy}(a,b)\,,\ C = f_{yy}(a,b)\,. \end{aligned}$$

We will ignore the remainder term $R_3(x,y)$ below. It can be shown similarly to the one variable case that under certain assumptions one can indeed neglect $R_3(x,y)$. We distinguish three special cases:

- If $Q(\Delta x, \Delta y) < 0$ for all sufficiently small Δx, Δy, then (a,b) is a local maximum.
- If $Q(\Delta x, \Delta y) > 0$ for all sufficiently small Δx, Δy, then (a,b) is a local minimum.
- If $Q(\Delta x, \Delta y)$ can take both positive and negative values for all sufficiently small Δx, Δy, then (a,b) is a saddle point.

In the special case $Q(\Delta x, \Delta y) = 0$ for all sufficiently small Δx, Δy, the remainder term $R_3(x,y)$ has to be analyzed.

Assume that $A \neq 0$. Then, we can rewrite Q as

$$\begin{aligned} Q(\Delta x, \Delta y) &= A\left((\Delta x)^2 + \frac{2B}{A}\Delta x\Delta y + \frac{C}{A}(\Delta y)^2\right) \\ &= A\left(\left(\Delta x + \frac{B}{A}\Delta y\right)^2 + \left(\frac{C}{A} - \frac{B^2}{A^2}\right)(\Delta y)^2\right) \\ &= A\Bigg(\underbrace{\left(\Delta x + \frac{B}{A}\Delta y\right)^2}_{\geq 0} + (AC - B^2)\,\underbrace{\frac{(\Delta y)^2}{A^2}}_{\geq 0}\Bigg)\,. \end{aligned} \tag{5.3}$$

Thus:

(a) If $AC - B^2 > 0$, then Q has the same sign as A for all (sufficiently small) Δx and Δy. We may discern the following cases:
- $AC - B^2 > 0$ and $A < 0 \Rightarrow (a, b)$ is a local maximum
- $AC - B^2 > 0$ and $A > 0 \Rightarrow (a, b)$ is a local minimum.

(b) If $AC - B^2 < 0$, consider

$$Q(\Delta x, 0) = A(\Delta x)^2 \,.$$

It has the same sign as A. On the other hand,

$$Q\left(-\frac{B}{A}\Delta y, \Delta y\right) = A \cdot \underbrace{(AC - B^2)}_{\leq 0} \cdot \underbrace{\left(\frac{\Delta y}{A}\right)^2}_{\geq 0}$$

has the sign opposite to the sign of A. Hence we conclude that if $AC - B^2 < 0$, we have a saddle point.

Remark. Notes:

(a) If $AC - B^2 > 0$, then $AC > B^2 \geq 0$, hence if A and C are not equal to 0 then A and C have the same sign in this case. Hence we can look at the sign of C in the above test.

(b) If $A = 0$ and $C \neq 0$, we can obtain a similar test by by dividing over C and completing the square. If $B \neq 0$, in this case we have that

$$AC - B^2 = -B^2 < 0 \,,$$

thus we have found a saddle point. A similar argument holds for the case $A \neq 0$, $C = 0$ and $B \neq 0$. In the case $A = B = 0$ but $C \neq 0$, we have for $C > 0$ a local minimum and for $C < 0$ a local maximum.

(c) If $A = C = 0$ and $B \neq 0$,

$$Q(\Delta x, \Delta y) = 2B\Delta x \Delta y$$

clearly changes sign, so we have a saddle point.

(d) In the special case $A = B = C = 0$, the test does not work!

Example 5.4

Find the stationary points of the following functions and determine their nature:
(*i*) $f(x, y) = 2x^2 - xy + y^2 + 7x$
(*ii*) $g(x, y) = xy - x^3y - xy^3$.

Solution. (*i*) $f_x = 4x - y + 7$, $f_y = -x + 2y$. A point (x, y) is a stationary point if

$$\left.\begin{array}{rcl} 4x - y + 7 & = & 0 \\ -x + 2y & = & 0 \end{array}\right\} \quad \Leftrightarrow \quad \left.\begin{array}{rcl} y & = & \frac{x}{2} \\ 7x + 14 & = & 0 \end{array}\right\} \quad \Leftrightarrow x = -2 \text{ and } y = -1 \,.$$

Hence, $(x, y) = (-2, -1)$ is the only stationary point.

Next, we calculate:

$$f_{xx} = 4 \,, \; f_{xy} = -1 \; f_{yy} = 2 \,,$$

from which we conclude that

$$AC - B^2 = f_{xx} \cdot f_{yy} - (f_{xy})^2 = 4 \cdot 21 = 7 > 0 \,.$$

Also, $A = f_{xx} = 4 > 0$, hence we conclude that $(-2, -1)$ is a local minimum.

(ii) From

$$f_x = y - 3x^2y - y^3 = y(1 - 3x^2 - y^2)$$
$$f_y = x - x^3 - 3xy^2 = x(1 - x^2 - 3y^2)\,,$$

and from the condition that (x, y) is stationary if $f_x(x, y) = f_y(x, y) = 0$, we conclude that there are three stationary points:

(a) $y = 0$ and $x(1 - x^2) = 0$, i.e. $y = 0$ and $x = -1, 0, 1$

(b) $x = 0$ and $y(1 - y^2) = 0$, i.e. $x = 0$ and $y = -1, 0, 1$

(c) $x \neq 0$ and $y \neq 0$, then

$$\left.\begin{array}{rcl} 1 - 3x^2 - y^2 & = & 0 \\ 1 - x^2 - 3y^2 & = & 0 \end{array}\right\},$$

and this pair of quadratic equations is solved by

$$x^2 = \frac{1}{4}\,,\ y^2 = \frac{1}{4} \quad \Rightarrow x = \pm\frac{1}{2}\,,\ y = \pm\frac{1}{2}\,.$$

Hence, f has 9 stationary points:

$$(0, \pm 1)\,,\ (0, 0)\,,\ (\pm 1, 0)\,,\ (\pm\tfrac{1}{2}, \pm\tfrac{1}{2})\,.$$

Next, calculating the double derivatives

$$f_{xx} = -6xy\,,\ f_{xy} = 1 - 3x^2 - 3y^2\,,\ f_{yy} = -6xy\,,$$

we may determine the nature of the various stationary points:

staionary point	A	B	C	$AC - B^2$	type
$(0, 0)$	0	1	0	< 0	saddle
$(1, 0)$	0	-2	0	< 0	saddle
$(\frac{1}{2}, \frac{1}{2})$	$-\frac{3}{2}$	$-\frac{1}{2}$	$-\frac{3}{2}$	> 0	local max.
$(-1, 0)$	0	1	0	< 0	sddale
$\vdots$					

■

In higher dimensions, for functions $f : \mathbb{R}^n \to \mathbb{R}$, stationary points can be found by putting all the first order derivatives to zero. However, determining the nature of the stationary points is much more complicated. Sometimes, the nature of the stationary points is obvious from the problem.

Example 5.5

Consider $f(x,y) = (x-2)^3 - (y-1)^3 + 2(x+3y-5)^4$. It can easily be checked that $(2,1)$ is a stationary point of f. Also, $A = B = C = 0$, so the second derivative test does not work. Letting $x = 2+\varepsilon$ and $y = 1$,

$$f(2+\varepsilon, 1) = \varepsilon^3 + 3\varepsilon^4 > 0 \quad \text{for} \quad \varepsilon > 0\,,$$

while for $x = 2$ and $y = 1+\varepsilon$

$$f(2, 1+\varepsilon) = -\varepsilon^3 + 2\cdot(3\varepsilon)^4 = -\varepsilon^3(1 - 2\cdot 3^4\cdot\varepsilon) < 0$$

for $0 < \varepsilon < 1/(2\cdot 3^4)$. Therefore, near $(2,1)$, f can be both positive and negative, so $(2,1)$ is a saddle point (note that $f(2,1) = 0$).

A similar trick can often be used to prove that a given stationary point is a saddle point – we only need to show that *some* points (x,y) near (x_0,y_0) give values smaller than $f(x_0,y_0)$ while *some* other points near the stationary points give values larger than $f(x_0,y_0)$. It is in general harder to show that a given stationary point is a local maximum or a local minimum, since we need to make a statement about *all* points in a neighborhood of the stationary point.

5.3 Constrained extrema - Lagrange multipliers

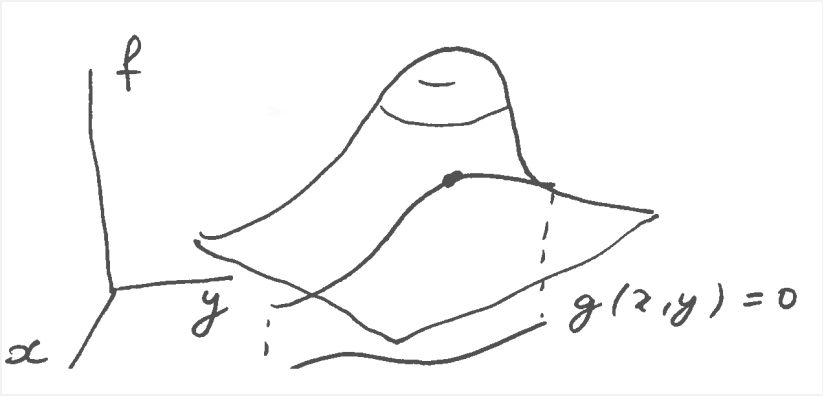
In certain problems we may need to minimize or maximize a given function subject to additional constraints, e.g. find the minimal distance from the origin to the plane $2x+3y-z = 5$. So we have to find a minimum of $f(x,y,z) = x^2 + y^2 + z^2$ subject to the constraint $2x + 3y - z = 5$. In this case, we can solve the constraint equation for one of the variables, say, $z = 2x + 3y - 5$, and then minimize

$$x^2 + y^2 + (2x + 3y - 5)^2\,,$$

which is a function of just two variables. But in general, it may be hard to solve the constraint equation.

Example 5.6

Maximize $f(x,y) > 0$ subject to $g(x,y) = 0$. We can think of the graph of f as a height function for some mountainous landscape:



The usual maximization looks for the top of the mountain, wwhile the constrained maximization looks for the highest point along a path given by $g(x,y) = 0$. Note that f_x and f_y are in general not zero at the constrained maximum.

Theorem 5.7

If (x_0, y_0) is a local maximum or local minimum of $f(x,y)$ subject to the constraint $g(x,y) = 0$ and if

$$(g_x(x_0,y_0), g_y(x_0,y_0)) \neq (0,0)\,, \tag{5.4}$$

then

$$\left.\begin{aligned} f_x(x_0,y_0) + \lambda g_x(x_0,y_0) &= 0 \\ f_y(x_0,y_0) + \lambda g_y(x_0,y_0) &= 0 \\ g(x_0,y_0) &= 0 \end{aligned}\right\} \tag{5.5}$$

for some $\lambda \in \mathbb{R}$. The above gives a system of three equations for three variables (i.e. for x_0, y_0 and λ). λ s called the *Lagrange multiplier*.

Remarks.

(1) If x_0, y_0 and λ do satisfy the above equations, there is no easy way (test) for determining the nature of the point, i.e. whether it is a local maximum, a local minimum or a saddle point.

(2) The first two equations in (5.5) are equivalent to looking for a stationary point of

$$h = f + \lambda g\,.$$

(3) If $f, g : \mathbb{R}^n \to \mathbb{R}$, then a local maximum or local minimum x_0 of $f(x)$ subject to the constraint $g(x) = 0$ satisfies a similar system of $n+1$ equations that can be written (after defining $h : \mathbb{R}^n \to \mathbb{R}$ by $h = f + \lambda g$) as:

$$\begin{aligned} \frac{\partial h}{\partial x_i}(x_0) = \frac{\partial f}{\partial x_i}(x_0) + \lambda \frac{\partial g}{\partial x_i}(x_0) = 0\,, \qquad i = 1, \ldots, n \\ g(x_0) = 0. \end{aligned}$$

Example 5.8

Minimize $f = x^2 + y^2 + z^2$ subject to the constraint $2x + 3y - z = 5$ and using the method of the Lagrange multiplier.

Solution. Put

$$h(x, y, z) = x^2 + y^2 + z^2 + \lambda\,(2x + 3y - z - 5)\ .$$

Then we obtain the equation system

$$\left.\begin{array}{rcrcl} h_x & = & 2x + 2\lambda & = & 0 \\ h_y & = & 2y + 3\lambda & = & 0 \\ h_z & = & 2z - \lambda & = & 0 \\ & & 2x + 3y - z - 5 & = & 0 \end{array}\right\}\ .$$

Hence

$$x = -\lambda\,,\ y = -\frac{3}{2}\lambda\,,\ z = -\frac{\lambda}{2}\,.$$

Substituting this into the last equation we ebtain:

$$\begin{aligned} 2(-\lambda) + 3(-\tfrac{3}{2}\lambda) - (\tfrac{1}{2}\lambda) - 5 = 0 & \\ \Rightarrow \qquad -7\lambda = 5 & \\ \Rightarrow \qquad \lambda = -\frac{5}{7}\,, & \end{aligned}$$

and therefore the minimum occours at

$$(x, y, z) = \left(\frac{5}{7}, \frac{15}{14}, -\frac{5}{14}\right)\ ,$$

and the minimal distance is

$$\sqrt{\left(\frac{5}{7}\right)^2 + \left(\frac{15}{14}\right)^2 + \left(-\frac{5}{14}\right)^2} = \frac{5}{\sqrt{14}}\,.$$

■

Example 5.9

Find the rectangular box of maximal volume with sides parallel to the coordinate planes which can be contained inside the ellipsoid described via the equaton

$$4x^2 + y^2 + 9z^2 = 1\,.$$

Solution. Suppose the box has the vertex (x, y, z) in the first octant, i.e. $x > 0$, $y > 0$ and $z > 0$.

Since the box has maximum volume, its vertices lie on the ellipsoid. The function we want to maximize is

$$V = 8x \cdot y \cdot z\,.$$

Set

$$h(x,y,z) = 8xyz + \lambda(4x^2 + y^2 + 9z^2 - 1)\,.$$

We get the equatons:

$$\begin{aligned} 8yz + \lambda 8x &= 0 \\ 8xz + 2\lambda y &= 0 \\ 8xy + 18\lambda z &= 0\,. \end{aligned}$$

Multiplying the first equation by x, the second by y and the third by z, we obtain

$$8xyz = -\lambda 8z^2 = -2\lambda y^2 = -18\lambda z^2\,.$$

If $\lambda = 0$, then the volume is zero. Therefore $\lambda \neq 0$ and we can divide by it to obtain

$$y^2 = 4x^2\,,\ z^2 = \frac{4}{9}x^2\,.$$

Substituting this into the constraint equation results in

$$4x^2 + 4x^2 + 4x^2 = 1 \quad \Leftrightarrow \quad x = \frac{1}{\sqrt{12}} = \frac{1}{2\sqrt{3}}\,,$$

hence

$$y = 2x = \frac{1}{\sqrt{3}}\,,\ z = \frac{2}{3}x = \frac{2}{3\sqrt{3}}\,,$$

and the maximal volume is

$$8xyz = 8 \cdot \frac{1}{2\sqrt{3}} \cdot \frac{1}{\sqrt{3}} \cdot \frac{2}{3\sqrt{3}} = \frac{8}{9\sqrt{3}}\,.$$

■

5.4 More examples

Example 5.10

Find the maximal and minimal value of

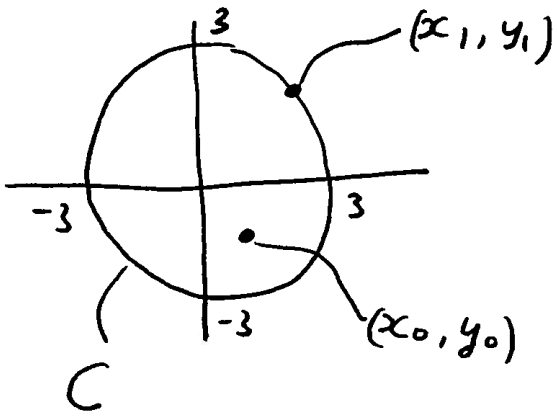
$$f(x,y) = x^2 - xy + y^2 - 3x$$

in the region $x^2 + y^2 \leq 9$.

Solution.

(1) If the maximum/minimum occurs at some point (x_0, y_0) inside the circle $\mathcal{C}$ described by the equation $x^2 + y^2 = 9$, then (x_0, y_0) must be an ordinary local maximum/minimum.

(2) If the maximum/minimum occurs at (x_0, y_0) on the circle $\mathcal{C}$, then (x_0, y_0) must be a maximum/minimum of f subject to the constraint $x^2 + y^2 = 9$.



Case (1) From the partical derivatives

$$f_x = 2x - y - 3\,,\ f_y = -x + 2y\,,$$

we may compute

$$\begin{cases} 2x - y - 3 &= 0 \\ -x + 2y &= 0 \end{cases} \quad \Rightarrow \quad x = 2\,,\ y = 1\,.$$

The nature of this stationary point may be determined as follows:

$$\begin{array}{c} f_{xx} = 2\,,\ f_{xy} = -1\,,\ f_{yy} = 2 \\ f(2,1) = -3 \end{array} \quad \Rightarrow \quad \begin{array}{c} f_{xx} \cdot f_{yy} - (f_{xy})^2 = 3 > 0 \\ f_{xx} > 0 \end{array}\,.$$

Thus, $(2, 1)$ is a local minimum.

Case (2) With

$$\begin{aligned} h(x,y) &= x^2 - xy + y^2 - 3x + \lambda(x^2 + y^2 - 9) \\ h_x &= 2x - y - 3 + 2\lambda x = 0 \\ h_y &= -x + 2y + 2\lambda y = 0\,, \end{aligned}$$

multiplying the first equation by y and the second equation by x and then taking the difference, we can get rid of λ:

$$2xy - y^2 - 3y - (-x^2 + 2xy) = 0 \quad \Leftrightarrow \quad x^2 = y^2 + 3y\,.$$

Combining this with the constraint equation

$$x^2 + y^2 = 9\,,$$

we obtain

$$2y^2 = 3y = 9 \quad \Leftrightarrow \quad (2y - 3)(y + 3) = 0 \quad \Leftrightarrow \quad y = \frac{3}{2} \text{ or } y = -3\,.$$

Solving for x,

$$\begin{aligned} y = \frac{3}{2} \quad &\Rightarrow \quad x^2 = 9 - \frac{9}{4} = \frac{27}{4} \quad \Rightarrow \quad x = \pm\frac{3\sqrt{3}}{2} \\ y = -3 \quad &\Rightarrow \quad x = 0\,, \end{aligned}$$

we conclude that the possible maxima/minima on the circle $\mathcal{C}$ are $(\pm\frac{3\sqrt{3}}{2}, \frac{3}{2})$ and $(0, -3)$. Together with the local minimum $(2, 1)$ computed previously, we have

$$\begin{aligned} f(2,1) &= -3 \\ f(0,-3) &= 9 \\ f(\tfrac{3\sqrt{3}}{2}, \tfrac{3}{2}) &= 9 - \frac{27\sqrt{3}}{2} \approx -2.69 \\ f(-\tfrac{3\sqrt{3}}{2}, \tfrac{3}{2}) &= 9 + \frac{27\sqrt{3}}{2} \approx 20.69 \end{aligned}$$

Hence the maximum of f is ≈ 20.69 and is achieved at $(-\frac{3\sqrt{3}}{2}, \frac{3}{2})$, while the minimum is -3 at $(2, 1)$.

■

Part 2 - Integration

6 The Riemann Integral

Let $f : [a, b] \to \mathbb{R}$ be a bounded function. The aim of this chapter is to give a rigorous definition of the *integral* of f over the interval $[a, b]$.

We begin by considering this as the area under the graph of f (more precisely, the area between the graph of f and the x-axis). However, integrals are much more general than just an area — we will say more about this at the end. Looking at Fig. 1, it is clear that the area under the graph of f lies between the areas of the upper and lower sets of rectangles drawn there, and we can work out what these areas are (the area of a rectangle is width×height).

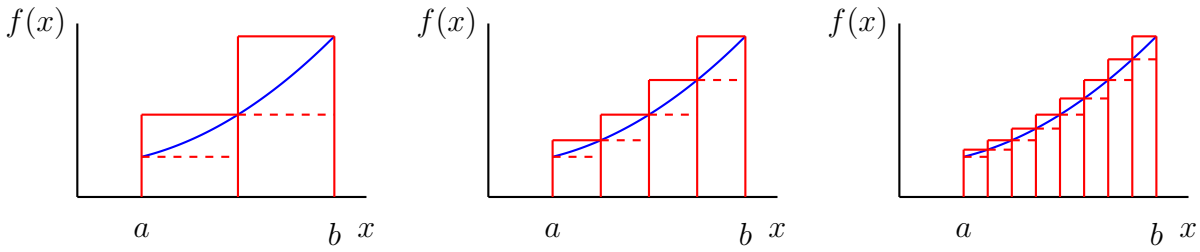


Figure 1: The area under the blue graph lies between the areas under the upper and lower sets of red rectangles.

To obtain the area under the graph, of a function f, we imagine splitting this area into narrower and narrower rectangles and taking the limit of the areas of the upper and lower sets of rectangles as the width tends to zero. We would guess that these upper and lower areas should tend to the same value, and this should be the area under the graph. In the following sections we try to make this precise.

6.1 Riemann sums

Definition 6.1

A *partition* P of $[a, b]$ is a finite set of points $P = \{x_0, x_1, x_2, \ldots, x_k\}$ in $[a, b]$ such that

$$a = x_0 < x_1 < x_2 < \cdots x_k = b.$$

For a partition P we also define

$$\delta_r := x_{r+1} - x_r, \quad \text{for each } r = 0, \ldots, k-1,$$
$$\delta_{\max}(P) := \max\{\delta_0, \ldots, \delta_{k-1}\}.$$

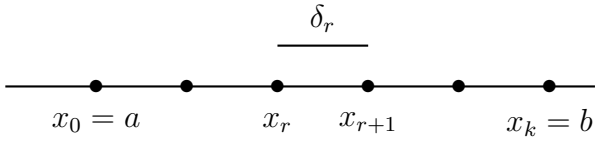


Figure 2: A partition, as in Definition 6.1.

Example 6.2

$P = \{0,\ 0.4,\ 0.75,\ 1\}$ is a partition of $[0, 1]$, with $\delta_0 = 0.4$, $\delta_1 = 0.35$, $\delta_2 = 0.25$.

Remarks.

- Partitions will do the splitting process in Fig. 1. A partition P splits the interval $[a, b]$ into k subintervals $[x_r, x_{r+1}]$, $r = 0, 1, \ldots, k-1$, of width δ_r, and these subintervals will be the bases of the rectangles, as in Fig. 1.
  This is the first step in the above process of calculating the area.
- The subintervals need not all have the same width (although they do in each of the figures in Fig. 1).
- $\delta_{\max}(P)$ is the maximum width of all the subintervals.
- Clearly, the sum of the widths of the subintervals equals the total width $b - a$ of the interval $[a, b]$, that is

$$\sum_{r=0}^{k-1} \delta_r = \sum_{r=0}^{k-1} (x_{r+1} - x_r) = (x_1 - x_0) + (x_2 - x_1) + \cdots + (x_k - x_{k-1}) = b - a. \quad (6.1)$$

We can use a given partition to define teh area of rectangles associated with the partition.

Definition 6.3

A *Riemann sum* with respect to the partition P is any sum of the form

$$S(P, f) = \sum_{r=0}^{k-1} f(x_r^*)\delta_r, \quad (6.2)$$

where, for $r = 0, \ldots, k-1$, x_r^* is a point in the subinterval $[x_r, x_{r+1}]$.

We now introduce a notation for the areas of the lower and upper sets of rectangles.

Definition 6.4

The *lower Riemann sum* $L(P, f)$ and the *upper Riemann sum* $U(P, f)$ of f with respect to the partition P are:

$$L(P, f) = \sum_{r=0}^{k-1} I_r(f)\delta_r, \qquad U(P, f) = \sum_{r=0}^{k-1} S_r(f)\delta_r, \quad (6.3)$$

where, for $r = 0, \ldots, k-1$,

$$\begin{aligned} I_r(f) &= \inf\{f(x) : x \in [x_r, x_{r+1}]\}, \\ S_r(f) &= \sup\{f(x) : x \in [x_r, x_{r+1}]\}. \end{aligned} \quad (6.4)$$

Remark. For a function f and partition P (as in Fig. 1):

- $L(P, f)$ is the area under the set of lower rectangles;
- $U(P, f)$ is the area under the set of upper rectangles.

- Clearly, the quantities I_r, S_r and δ_r depend on the partition P, so we should really write $I_r(P)$, $S_r(P)$ and $\delta_r(P)$, but this would make all the formulae fairly unwieldy and hard to read (and write), so we have omitted the dependence on P in these quantities, although we have kept it in the notation $L(P, f)$ and $U(P, f)$.

Example 6.5

Let $f(x) = x^2$ and let $P = \{0, 1/2, 3/4, 1\}$.

$$S_0(f) = \sup\{x^2 : x \in [0, 1/2]\} = \left(\frac{1}{2}\right)^2 = \frac{1}{4}, \qquad \delta_0 = \frac{1}{2},$$

$$S_1(f) = \sup\{x^2 : x \in [1/2, 3/4]\} = \left(\frac{3}{4}\right)^2 = \frac{9}{16}, \quad \delta_1 = \frac{1}{4},$$

$$S_2(f) = \sup\{x^2 : x \in [3/4, 1]\} = 1^2 = 1, \qquad \delta_2 = \frac{1}{4},$$

$$U(P, f) = \frac{1}{4} \times \frac{1}{2} + \frac{9}{16} \times \frac{1}{4} + 1 \times \frac{1}{4} = \frac{33}{64}.$$

A similar calculation shows that

$$I_0(f) = 0, \qquad I_1(f) = \frac{1}{4}, \qquad I_2(f) = \frac{9}{16}.$$

$$L(P, f) = 0 \times \frac{1}{2} + \frac{1}{4} \times \frac{1}{4} + \frac{9}{16} \times \frac{1}{4} = \frac{13}{64}.$$ ■

We now define the 'trivial' partition $T_0 := \{a, b\}$ (that is T_0 just contains the end points of the interval $[a, b]$, and no interior points), and let

$$I(f) := \inf\{f(x) : x \in [a, b]\}, \ S(f) := \sup\{f(x) : x \in [a, b]\}. \tag{6.5}$$

It is clear from the definitions that

$$L(T_0, f) = I(f)(b - a), \quad U(T_0, f) = S(f)(b - a).$$

Lemma 6.6

For any partition P of $[a, b]$,

$$L(T_0, f) \leqslant L(P, f) \leqslant U(P, f) \leqslant U(T_0, f). \tag{6.6}$$

The proof is an immediate consequence of the definitions.

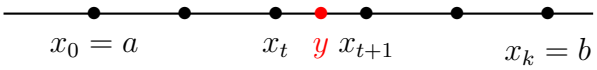
Next, suppose that we have two partitions P and R, with $P \subset R$, that is, all the points in P are also in R. In this case we say that R is a *refinement* of P. In effect, we refine a partition P by adding points to it to give the new partition R, so that all the gaps have either stayed the same, or got smaller.

Lemma 6.7

Let P and R be partitions of $[a, b]$ such that R is a refinement of P. Then

$$L(P, f) \leqslant L(R, f), \quad U(P, f) \geqslant U(R, f). \tag{6.7}$$

Proof. We first suppose that R contains just one more point than P, say $y \in (x_t, x_{t+1})$, for some integer t with $0 \leqslant t \leqslant k-1$.



Since nothing else changes, we only need to look at the interval $[x_t, x_{t+1}]$, with the trivial partition $T_0 = \{x_t, x_{t+1}\}$ and the refinement $R = \{x_t, y, x_{t+1}\}$. Now, by Lemma 6.6,

$$L(T_0, f) \leqslant L(R, f) \leqslant U(R, f) \leqslant U(T_0, f),$$

and so

$$L(P, f) \leqslant L(R, f) \leqslant U(R, f) \leqslant U(P, f)$$

(since the contributions to each of these summations outside the interval $[x_t, x_{t+1}]$ are all the same). That is, (6.7) holds for a single point refinement.

We can now prove the general case as follows:

- we can obtain the refined partition R from the partition P in a step by step manner by adding a single point at a time;
- the required inequalities (6.7) hold at each step in this process, so they hold at the end of the process. ■

The next theorem deals with any two partitions P and Q, neither of which need be a refinement of the other one.

Theorem 6.8

If P and Q are any partitions of $[a, b]$ then

$$L(P, f) \leqslant U(Q, f).$$

Proof. We first note that the set $R = P \cup Q$ is a partition, and is a refinement of both the partitions P and Q (since $P \subset P \cup Q$ and $Q \subset P \cup Q$). Hence, by Lemma 6.6 and Lemma 6.7,

$$L(P, f) \leqslant L(R, f) \leqslant U(R, f) \leqslant U(Q, f).$$ ■

6.2 The Riemann integral of f

For any given partition P we now have the idea of the upper and lower Riemann sums of f, with respect to P, which give us the total areas of the upper and lower rectangles in Fig. 1. What we now want to do is take some sort of limit as the widths of the gaps in the partitions shrinks to zero — and we hope that these upper and lower areas will tend to the same limit! Unfortunately, it is not easy to define what we mean by the limit of the set of partitions, so instead we will first look at the 'smallest' upper sum, and the 'largest' lower sum. That is, we will look at the inf and the sup of these sums.

It follows from Lemma 6.6 that the sets of values of lower and upper Riemann sums

$$\mathcal{L}(f) := \big\{L(P,f) : P \text{ is a partition of } [a,b]\big\},$$
$$\mathcal{U}(f) := \big\{U(P,f) : P \text{ is a partition of } [a,b]\big\},$$

are bounded, so we can define

$$\overline{L}(f) := \sup \mathcal{L}(f), \quad \underline{U}(f) := \inf \mathcal{U}(f).$$

Theorem 6.9

For any partitions P, Q,

$$L(P,f) \leqslant \overline{L}(f) \leqslant \underline{U}(f) \leqslant U(Q,f). \tag{6.8}$$

Proof. The outer inequalities in (6.8) follow immediately from the definitions of $\overline{L}(f)$ and $\underline{U}(f)$. The middle one follows from Theorem 6.8 but is not so obvious. To prove it we note that:

- for any P, Q, $L(P,f) \leqslant U(Q,f)$, so $U(Q,f)$ is an upper bound for the set $\mathcal{L}(f)$
- $\overline{L}(f)$ is the **least** upper bound for $\mathcal{L}(f)$, so $\overline{L}(f) \leqslant U(Q,f)$ for any partition Q
- hence $\overline{L}(f)$ is a lower bound for the set $\mathcal{U}(f)$
- $\underline{U}(f)$ is the **greatest** lower bound for $\mathcal{U}(f)$, so $\overline{L}(f) \leqslant \underline{U}(f)$. ■

Definition 6.10

A bounded function $f : [a,b] \to \mathbb{R}$ is *(Riemann) integrable on* $[a,b]$ if

$$\overline{L}(f) = \underline{U}(f).$$

If f is integrable, then this common value is defined to be the *(Riemann) integral of f over* $[a,b]$ and is denoted by

$$\int_a^b f \quad \text{or} \quad \int_a^b f(x)\,dx$$

(the latter notation is useful if we need to emphasize which variable is being integrated, or we want to put in a specific function, e.g. $\int_0^1 x^2\,dx$; the former is shorter and simpler if we are doing an abstract integration and haven't defined the letter we are using for the variable of integration).
It is also convenient (notationally) to define the integral of f when the lower limit is greater than, or even equal to, the upper limit. So, we define

$$\int_b^a f := -\int_a^b f, \quad \int_a^a f := 0, \quad a < b. \tag{6.9}$$

Remark. It follows from the definition above that if a function $f : \mathbb{R} \to \mathbb{R}$ si integrable then its Riemann integral can be computed as the limit as the size of any sequence of partitions

goes to zero, of any Riemann sums - it needs not be the lower or upper one, as indeed for any $x_r^* \in I_r := [x_r, x_{r+1}]$, it holds that

$$\inf_{I_r} f(x) \leq f(x_r^*) \leq \sup_{I_r} f(x)$$

.

Remark. The above constructions have been motivated by finding the area between the graph of f and the x-axis, and worked out the areas of rectangles by multiplying 'heights' by 'widths', where the 'heights' are given by function values. So, if any function values are negative the corresponding 'areas' are negative. That is, the Riemann integral counts 'areas' below the x-axis as negative.

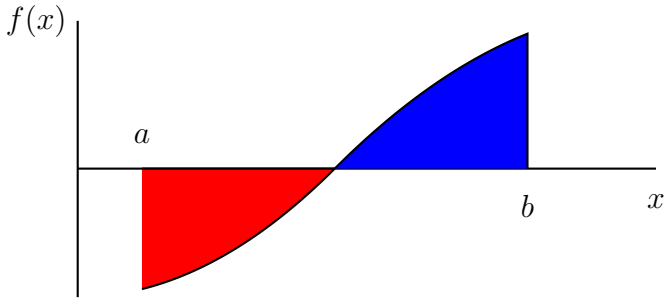


Figure 3: Positive (blue) and negative (red) areas 'under' the graph of f.

We first observe that not every bounded function is integrable.

Example 6.11

Consider the function $f : [0, 1] \to \mathbb{R}$ defined by

$$f(x) = \begin{cases} 0, & \text{if } x \in [0,1] \text{ is irrational,} \\ 1, & \text{if } x \in [0,1] \text{ is rational.} \end{cases}$$

For any partition P, every interval $[x_r, x_{r+1}]$ contains both rational and irrational numbers, so we see that

$$I_r(f) = \inf\{f(x) : x \in [x_r, x_{r+1}]\} = 0\,, \quad S_r(f) = \sup\{f(x) : x \in [x_r, x_{r+1}]\} = 1,$$

so

$$L(P, f) = 0, \quad U(P, f) = 1 \implies \overline{L}(f) = 0, \quad \underline{U}(f) = 1 \implies f \text{ is not integrable.}$$

■

Remark. We would now like to answer the following questions.

- In view of Example 6.11, are there any simple criteria for when a function is integrable?
- How do we calculate the integral of an integrable function?

To answer these questions, we first give a simpler characterisation of an integrable function, in terms of sequences of partitions, rather than the inf/sup definition in Definition 6.10.

Theorem 6.12

For a bounded function $f : [a, b] \to \mathbb{R}$, the following condition are equivalent:

(a) f is integrable on $[a, b]$;

(b) there exists a sequence of partitions (P_n) such that

$$\lim_{n\to\infty} U(P_n, f) = \lim_{n\to\infty} L(P_n, f); \tag{6.10}$$

(c) there exists a sequence of partitions (P_n) such that

$$\lim_{n\to\infty} \big(U(P_n, f) - L(P_n, f)\big) = 0. \tag{6.11}$$

If any of these conditions hold then the integral $\displaystyle\int_a^b f$ exists and is given by the limits in (6.10).

Proof. $((a) \Rightarrow (b))$ By a theorem proved in part A, on finding sups and infs, there exists sequences of partitions (Q_n^u), (Q_n^l) such that

$$\lim_{n\to\infty} U(Q_n^u, f) = \underline{U}(f) = \overline{L}(f) = \lim_{n\to\infty} L(Q_n^l, f) \tag{6.12}$$

(we have $\underline{U}(f) = \overline{L}(f)$ here, since f is integrable). The sequences (Q_n^u), (Q_n^l) need not be the same, but if we set $R_n = Q_n^u \cup Q_n^l$, $n = 1, 2, \ldots$ (so that R_n is a refinement of each of the partitions Q_n^u, Q_n^l), then

$$\begin{aligned}
\underline{U}(f) &\leqslant U(R_n, f) && \text{(by (6.8), with } Q = R_n\text{)}\\
&\leqslant U(Q_n^u, f) && \text{(by (6.7), since } R_n \text{ is a refinement of } Q_n^u\text{)}\\
&\to \underline{U}(f) && \text{(by (6.12))}
\end{aligned}$$

so, by the sandwich theorem $U(R_n, f) \to \underline{U}(f)$. Similarly, $L(R_n, f) \to \overline{L}(f)$. Now, setting $P_n = R_n$, $n = 1, 2, \ldots$, and using $\underline{U}(f) = \overline{L}(f)$, from (6.12), gives (6.10).

$((b) \Rightarrow (c))$ This is obvious.

$((c) \Rightarrow (a))$ Now,

$$\begin{aligned}
0 \leqslant \underline{U}(f) - \overline{L}(f) &\leqslant U(P_n, f) - L(P_n, f) && \text{(by (6.8))}\\
&\to 0 && \text{(by (6.11))}\\
&\Longrightarrow \underline{U}(f) - \overline{L}(f) = 0 && \text{(by the sandwich theorem)}\\
&\Longrightarrow f \text{ is integrable} && \text{(by Definition 6.10).}
\end{aligned}$$

Combining these results proves the equivalence of the conditions (a)-(c).

To prove the final result, suppose that (P_n) is a sequences of partitions as in (b) or (c). Then,

$$0 \leqslant \overline{L}(f) - L(P_n, f) \leqslant U(P_n, f) - L(P_n, f) \to 0,$$

so the result follows from the definition of the integral in Definition 6.10. ■

Remark 6.1. Note that

- The implication $(b) \Rightarrow (c)$ in Theorem 6.12 is obvious, but the opposite implication $(b) \Leftarrow (c)$ is not. At first sight (6.11) does not imply that the separate limits in (6.10) exist.
- Condition (c) in Theorem 6.12 is a useful test for integrability of a function f. It says that a function f is integrable $\iff$ the areas between the upper and lower sets of rectangles in Fig. 1 shrinks to zero, see Fig. 4.

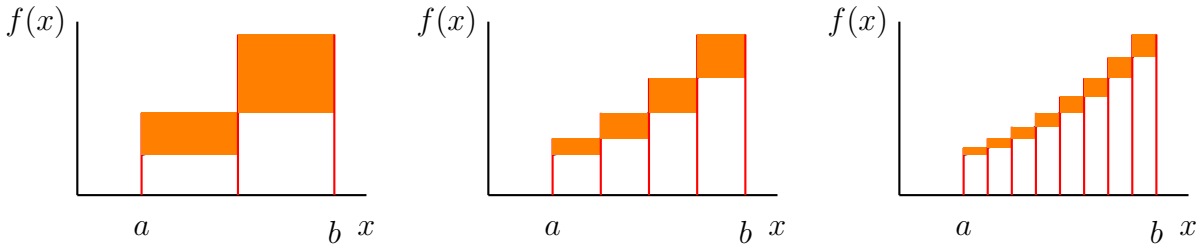


Figure 4: The areas between the upper and lower sets of rectangles in Fig. 1. These are shrinking to zero as the partitions get finer.

Example 6.13

Define $f : [0,5] \to \mathbb{R}$ by

$$f(x) = \begin{cases} 4, & \text{if } 0 \leqslant x < 3, \\ 2, & \text{if } x = 3, \\ 1, & \text{if } 3 < x \leqslant 5, \end{cases}$$

see the black graph in Fig. 5. The area under this graph looks like two large rectangles of area $4 \times 3 = 12$ and $1 \times 2 = 2$, together with an 'infinitely thin' rectangle over the point $x = 3$, with zero area. So, the area 'ought to be' $12 + 2 = 14$, but we need to do this properly (what does 'infinitely thin' mean??).
To deal with the jump at $x = 3$, for each $n \geqslant 1$, let P_n be the partition given by

$$P_n = \left\{0, 3 - \frac{1}{n}, 3 + \frac{1}{n}, 5\right\} .$$

Find $U(P_n, f)$ and $L(P_n, f)$, and hence show that f is integrable on $[0,5]$ and find $\int_0^5 f$.

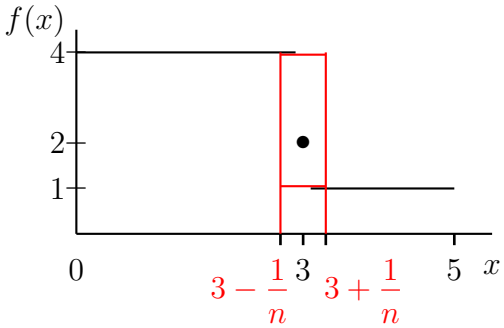


Figure 5:

Solution. We see that

$$\sup\left\{f(x) : x \in \left[0, 3-\frac{1}{n}\right]\right\} = 4,$$
$$\sup\left\{f(x) : x \in \left[3-\frac{1}{n}, 3+\frac{1}{n}\right]\right\} = 4,$$
$$\sup\left\{f(x) : x \in \left[3+\frac{1}{n}, 5\right]\right\} = 1,$$

so that

$$U(P_n, f) = 4\left(3-\frac{1}{n}\right) + 4\left(3+\frac{1}{n}-\left(3-\frac{1}{n}\right)\right) + 1\left(5-\left(3+\frac{1}{n}\right)\right) = 14+\frac{3}{n}\cdot$$

Similarly,

$$L(P_n, f) = 4\left(3-\frac{1}{n}\right) + 1\left(3+\frac{1}{n}-\left(3-\frac{1}{n}\right)\right) + 1\left(5-\left(3+\frac{1}{n}\right)\right) = 14-\frac{3}{n}\cdot$$

Thus

$$\lim_{n\to\infty}\left(U(P_n, f) - L(P_n, f)\right) = \lim_{n\to\infty}\frac{4}{n} = 0.$$

Hence f is integrable on $[0, 5]$ and

$$\int_0^5 f = \lim_{n\to\infty} U(P_n, f) = \lim_{n\to\infty}\left(14+\frac{3}{n}\right) = 14. \qquad \blacksquare$$

6.3 Properties of the integral

We state some of the standard properties of integrals. The proof is left as an exercise, as it is just an application of the definitions.

Theorem 6.14

Suppose that $f, g : [a, b] \to \mathbb{R}$ are both integrable and C is a constant.

(a) $f+g$ is integrable, with $\displaystyle\int_a^b (f+g) = \int_a^b f + \int_a^b g.$

(b) Cf is integrable, with $\displaystyle\int_a^b Cf = C\int_a^b f\ .$

(c) fg is integrable, with $\displaystyle\left(\int_a^b fg\right)^2 \leqslant \left(\int_a^b f^2\right)\left(\int_a^b g^2\right)$ (Cauchy's inequality).

(d) $f(x) \leqslant g(x)$ for all $x \in [a, b] \implies \displaystyle\int_a^b f \leqslant \int_a^b g.$

(e) $f(x) = k$, for some constant k, for all $x \in [a, b] \implies \displaystyle\int_a^b f = k(b-a).$

(f) $|f|$ is integrable, with $\displaystyle\left|\int_a^b f\right| \leqslant \int_a^b |f|\ .$

Next, we state the result implying that if f is integrable on $[a,b]$ then it is integrable on any subinterval of $[a,b]$, and also that we can split the integration interval $[a,b]$ into two subintervals $[a,c]\cup[c,b]$ (with $a<c<b$) then integrate separately over the two parts and add the results to get the same answer as integrating over $[a,b]$.

Theorem 6.15

Suppose that $f:[a,b]\to\mathbb{R}$ is integrable.

(a) If $[c,d]\subset[a,b]$ then f is integrable on $[c,d]$.

(b) If $c\in(a,b)$ then

$$\int_a^b f=\int_a^c f+\int_c^b f.$$

6.4 Some classes of integrable functions

We now have two characterisations of integrable functions, in terms of partitions (Definition 6.10 and Theorem 6.12). However, we really don't want to have to go back to these every time we want to know if a function is integrable. The following two theorems show that two broad classes of (bounded) functions are integrable, and these results show that most functions that we normally meet are integrable.

Theorem 6.16

If f is monotone on $[a,b]$ then it is integrable.

Proof. Suppose that f is increasing (the proof when f is decreasing is similar). Then, by definition, for any partition P,

$$\begin{aligned}
U(P,f)-L(P,f)&=\sum_{r=0}^{k-1}(S_r(f)-I_r(f))\delta_r\\
&\leqslant\delta_{\max}(P)\sum_{r=0}^{k-1}\big(f(x_{r+1})-f(x_r)\big)\qquad\text{(since } f\text{ is increasing – see Fig. 6)}\\
&=\delta_{\max}\big[(f(x_1)-f(x_0))+(f(x_2)-f(x_1))+\cdots+(f(x_{x_k})-f(x_{k-1}))\big]\\
&=\delta_{\max}(P)\big(f(b)-f(a)\big)\qquad\text{(since } a=x_0,\ b=x_k).
\end{aligned}$$

Now let (P_n) be a sequence of partitions with $\delta_{\max}(P_n)\to0$. Then

$$0\leqslant U(P_n,f)-L(P_n,f)<\delta_{\max}(P_n)(f(b)-f(a))\to0,$$

so f is integrable, by Theorem 6.12. ∎

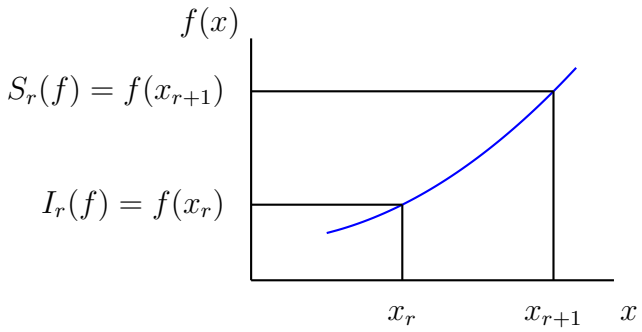


Figure 6: $S_r(f)$ and $I_r(f)$ for increasing f.

Theorem 6.17

If f is continuous on $[a, b]$ then it is integrable.

Proof. Since f is continuous on a closed interval, f is in fact uniformly continuous on $[a, b]$, so for each integer $n \geqslant 1$, there exists $\delta_n > 0$ such that

$$|x - y| < \delta_n \implies |f(x) - f(y)| < \frac{1}{n}.$$

Now, for any partition P_n with $\delta_{\max}(P_n) < \delta_n$, it follows from the Boundedness Theorem that for each $r = 0, \ldots, k-1$, there exists y_r, z_r, such that $f(y_r) = I_r(f)$, $f(z_r) = S_r(f)$, and so

$$S_r(f) - I_r(f) = f(z_r) - f(y_r) < \frac{1}{n}.$$

Hence,

$$0 \leqslant U(P_n, f) - L(P_n, f) = \sum_{r=0}^{k-1} (S_r(f) - I_r(f))\delta_r \leqslant \frac{1}{n} \sum_{r=0}^{k-1} \delta_r = \frac{1}{n}(b - a) \to 0$$

so f is integrable, by Theorem 6.12. ■

The proofs of Theorem 6.16 and Theorem 6.17 also prove the following result.

Corollary 6.18

Suppose that f is either monotone or continuous on $[a, b]$. Let (P_n) be a sequence of partitions such that $\delta_{\max}(P_n) \to 0$. Denote these partitions by

$$P_n = \{x_{n,0}, \ldots, x_{n,k_n}\}, \quad n \geqslant 1,$$

and for each $n \geqslant 1$ and $r = 0, 1, \ldots, k_{n-1}$, suppose that

$$\begin{aligned} &\xi_{n,r} \text{ denotes an arbitrary point in } [x_{n,r}, x_{n,r+1}], \\ &\delta_{n,r} = x_{n,r+1} - x_{n,r}. \end{aligned}$$

Then

$$\int_a^b f = \int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{r=0}^{k-1} f(\xi_{n,r})\delta_{n,r}. \tag{6.13}$$

7 The Fundamental Theorem of Calculus

Theorem 6.16 and Theorem 6.17 give us a large class of integrable functions, but they don't tell us how to calculate integrals. We can now (finally) relate the above idea of an integral to the derivative and show that, in a sense, integration is the reverse of differentiation (in 1-dimension).

Theorem 7.1

(a) Suppose that $g : [a, b] \to \mathbb{R}$ is continuous, and define $G : [a, b] \to \mathbb{R}$ by

$$G(x) := \int_a^x g(t)\,dt, \quad x \in [a, b].$$

Then G is differentiable on (a, b), and $G'(x) = g(x)$ for all $x \in (a, b)$.

(b) Suppose that I is an open interval and $F : I \to \mathbb{R}$ is differentiable on I. Suppose also that $[a, b] \subset I$ and F' is continuous on $[a, b]$. Then

$$\int_a^x F'(t)\,dt = F(x) - F(a), \quad x \in [a, b].$$

Remarks.

(a) Theorem 7.1 (a) says that if you integrate a continuous function g then differentiate, you get back to where you started from.

(b) Theorem 7.1 (b) says that if you differentiate a function F (and the derivative F' is continuous) and then integrate, you get back to where you started from, apart from the 'constant of integration' $-F(a)$.
Or you can rewrite the result as $F(x) = F(a) + \int_a^x F'(t)\,dt$, which you can regard as saying: starting with the value $F(a)$ and integrating F' from a to x gives you $F(x)$.

Proof. (a) Let $x \in (a, b)$ and let $\epsilon > 0$. Since g is continuous at x, there exists $\delta > 0$ such that

$$|t - x| < \delta \implies |g(t) - g(x)| < \epsilon. \tag{7.1}$$

Then, if $0 < h < \delta$,

$$\begin{aligned}
\left|\frac{G(x+h) - G(x)}{h} - g(x)\right| &= \left|\frac{1}{h}\left(\int_0^{x+h} g(t)\,dt - \int_0^x g(t)\,dt\right) - g(x)\right| \\
&= \left|\frac{1}{h}\int_x^{x+h} g(t)\,dt - g(x)\right| && \text{(by Theorem 6.15)} \\
&= \left|\frac{1}{h}\int_x^{x+h} (g(t) - g(x))\,dt\right| && \text{(by Theorem 6.14 } (e)) \\
&\leqslant \frac{1}{h}\int_x^{x+h} \epsilon\,dt && \text{(by (7.1) and Theorem 6.14 } (f)) \\
&= \epsilon && \text{(by Theorem 6.14) } (e))
\end{aligned}$$

so that

$$\lim_{h\to 0^+} \frac{G(x+h)-G(x)}{h} = g(x)\,.$$

We can also prove the same result for the left-sided limit $h \to 0^-$ (recall the definition, (6.9), of the integral when the lower limit of integration is larger than the upper limit). Hence, G is differentiable at x and $G'(x) = g(x)$.

(b) Define

$$H(x) = \int_a^x F'(t)\,dt, \quad x \in [a,b].$$

Then by part (a), H is differentiable on (a,b), and $H'(x) = F'(x)$, $x \in (a,b)$. Hence $H - F$ is constant on $[a,b]$, so $H(a) - F(a) = H(x) - F(x)$. Thus

$$F(x) - F(a) = H(x) - H(a) = \int_a^x F'(t)\,dt\,. \qquad \blacksquare$$

n

8 Double integrals

8.1 Double integrals: general definition and properties

We now generalise the ideas used to define the Riemann integral to the case of more variables.

Consider a region on the plane $D \subset \mathbb{R}^2$. We assume that D is bounded in the sense of the following definition.

Definition 8.1

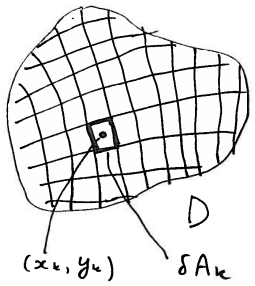
A set $D \subset \mathbb{R}^2$ is called bounded if

$$\exists\, R > 0 \text{ such that } D \subset D_R = \{(x, y) | x^2 + y^2 < R^2\}\,.$$

In simple words, D is bounded if it fits inside a sufficiently large disc.

We also assume that the boudnary of D is at most a finite number of continuous curves.

Suppose that a function of two variables $f = f(x, y)$ is defined on D. We would like to defined a two-dimensional (or double) integral of f over D. Generalising the one-dimensional integral construction we would like to subdivide D into smaller regions. It would be nice to have the subdividing blocks to be of standard shapes, say rectangles. However if the boundary of D does not consist of straight lines a subdivision into rectangles only is impossible. Thus we assume that we can subdivide D into finitely many blocks of unspecified shapes that we denote as $B_k, k = 1, \ldots, N$. We assume that each block is closed, in the sense that it contains its own boundary and that any two blocks either have no points in common or share finitely many curves. Suppose further that each block has an area δA_k and that in each block B_k we choose a point with coordinates (x_k, y_k).



We set up a Riemann sum approximating a double integral as

$$S(B_k, (x_k, y_k), f) = \sum_{k=1}^{N} f(x_k, y_k)\delta A_k\,.$$

As a measure of how small each block is we take the block's diameter that is given by the maximum distance between any two points inside the block:

$$\text{diam}(B) = \max_{(x,y)\in B,\,(x',y')\in B} \sqrt{(x - x')^2 + (y - y')^2}\,.$$

For example for a rectangle with side lengths a and b its diameter equals the length of the diagonal: $\sqrt{a^2 + b^2}$, for a disc of radius r it is the same as its usual diameter: $2r$. Now set

$$\lambda(B_1, \ldots, B_N) = max\{\text{diam}(B_1), \text{diam}(B_2), \ldots \text{diam}(B_N)\}\,.$$

This is a quantity measuring how small the blocks are, or how fine our subdivision is. If $\lambda(B_1, \ldots, B_N) < \delta$ then the distance between any two points inside any block is less than δ.

We can now formulate the following definition.

Definition 8.2 (Double integral)

Consider a bounded region on the plane: $D \subset \mathbb{R}^2$. We say that a function $f = f(x, y)$ defined on D is *integrable on* D if the following limit

$$\lim_{\lambda(B_1,\ldots B_N)\to 0} S(B_k, (x_k, y_k), f) = I\,.$$

exists and is independent of the choice of partition in two-dimensional blocks (so it si equal no matter how we choose the partition).

We call this value I the double integral of f over D and denote

$$\iint_D f(x, y)dxdy\,.$$

The above limit means that for any $\epsilon > 0$ there exists $\delta > 0$ such that for any subdivision $\{B_1, \ldots, B_N\}$ such that $\lambda(B_1, \ldots B_N) < \delta$ and any choice of points (x_k, y_k) inside the blocks we have $|I - S(P, (x_k, y_k), f)| < \epsilon$. In simple words this means that Riemann sums get arbitrarily close to the value of the integral for a sufficiently fine subdivision of D into smaller blocks.

As long as the blocks B_k constitute a well defined partition, namely

- Each B_k is closed
- If $B_k \cap B_j \neq$, then the intersection can only be a finite number of curves.

For example, all B_k's could be rectangular except at the boundary of D.

Then we could also use the notation

$$\iint_D f(x, y)dxdy \text{ or } \iint_D f(x, y)dydx\,, \tag{8.1}$$

i.e. if the function f is integrable, the order $dxdy$ or $dydx$ does not matter.

The definition may sound like it places a lot of constraints on the function f to be integrable, but in fact there is a rather large class of integrable functions described in the following theorem, which mirrors the analogous tehorem in one dimension (and is proved in analogous way).

Theorem 8.3

Given a bounded region D on the plane, functions which are continuous on D except for possibly finitely many curves, are integrable on D.

From the above definition and the basic properties of limits the following general rules for double integrals can be derived

Theorem 8.4

For any $f = f(x,y)$ integrable on a bounded region $D \subset \mathbb{R}^2$ and for any constant $C \in \mathbb{R}$ we have

$$\iint_D Cf(x,y)dxdy = C\iint_D f(x,y)dxdy\,. \tag{8.2}$$

Moreover, if $f = f(x,y)$ and $g = g(x,y)$ are both integrable on D then

$$\iint_D (f(x,y)+g(x,y))dxdy = \iint_D f(x,y)dxdy + \iint_D g(x,y)dxdy\,. \tag{8.3}$$

If in addition $f(x,y) \leqslant g(x,y)\ \forall (x,y) \in D$ then

$$\iint_D f(x,y)dxdy \leqslant \iint_D g(x,y)dxdy\,. \tag{8.4}$$

Theorem 8.5

Consider two bounded regions $D_1 \subset \mathbb{R}^2$, $D_2 \subset \mathbb{R}^2$ such that D_1 and D_2 either have no points in common or share a collection of curves. Then for any function $f = f(x,y)$ integrable on D_1 and on D_2, f is integrable on $D_1 \cup D_2$ and we have

$$\iint_{D_1\cup D_2} f(x,y)dxdy = \iint_{D_1} f(x,y)dxdy + \iint_{D_2} f(x,y)dxdy \tag{8.5}$$

8.2 How to calculate double integrals

Assume f is an integrable function, so that the order in which we describe the region of integration does not affect the value of the integral. The double integral of f can then be calculated by reducing it to repeated integrals. We illustrate this first on a rectangular region.

Theorem 8.6

Let

$$R = \{(x, y) | a \leqslant x \leqslant b\,, c \leqslant y \leqslant d\}$$

be a rectangle, and let $f = f(x, y)$ be a function integrable on R. Assume that the one-dimensional integral

$$\int_a^b f(x, y)dx \tag{8.6}$$

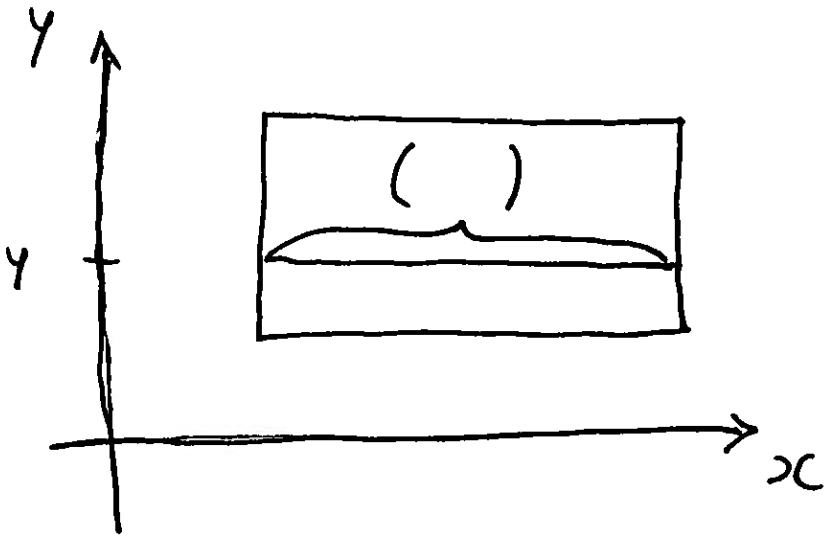
exits for all values of $y \in [c, d]$. Then the repeated integral

$$\int_c^d \left(\int_a^b f(x, y)dx \right) dy$$

also exists and

$$\iint_R f dxdy = \int_c^d \left(\int_a^b f(x, y)dx \right) dy\,. \tag{8.7}$$

The right hand side in (8.7) is called a *repeated* or *iterated integral* – it is a succession of two ordinary integrals performed one after the other. The integral inside the brackets $(\dots)$ is the result of integrating in the x-direction along the line of fixed height y. The outcome is a function of y, which is then integrated in the y direction.



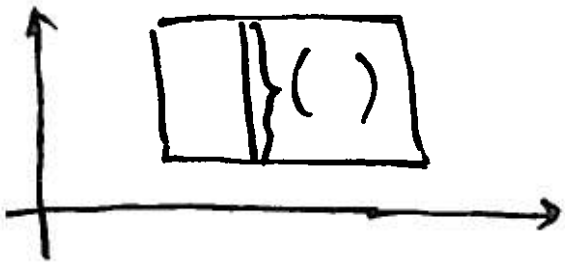
The roles of x and y in the above theorem can be interchanged. Given that a one-dimensional integral

$$\int_c^d f(x, y)dy$$

exits for any value of $x \in [a,b]$ we can represent the double integral of f over R as

$$\iint_R f\,dxdy = \int_a^b \left(\int_c^d f(x,y)dy \right) dx\,. \tag{8.8}$$

which is an iterated integral where we integrate in the y-direction first, and then in the x-direction.



Thus we see that in theory we may perform the x- and y-integrations in either order and for an integrable function the result should be the same. However, we shall see below that in practice it is sometimes much easier to perform the double integral in one way rather than the other.

Example 8.7

Let $f(x,y) = 3xy^2 + 2x^3$, and let

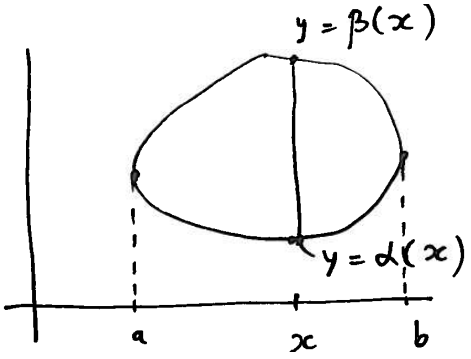
$$R = \{(x,y) \mid 1 \le x \le 2\,,\ 0 \le y \le 1\} \subset \mathbb{R}^2\,.$$

$$\begin{aligned}
\iint_R f(x,y)dxdy &= \int_1^2 \left(\int_0^1 (3xy^2 + 2x^3)dy \right) dx \\
&= \int_1^2 \left(\left[xy^3 + 2x^3 y \right]_{y=0}^{y=1} \right) dx \\
&= \int_1^2 \left(x + 2x^3 \right) dx = \left[\frac{1}{2}x^2 + \frac{1}{2}x^4 \right]_{x=1}^{x=2} \\
&= \frac{1}{2}\cdot 2^2 + \frac{1}{2}\cdot 2^4 - \frac{1}{2}\cdot 1^2 - \frac{1}{2}\cdot 1^4 = 9\,.
\end{aligned}$$

Alternatively,

$$\begin{aligned}
\iint_R f(x,y)dxdy &= \int_0^1 \left(\int_1^2 \left(3xy^2 + 2x^3 \right) dx \right) dy \\
&= \int_0^1 \left(\left[\frac{3}{2}x^2y^2 + \frac{1}{2}x^4 \right]_{x=1}^{x=2} \right) dy \\
&= \int_0^1 \left(6y^2 + 8 - \frac{3}{2}y^2 - \frac{1}{2} \right) = \int_0^1 \left(\frac{9}{2}y^2 + \frac{15}{2} \right) dy \\
&= \left[\frac{3}{2}y^3 + \frac{15}{2}y \right]_{y=0}^{y=1} = 9\,.
\end{aligned}$$

Next we would like to calculate double integrals over more complicated regions on the plane by reducing them to iterated (repeated) integrals. Defining *iterated* integrals over general regions D is a bit more complicated. Suppose that the boundary of D consists of the curves $y = \alpha(x)$ and $y = \beta(x)$, with $a \leq x \leq b$:



Then we can set up a repeated integral as

$$\int_a^b \left(\int_{\alpha(x)}^{\beta(x)} f(x,y)dy \right) dx \,. \tag{8.9}$$

The inner integral w.r.t. y is taken along the line at constant x, between the bottom and the top region given by $\alpha(x)$ and $\beta(x)$. The following theorem holds

Theorem 8.8

Let D be a region bounded by continuous curves $y = \alpha(x)$ and $y = \beta(x)$, with $a \leq x \leq b$, and let $f = f(x,y)$ be a function integrable on D. Assuming that a one-dimensional integral

$$\int_{\alpha(x)}^{\beta(x)} f(x,y)dy \tag{8.10}$$

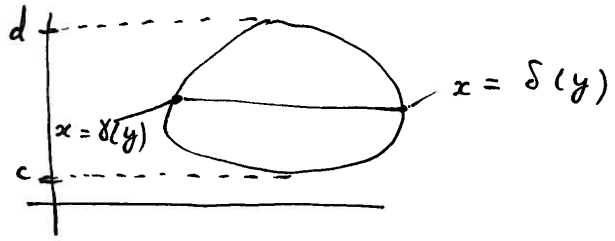
exists $\forall x \in [a,b]$ then the iterated integral

$$\int_a^b \left(\int_{\alpha(x)}^{\beta(x)} f(x,y)dy \right) dx$$

also exists and

$$\iint_D f(x,y)dxdy = \int_a^b \left(\int_{\alpha(x)}^{\beta(x)} f(x,y)dy \right) dx \,. \tag{8.11}$$

Similarly, we can specify the region D via the curves $x = \gamma(y)$ and $x = \delta(y)$, with $c \leq y \leq d$:



Then, under assumptions similar to those in theorem 8.8

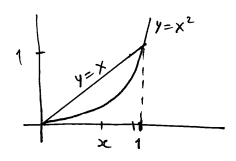
$$\iint_D f(x,y)dxdy = \int_c^d \left(\int_{\gamma(y)}^{\delta(y)} f(x,y)dx \right) dy\,, \tag{8.12}$$

where we integrate w.r.t. x first and then w.r.t. y.

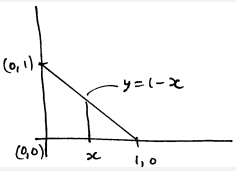
Example 8.9

Determine $\iint_D xy\ dxdy$, where D is

(a) the area in the first quadrant between the curves $y = x$ and $y = x^2$:



(b) the triangle with vertives $(0,0)$, $(1,0)$ and $(0,1)$:



Solution.

(a) The x-range for D is $0 \le x \le 1$. For x fixed, the y-range is $x^2 \le y \le x$. Thus $\alpha(x) = x^2$ and $\beta(x) = x$.

$$\begin{aligned}\iint_D xy\ dxdy &= \int_0^1 \left(\int_{x^2}^{x} xy\ dy \right) dx = \int_0^1 \left(\left[\frac{1}{2}xy^2 \right]_{y=x^2}^{y=x} \right) dx \\ &= \int_0^1 \left(\frac{1}{2}x^3 - \frac{1}{2}x^5 \right) dx = \left[\frac{1}{8}x^4 - \frac{1}{12}x^6 \right]_0^1 = \frac{1}{24}\,.\end{aligned}$$

(b) The x-range for D is $0 \le x \le 1$. For x fixed the y-range is $0 \le y \le 1 - x$:

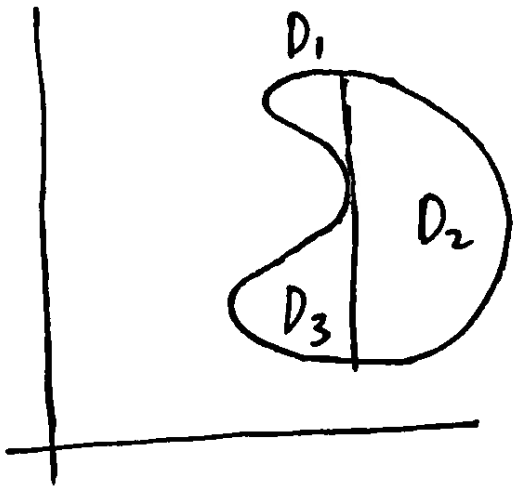
$$\begin{aligned}\iint_D xy\ dxdy &= \int_0^1 \left(\int_0^{1-x} xy\ dy \right) dx \\ &= \int_0^1 \left(\left[\tfrac{1}{2}xy^2 \right]_{y=0}^{y=1-x} \right) dx = \int_0^1 \tfrac{1}{2}x(1-x)^2 dx \\ &= \tfrac{1}{2}\int_0^1 \left(x - 2x^2 + x^3 \right) dx = \tfrac{1}{2}\left[\tfrac{1}{2}x^2 - \tfrac{2}{3}x^3 + \tfrac{1}{4}x^4 \right]_0^1 = \frac{1}{24}\,.\end{aligned}$$

(Note: it is a pure coincidence that the answer is the same as in part **(a)**!)

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For regions that cannot be described in terms of two bounding curves we can break them

up into such subregions as illustrated on the picture



and then use theorem 8.5 to write

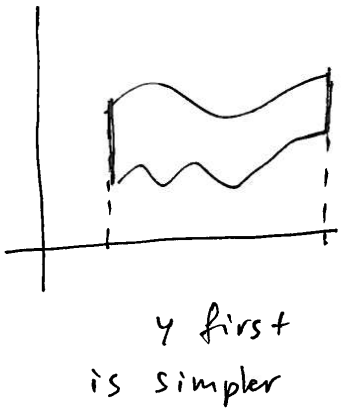
$$\iint_D f dxdy = \iint_{D_1} f dxdy + \iint_{D_2} f dxdy + \iint_{D_3} f dxdy\,.$$

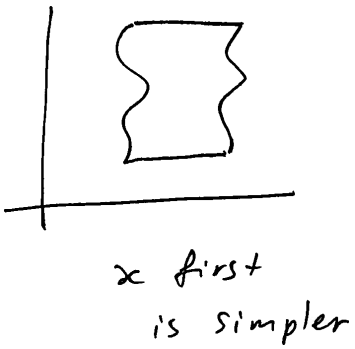
8.3 Interchanging the integration order

Although

$$\int \left(\int f(x,y) dy \right) dx = \int \left(\int f(x,y) dx \right) dy\,, \tag{8.13}$$

it is sometimes much easier to do the integrals in one rather than the other order. This may depend on the shape of the region. For example the following rule of thumb holds

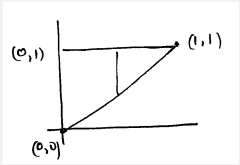




Or it may depend on the function we are integrating, as the following examples illustrate

Example 8.10

Determine $\iint_T e^{-y^2}dxdy$, where T is the triangular region with vertices $(0,0)$, $(1,1)$ and $(0,1)$:



Solution The x-range is $0 \le x \le 1$. For x fixed, the y-range is $x \le y \le 1$. Hence

$$\iint_T e^{-y^2}dxdy = \int_0^1 \left(\underbrace{\int_x^1 e^{-y^2}dy}_{\text{impossible to write out!}} \right) dx\,.$$

However, if we do the integrations in the other order, which means for the y-range $0 \le y \le 1$ and for the x-range with y fixed $0 \le x \le y$, we obtain:

$$\begin{aligned}\iint_T e^{-y^2}dxdz &= \int_0^1 \left(\int_0^y e^{-y^2}dx \right) dy \\ &= \int_0^1 \left(\left[xe^{-y^2} \right]_{x=0}^{x=y} \right) dy = \int_0^1 ye^{-y^2}dy \\ &= \left[-\tfrac{1}{2}e^{-y^2} \right]_0^1 = \tfrac{1}{2}\left(1 - e^{-1}\right)\,.\end{aligned}$$

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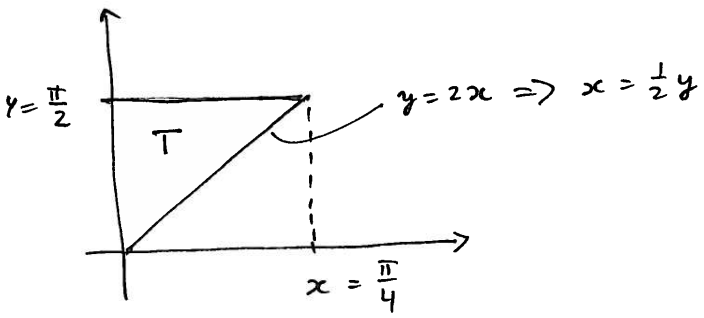
Example 8.11

Evaluate

$$\int_0^{\pi/4} \int_{2x}^{\pi/2} \frac{\cos(y)}{y} dydx\,.$$

(Note: the limits on each integral sign indicate that this is an iterated integral; even though the brackets are omitted, the order of the integration, i.e. $dydx$, is such that the y-integration is performed first!)

Solution We cannot integrate $\int \frac{\cos(y)}{y}$, so we will try to change the order of integration first.



The integration region is the triangle T depicted above. The y-range is $0 \le y \le \pi/2$. For y fixed, the y-range is $0 \le x \le \frac{y}{2}$. We thus obtain for the integral:

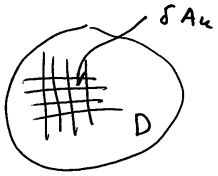
$$\int_0^{\pi/2} \left(\int_0^{y/2} \frac{\cos y}{y} dx \right) dy = \int_0^{\pi/2} \left(\left[x \frac{\cos y}{y} \right]_{x=0}^{x=y/2} \right) dy$$
$$= \int_0^{\pi/2} \tfrac{1}{2} \cos y dy = \frac{1}{2} .$$

■

8.4 Calculating areas using double integrals

Consider a region D. Cover it with small blocks of area δA_k. Then the area of the region D is approximately given by $D \approx \sum_k \delta A_k$. The precise area is obtained by

$$\boxed{\text{Area of } D = \iint_D 1 dA = \iint_D 1 dx dy .} \tag{8.14}$$



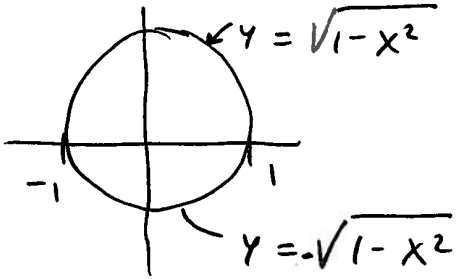
Example 8.12

Find the area of the region inside the unit circle

$$x^2 + y^2 = 1 .$$

Solution The x-range for D is $-1 \le x \le 1$. For fixed x, the y-range is

$$-\sqrt{1-x^2} \le y \le \sqrt{1-x^2} .$$



We thus obtain:

$$\begin{aligned}\text{Area of } D &= \iint_D 1 dxdy = \int_{-1}^{1}\left(\int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} dy\right) dx \\ &= 2\int_{-1}^{1}\sqrt{1-x^2}dx \qquad \text{use substituation } x=\sin\theta,\ dx=\cos\theta d\theta \\ &= 2\int_{-\pi/2}^{\pi/2}\cos^2\theta d\theta \\ &= 2\int_{-\pi/2}^{\pi/2}\frac{(1+\cos(2\theta))}{2}d\theta = \left[\theta+\tfrac{1}{2}\sin(2\theta)\right]_{-\pi/2}^{\pi/2} = \pi\,.\end{aligned}$$

■

8.5 Change of variables in double integrals - the Jacobian

Recall that for one-variable integrals such as $\int_a^b f(x)dx$, integrals may be transformed by changing the variable: $x = x(u)$, $dx = \frac{dx}{du}du$,

$$\int_a^b f(x)dx = \int_c^d f(x(u))\frac{dx}{du}du\,, \quad \text{where } a = x(c)\,,\ b = x(d)\,. \tag{8.15}$$

We would like to perform similar transformations when evaluating double integrals.

Definition 8.13 (Jacobian)

Suppose that $x = x(u,v)$ and $y = y(u,v)$ for some variables u, v, and suppose that the region D' in the uv-plane is mapped to the region D in the xy-plane via the transformation

$$(u,v) \mapsto (x(u,v), y(u,v))\,. \tag{8.16}$$

The *Jacobian* of x and y with respect to u and v is the determinant $det\frac{\partial(x,y)}{\partial(u,v)}$ of the matrix of partial derivatives:

$$\left|\frac{\partial(x,y)}{\partial(u,v)}\right| := \left|\det\begin{pmatrix}\frac{\partial x}{\partial u} & \frac{\partial x}{\partial v}\\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v}\end{pmatrix}\right|\,. \tag{8.17}$$

The Jacobian describes how the infinitesimal element of area changes due to a change of variables:

$$dxdy \sim \frac{\partial(x,y)}{\partial(u,v)}dudv.$$

Hence the Jacobian is crucial when changing variables in a double integral.

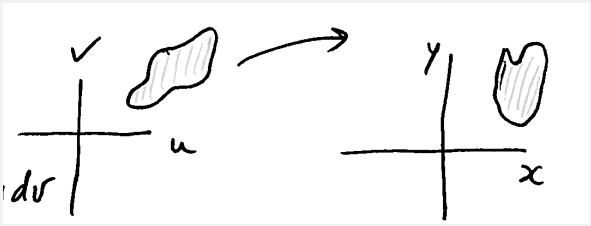
Theorem 8.14

Consider a double integral

$$\iint_D f(x,y)dxdy\,.$$

Suppose that $x = x(u,v)$ and $y = y(u,v)$ for some variables u, v, and suppose that the region D' in the uv-plane is mapped to the original region of integration D in the xy-plane via the transformation

$$(u,v) \mapsto (x(u,v), y(u,v))\,. \tag{8.18}$$



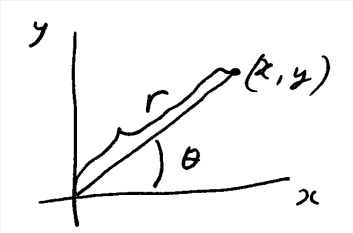
Then

$$\iint_D f(x,y)dxdy = \iint_{D'} f(x(u,v), y(u,v)) \left|\frac{\partial(x,y)}{\partial(u,v)}\right| dudv\,. \tag{8.19}$$

We do not give a formal proof of this theorem, but illustrate it through many examples below.

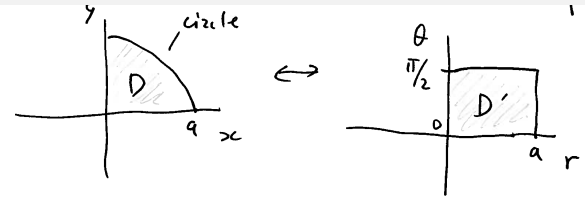
Example 8.15 (Polar coordinates)

When the integration region for a double integral is of circular shape, or if the integrand involves the combination $x^2 + y^2$, it is often useful to employ polar coordinates:



$$x = r\cos\theta\,,\ y = r\sin\theta\,, \tag{8.20}$$

e.g.



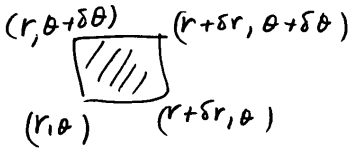
that is a quarter circle in the xy-plane corresponds to a rectangle in the $r\theta$-plane. The Jacobian of x and y with respect to r and θ reads

$$\frac{\partial(x,y)}{\partial(r,\theta)} = det\begin{pmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{pmatrix} = det\begin{pmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{pmatrix} = r\,. \tag{8.21}$$

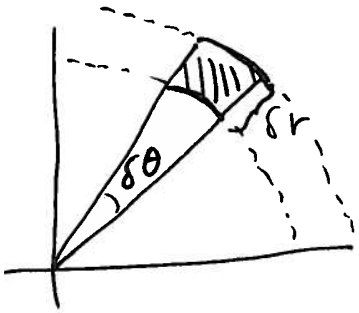
Hence

$$\iint_D f(x,y)dxdy = \iint_{D'} f(x(r,\theta), y(r,\theta))rdrd\theta\,. \tag{8.22}$$

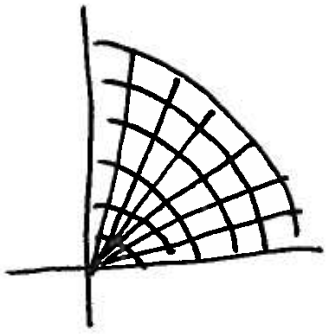
Remark. Note: Consider a small rectangle in the $r\theta$-plane:



It corresponds in the xy-plane to the following region:



The length of the curved edge is $\approx r\delta\theta$, so that the area of the region is $r\delta r\delta\theta$. Thus, when we compute a duoble integral in polar coordinates, we, in effect, cover the integration region with a mesh of little subregions of area $r\delta r\delta\theta$ like this:



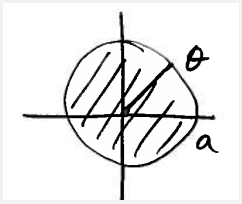
Example 8.16

Compute

$$\iint_D e^{-(x^2+y^2)}dxdy$$

with integration region

$$D = \left\{(x,y) \in \mathbb{R}^2 \;\middle|\; x^2+y^2 \le a^2\right\}$$



Solution. In polar coordinates, the range for θ is $0 \le \theta \le 2\pi$, while for fixed θ we have $0 \le r \le a$. Hence:

$$\begin{aligned}\iint_D e^{-(x^2+y^2)}dxdy &= \int_0^{2\pi}\left(\int_0^a e^{-r^2}rdr\right)d\theta = \left(\int_0^a e^{-r^2}rdr\right)\cdot\left(\int_0^{2\pi}d\theta\right)\\ &= \left[-\tfrac{1}{2}e^{-r^2}\right]_{r=0}^{r=a}\cdot 2\pi\\ &= \pi(1-e^{-a^2})\,.\end{aligned}$$

■

8.6 Polar coordinates – continued

Polar coordinates are also useful in representing circles other than those centered at the origin $(0,0)$. Consider the equation

$$r = a\cos\theta \quad\Leftrightarrow\quad r = \frac{ax}{r} \quad\Leftrightarrow\quad r^2 = ax\,, \tag{8.23}$$

hence

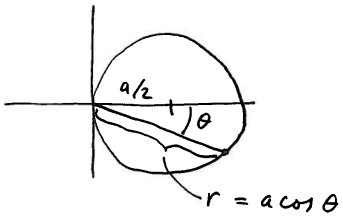
$$x^2 + y^2 = ax\,. \tag{8.24}$$

Completing the square, we obtain

$$\left(x - \tfrac{a}{2}\right)^2 + y^2 = \tfrac{a^2}{4}\,. \tag{8.25}$$

This means that $r = a\cos\theta$ represents a circle with center $(\frac{a}{2}, 0)$ and radius $\frac{a}{2}$. If D' denotes the disc bounded by the circle, we have the explicit parametrization

$$D' = \left\{(r,\theta) \;\middle|\; 0 \le r \le a\cos\theta\,,\; -\tfrac{\pi}{2} \le \theta \le \tfrac{\pi}{2}\right\}\,. \tag{8.26}$$



Similarly, $r = a\sin\theta$ represents the circle centered at $(0, \frac{a}{2})$ with radius $\frac{a}{2}$. The disc bounded by this circle is described via the explicit parametrization $0 \le \theta \le \pi$ and $0 \le r \le a\sin\theta$.

Example 8.17

Sketch the region of integration for

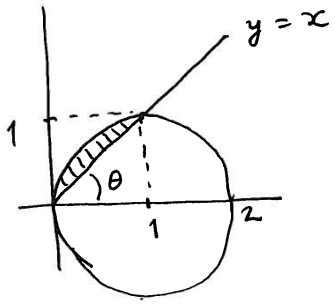
$$I = \int_0^1 \int_x^{\sqrt{2x-x^2}} \sqrt{4 - x^2 - y^2}\; dydx$$

and then evaluate the integral by using polar coordinates.

Solution. We first compute

$$y = \sqrt{2x - x^2} \quad \Rightarrow \quad y^2 = 2x - x^2 \quad \Leftrightarrow \quad (x-1)^2 + y^2 = 1\,.$$

Thus, the region of integration is the region between the line $y = x$ and the circle $(x-1)^2 + y^2 = 1$:



In polar coordinates, this region is described as

$$\tfrac{\pi}{4} \le \theta \le \tfrac{\pi}{2}\,,\; 0 \le r \le 2\cos\theta\,.$$

Therefore,

$$
\begin{aligned}
I &= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \int_0^{2\cos\theta} \sqrt{4-r^2}\, r dr d\theta \\
&= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \left[-\tfrac{1}{3}(4-r^2)^{\frac{3}{2}} \right]_{r=0}^{r=2\cos\theta} d\theta \\
&= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \tfrac{1}{3}(8 - 8\sin^3\theta) d\theta \\
&= \tfrac{8}{3}\left[\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} d\theta - \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (1-\cos^2\theta)\sin\theta d\theta \right] \qquad u = cos\theta\,,\ du = -\sin\theta d\theta \\
&= \tfrac{8}{3}\left[\tfrac{\pi}{4} + \int_{\frac{1}{\sqrt{2}}}^{0} (1-u^2) du \right] \\
&= \tfrac{8}{3}\left[\tfrac{\pi}{4} - \tfrac{1}{\sqrt{2}} + \tfrac{1}{3}\left(\tfrac{1}{\sqrt{2}}\right)^2 \right].
\end{aligned}
$$

■

8.7 Other substitutions

Sometimes the form of the integrand suggests a suitable substitution.

Example 8.18

Find

$$\iint_D \left(\frac{x-y}{x+y}\right)^2 dA\,,$$

where D is the region bounded by the line $x+y=1$ and the coordinate axes.

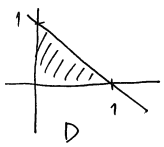
Solution. Let $u = x+y$ and $v = x-y$, such that

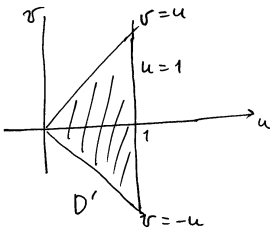
$$x = \tfrac{1}{2}(u+v)\,,\ y = \tfrac{1}{2}(u-v)\,.$$

The Jacobian for the change of coordinates reads

$$\frac{\partial(x,y)}{\partial(u,v)} = det\begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{pmatrix} = det\begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix} = -\frac{1}{2}\,.$$

Next, we need to know the region D' in the (u,v) variables that corresponds to the region D in the (x,y) variables, which is described via $x+y \le 1$, $x \ge 0$ and $y \ge 0$:





This may be achieved by simply expressing the above equations in terms of the (u, v) variables, that is

$$u \leq 1\,,\ u+v \geq 0\,,\ u-v \geq 0\,.$$

To compute the double integral, it is most convenient to perform the integral over v first. It is easy to check that

$$\frac{\partial(x,y)}{\partial(v,u)} = -\frac{\partial(x,y)}{\partial(u,v)} = \tfrac{1}{2}\,,$$

hence we obtain:

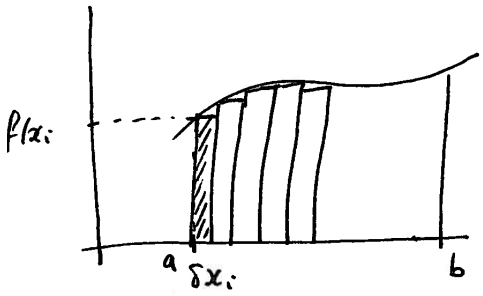
$$\begin{aligned} I &= \int_0^1 \int_{-u}^{u} \left(\frac{v}{u}\right)^2 \frac{\partial(x,y)}{\partial(v,u)} dv du \\ &= \tfrac{1}{2} \int_0^1 \left[\tfrac{1}{3}\frac{v^3}{u^2}\right]_{v=-u}^{v=u} du = \tfrac{1}{6}\int_0^1 2u du \\ &= \frac{1}{6}\,. \end{aligned}$$

■

8.8 Calculating volumes using double integrals

Consider a function $f : \mathbb{R}^1 \to \mathbb{R}^1$ with $f \geq 0$. We may compute

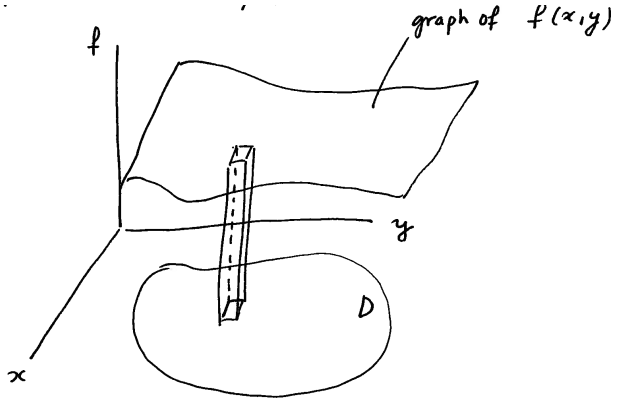
$$\sum_i f(x_i)\delta x_i = \text{sum of areas of rectangles of width } \delta x_i \text{ and height } f(x_i)\,. \tag{8.27}$$



So

$$\int_a^b f(x)dx = \text{area under the graph of } f\,. \tag{8.28}$$

Consider next a function $f : \mathbb{R}^2 \to \mathbb{R}^1$ with $f \geq 0$.



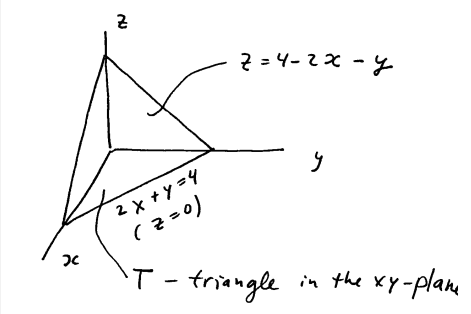
$$\sum_i \sum_j f(x_i, y_j)\delta x_i \delta y_j = \text{sum of volumes of boxes with base area } \delta x_i \delta y_j \text{ and height } f(x_i, y_j)\,. \tag{8.29}$$

Therefore,

$$\iint_D f(x,y)dxdy = \text{volume of the region between } D \text{ and the graph of } f\,. \tag{8.30}$$

Example 8.19

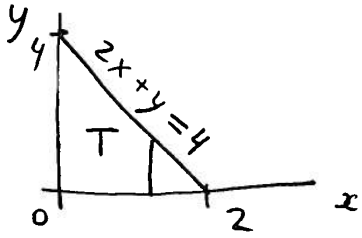
Find the volume bounded by the plane $2x + y + z = 4$ and the coordinate planes.



Solution. The volume V we wish to compute is the volume between the triangle T and the surface $z = 4 - 2x - y$, hence

$$V = \iint_T (4 - 2x - y)dxdy\,.$$

The x-range for T is $0 \le x \le 2$. For x fixed, the y-range is $0 \le y \le 4 - 2x$.



$$\begin{aligned} V &= \int_0^2 \left(\int_0^{4-2x} (4 - 2x - y)dy \right) dx \\ &= \int_0^2 \left[(4-2x)y - \tfrac{1}{2}y^2 \right]_{y=0}^{y=4-2x} dx \\ &= \tfrac{1}{2} \int_0^2 (4-2x)^2 dx = \tfrac{1}{2} \left[\frac{(4-2x)^3}{3 \cdot (-2)} \right]_0^2 \\ &= \frac{16}{3}\,. \end{aligned}$$

■

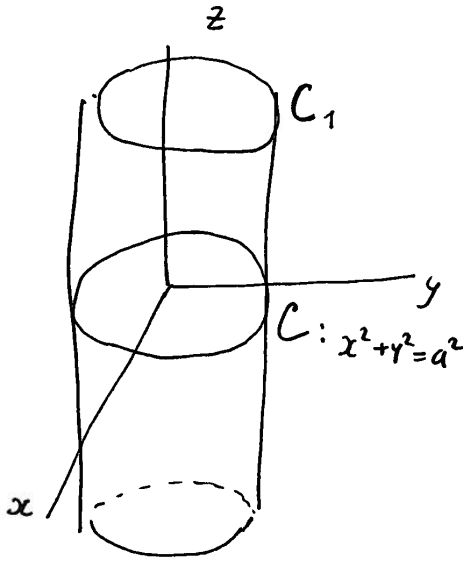
Example 8.20

Find the volume of the region inside the intersection of two cylnders, which are described by

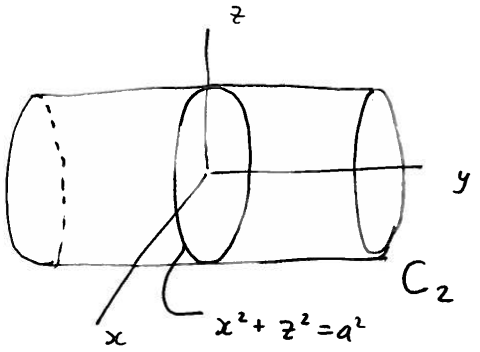
$$x^2 + y^2 = a^2\,,\ x^2 + z^2 = a^2\,.$$

Solution. Call the two cylinders C_1 and C_2. C_1 is parallel to the z-axis. Its intersection with the xy-plane is a circle

$$C: \; x^2 + y^2 = a^2 \, .$$



The cylinder C_2 is parallel to the y-axis.

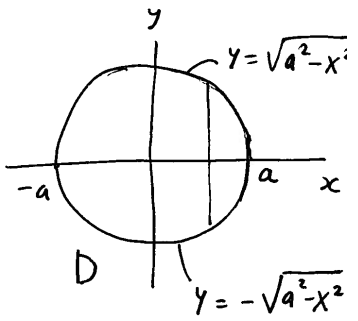


Let D be the region in the xyplane inside the circle C. If $(x, y) \in D$, then the height of C_2 above (x, y) is $z = \sqrt{a^2 - x^2}$. The region below the circle C must have the same volume as the region above it because C_2 is symmetric under $z \to -z$. Hence

$$V = 2 \cdot \iint_D \sqrt{a^2 - x^2} dx dy \, .$$

The x-range is $-a \le x \le a$. For x fixed, the y-range is

$$-\sqrt{a^2 - x^2} \le y \le \sqrt{a^2 - x^2} \, ,$$



hence we obtain:

$$\begin{aligned} V &= 2\int_{-a}^{a}\left(\int_{-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}}\sqrt{a^2-x^2}dy\right)dx \\ &= 2\int_{-a}^{a}\left[y\sqrt{a^2-x^2}\right]_{y=-\sqrt{a^2-x^2}}^{y=\sqrt{a^2-x^2}}dx \\ &= 4\int_{-a}^{a}(a^2-x^2)dx = 4\left[a^2x-\tfrac{1}{3}x^3\right]_{-a}^{a} \\ &= \tfrac{16}{3}a^3\,. \end{aligned}$$

■

9 Triple integrals

Definition 9.1

A 3-dimensional region $D \subset \mathbb{R}^3$ is called bounded if

$$\exists\, R > 0 \text{ such that } D \subset B_R = \{(x, y, z) | x^2 + y^2 + z^2 < R^2\}$$

This means that D fits inside an open ball B_R of sufficiently large radius R.

Let $f = f(x, y, z)$ be a function defined on D. We can define a triple integral of f over D in much the same way we defined a double integral in definition 8.2. Namely, consider a subdivision of D into small three-dimensional blocks $B_1, \ldots, B_N$. Let each block be a close set (containing its own boundary) and let δV_k denote the volume of block B_k. Furthermore, in each block choose a point with coordinates $(x_k, y_k, z_k) \in B_k$. We set up a Riemann sum as

$$S(B_k, (x_k, y_k, z_k), f) = \sum_{k=1}^{N} f(x_k, y_k, z_k)\delta V_k \,.$$

As a measure of how small each block is we take the block's diameter that is given by the maximum distance between any two points inside the block:

$$\text{diam}(B) = \max_{(x,y,z)\in B,\,(x',y',z')\in B} \sqrt{(x-x')^2 + (y-y')^2 + (z-z')^2}\,.$$

For example for a rectangular box with side lengths a, b, c its diameter equals the length of the long diagonal: $\sqrt{a^2 + b^2 + c^2}$, for a three-dimensional ball of radius r the diameter equals $2r$. Now set

$$\lambda(B_1, \ldots, B_N) = max\{\text{diam}(B_1), \text{diam}(B_2), \ldots \text{diam}(B_N)\}\,.$$

This is a quantity measuring how small the blocks are, or how fine our subdivision is. If $\lambda(B_1, \ldots, B_N) < \delta$ then the distance between any two points inside any block is less than δ.

Definition 9.2 (Triple integral)

Consider a bounded region on in three-dimensional space: $D \subset \mathbb{R}^3$. We say that a function $f = f(x, y, z)$ defined on D is *integrable on* D if the following limit exists

$$\lim_{\lambda(B_1,\ldots B_N)\to 0} S(B_k, (x_k, y_k, z_k), f) = I\,.$$

We call this value I the triple integral of f over D and denote

$$\iiint_D f(x, y, z)dxdydz\,.$$

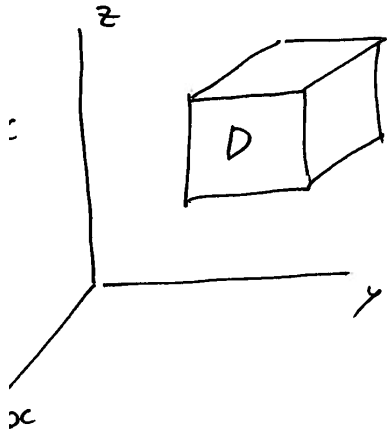
As in the case of double integral notation the order of the differentials $dxdydz$ does not matter and can be replaced by any of the other 6 possible orderings.

Another *notation* frequently used is

$$\iiint_D f(x, y, z)dV\,. \tag{9.1}$$

Triple integrals can also be expressed as iterated integrals. Suppose D is a rectangular box, i.e.

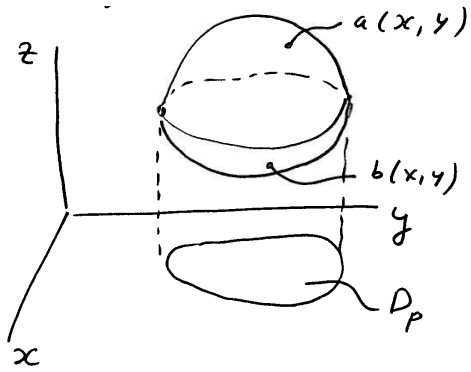
$$a \leq x \leq b\,,\ c \leq y \leq d\,,\ h \leq z \leq k\,.$$



Then, the triple integral can be calculated as

$$\iiint_D f(x,y,z)dV = \int_a^b \underbrace{\left[\int_c^d \overbrace{\left(\int_h^k f(x,y,z)dz\right)}^{\text{function of } x \text{ and } y} dy\right]}_{\text{function of } x \text{ only}} dx\,. \tag{9.2}$$

Here, we do the z-integration first, then the y-integration and finally the x-integration. As was the case for double integrals, we can change the order of integration. We can also consider integration regons with more complicated boundaries, such that the limits in the inner integrals depend on the variables in the outer integrals. Suppose that the top and bottom surfaces forming the boundary of D are specified by the functions $a(x,y)$ and $b(x,y)$ for $(x,y) \in D_p$, where D_p is the *projection* of D onto the xy-plane:

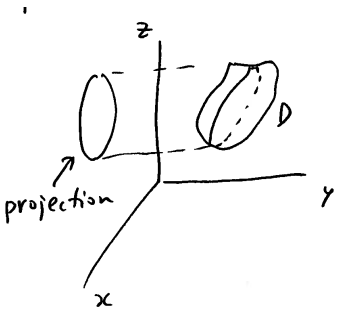


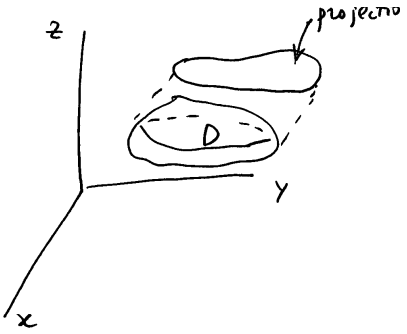
Then,

$$\iiint_D f dV = \iint_{D_p} \left(\int_{b(x,y)}^{a(x,y)} f(x,y,z)dz\right) dxdy\,. \tag{9.3}$$

To evaluate the double integral over D_p, we can turn it into an iterated integral so that in total we have to perform three iterated integrals.

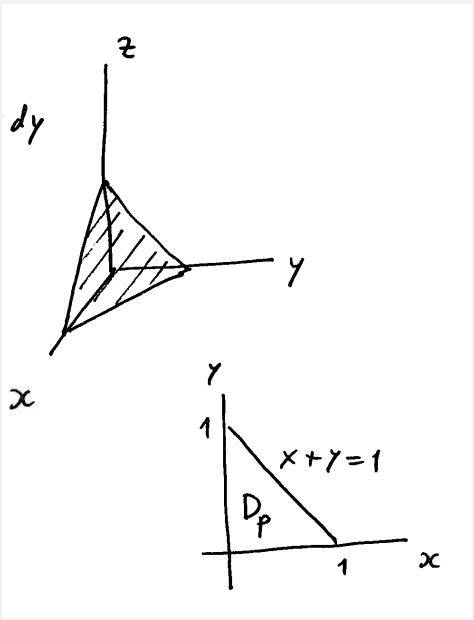
Note that we can also project D onto the xy- or yz-planes, which occasionally simplifies the computations:





Example 9.3

Let D be the region in the first octant bounded by the coordinate planes and the plane $x + y + z = 1$. Find $\iiint_D x\, dV$.



Solution.

$$\begin{aligned}
\iiint_D x\, dV &= \iint_{D_p} \left(\int_0^{1-x-y} x\, dz \right) dxdy \\
&= \iint_{D_p} [xz]_{z=0}^{z=1-x-y} = \iint_{D_p} x(1-x-y)dxdy \\
&= \int_0^1 \int_0^{1-x} (x - x^2 - xy)dydx \\
&= \int_0^1 \left[x(1-x)y - \frac{xy^2}{2} \right]_{y=0}^{y=1-x} dx \\
&= \int_0^1 \frac{x - 2x^2 - 4x^3}{2} dx = \left[\frac{x^2}{4} - \frac{x^3}{3} + \frac{x^4}{8} \right]_0^1 = \frac{1}{24}\,.
\end{aligned}$$

■

Example 9.4

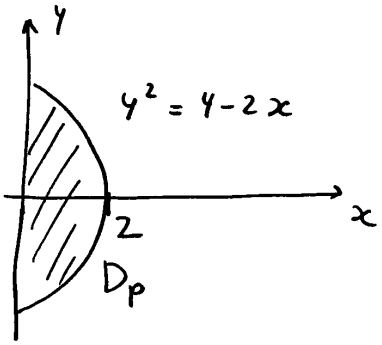
Express $\iiint_D f(x,y,z)dV$ as an iterated integral in the form

$$\int \left[\int \left(\int f(x,y,z)dz \right) dy \right] dx$$

if D is the region bounded by the surfaces $z = 0$, $z = x$ and $y^2 = 4 - 2x$ (where the first two equations describe planes, while the third one describes a half-cylinder) and with $x \geq 0$.

Solution. The projection D_p onto the xy-plane reads

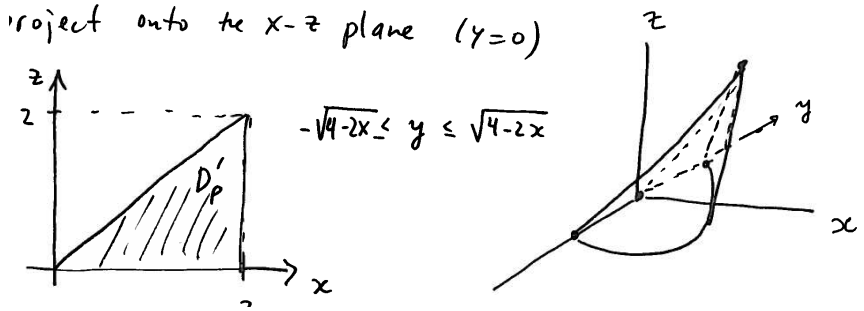
$$D_p = \left\{ (x,y) \in \mathbb{R}^2 \;\middle|\; x \geq 0 \text{ and } y^2 \leq 4 - 2x \right\} \subset \mathbb{R}^2 .$$



Also, $0 \leq z \leq x$, so that

$$\begin{aligned} \iiint_D f(x,y,z)dV &= \iint_{D_p} \left(\int_0^x f(x,y,z)dz \right) dxdy \\ &= \int_0^2 \left[\int_{-\sqrt{4-2x}}^{\sqrt{4-2x}} \left(\int_0^x f(x,y,z)dz \right) dy \right] dx . \end{aligned}$$

Alternatively, we could choose to project onto the xz-plane ($y = 0$):



$$\begin{aligned} \iiint_D f(x,y,z)dV &= \iint_{D'_p} \left(\int_{-\sqrt{4-2x}}^{\sqrt{4-2x}} f(x,y,z)dy \right) dxdz \\ &= \int_0^2 \left[\int_z^2 \left(\int_{-\sqrt{4-2x}}^{\sqrt{4-2x}} f(x,y,z)dy \right) dx \right] dz . \end{aligned}$$

■

Note that for any triple integral of $f = 1$,

$$\iiint_D 1dV \approx \sum_i \delta V_i \,,$$

or more precisely

$$\boxed{\iiint_D 1dV = Vol(D) \text{ – the volume of the 3-dimensional region } D.} \tag{9.4}$$

Example 9.5

Set $f = 1$ in the previous example. In the first representation we obtain:

$$V = \int_0^2 \left[\int_{-\sqrt{4-2x}}^{\sqrt{4-2x}} \left(\int_0^x dz \right) dy \right] dx = \int_0^2 \int_{-\sqrt{4-2x}}^{\sqrt{4-2x}} x dy dx$$
$$= 2 \int_0^2 \overbrace{x}^{f} \cdot \overbrace{\sqrt{4-2x}}^{g} dx \,.$$

Integrating by parts we obtain:

$$V = 2 \cdot \left(f \int g - \int \left[\int g f' \right] \right)$$
$$= 2 \left[\frac{x \cdot (4-2x)^{3/2}}{(-2) \cdot 3/2} \right]_0^2 + \int_0^2 \frac{2 \cdot (4-2x)^{3/}}{3} dx$$
$$= \left[\frac{2}{3} \frac{(4-2x)^{5/2}}{(-2) \cdot 5/2} \right]_0^2 = \frac{64}{15} \,.$$

Using the second representation of the integral we have:

$$V = \int_0^2 \left[\int_z^2 \left(\int_{-\sqrt{4-2x}}^{\sqrt{4-2x}} dy \right) dx \right] dz$$
$$\int_0^2 \int_z^2 2\sqrt{4-2x} dx dz = 2 \int_0^2 \left[\frac{(4-2x)^{3/2}}{(-2) \cdot 3/2} \right]_{x=z}^{x=2} dz = \frac{64}{15}$$

as above.

9.1 Change of variables in triple integrals

Consider a function $f : \mathbb{R}^3 \to \mathbb{R}$, i.e. $f = f(x, y, z)$, and suppose that

$$x = x(u, v, w) \,, \; y = y(u, v, w) \,, \; z = z(u, v, w) \,. \tag{9.5}$$

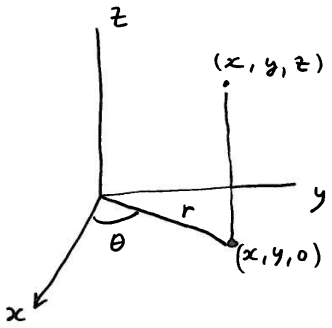
Then

$$\iiint_V f(x, y, z) dx dy dz = \iiint_{V'} f(x(u, v, w), y(u, v, w), z(u, v, w)) |J| du dv dw \,, \tag{9.6}$$

where $|J|$ is the absolute value of the Jacobian,

$$J = \frac{\partial(x,y,z)}{\partial(u,v,w)} = det \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{pmatrix} . \tag{9.7}$$

9.1.1 Cylindrical polar coordinates (r, θ, z)



$$\begin{aligned} x &= r\cos\theta \\ y &= r\sin\theta \\ z &= z\,, \end{aligned} \tag{9.8}$$

hence z is unchanged, but we use 2d polar coordinates for the (x, y) variables. The Jacobian for this change of coordinates reads

$$\begin{aligned} \frac{\partial(x,y,z)}{\partial(r,\theta,z)} &= det \begin{pmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} & \frac{\partial x}{\partial z} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} & \frac{\partial y}{\partial z} \\ \frac{\partial z}{\partial r} & \frac{\partial z}{\partial \theta} & \frac{\partial z}{\partial z} \end{pmatrix} = det \begin{pmatrix} \cos\theta & -r\sin\theta & 0 \\ \sin\theta & r\cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ &= r\cos^2\theta + r\sin^2\theta = r\,. \end{aligned} \tag{9.9}$$

In summary, denoting by V' the region in (r, θ, z) variables that corresponds to the region V in (x, y, z) variables, we obtain the useful formula

$$\boxed{\iiint_V f dx dy dz = \iiint_{V'} f r dr d\theta dz\,.} \tag{9.10}$$

Example 9.6

Find the volume of the cone of radius a and height h.



Solution. A general point on the surface of the cone has

$$\frac{r}{z} = \frac{a}{h} \quad \Leftrightarrow \quad r = \left(\frac{a}{h}\right) \cdot z\,.$$

Hence for the points in the interior and on the surface of the cone, we have

$$0 \le \theta \le 2\pi\,,\ 0 \le z \le h\,,\ 0 \le r \le \left(\frac{a}{h}\right) z\,.$$

Hence the volume of the cone is

$$V = \int_0^{2\pi} \left[\int_0^h \left(\int_0^{(\frac{a}{h})z} r dr \right) dz \right] d\theta = \tfrac{1}{3}\pi a^2 h\,.$$

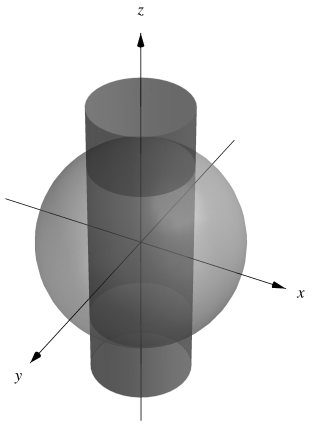
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Example 9.7

Find the volume of the region in the intersection of the solid cylinder $x^2 + y^2 \le 1$ (i.e. the solid cylinder of radius 1 around the z-axis) and the solid sphere described by

$$x^2 + y^2 + z^2 \le 4$$

(i.e. the solid sphere of radius 4 around the origin):



Solution.

$$\begin{aligned} x^2 + y^2 \le 1 \quad &\Rightarrow \quad r \le 1 \\ x^2 + y^2 + z^2 \le 4 \quad &\Rightarrow \quad z^2 \le 4 - r^2\,. \end{aligned}$$

Thus we have for the θ range $0 \le \theta \le 2\pi$, for the r range $0 \le r \le 1$, and for the z range for fixed (r, θ)

$$-\sqrt{4 - r^2} \le z \le \sqrt{4 - r^2}\,.$$

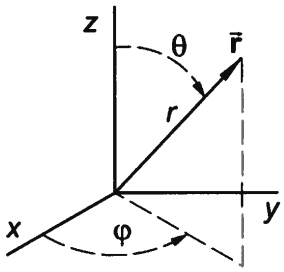
Hence the volume of the intersection of the solid cylinder and of the solid sphere may be computed as

$$\begin{aligned} V &= \int_0^{2\pi} \left[\int_0^1 \left(\int_{-\sqrt{4-r^2}}^{\sqrt{4-r^2}} dz \right) r dr \right] d\theta \\ &= \int_0^{2\pi} \left[\int_0^1 2r\sqrt{4-r^2} dr \right] d\theta \\ &= \tfrac{4\pi}{3}(8 - 3\sqrt{3}) \,. \end{aligned}$$

■

9.1.2 Spherical polar coordinates (r, θ, ϕ)

The spherical polar coordinates (r, θ, ϕ) of a point $\mathbf{r}$ are defined as shown in the figure:



The relationship between Cartesian (x, y, z) and spherical polar coordinates (r, θ, ϕ) of $\mathbf{r}$ is given by

$$\begin{aligned} x &= r \sin\theta \cos\phi \\ y &= r \sin\theta \sin\phi \\ z &= r \cos\theta \,, \end{aligned} \tag{9.11}$$

with the angular ranges

$$0 \le \theta \le \pi \,, \; 0 \le \phi \le 2\pi \,. \tag{9.12}$$

These formulae can be obtained as follows. Let T be the vertical triangle in the figure, with base in the xy-plane and top corner at the point $\mathbf{r}$. Firstly, the height of T is $z = r\cos\theta$. Next, the x and y coordinates of $\mathbf{r}$ are the same as the x and y coordinates of the point in the base of T at the right angle (i.e. at the foot of the vertical line in T), and we can obtain these coordinates by applying trigonometry (in the xy-plane) to this point. The length of the base of T is $r_b = r\sin\theta$, and the point lies at an angle ϕ from the x-axis, so we have

$$\begin{aligned} x &= r_b \cos\phi = r\sin\theta\cos\phi \\ y &= r_b \sin\phi = r\sin\theta\sin\phi \,. \end{aligned} \tag{9.13}$$

The angular ranges are obvious from the figure.

Remark. Note: The angle between the point and the x-axis was denoted θ when working in cylindrical polar coordinates, but has now become ϕ in the spherical polar coordinate system. This might be rather irritating, but is the standard notation, so we have to stick with it.

9.1.3 Jacobian

The *Jacobian* for the change from Cartesian (x, y, z) to spherical polar coordinates (r, θ, ϕ) is

$$\frac{\partial(x, y, z)}{\partial(r, \theta, \phi)} = r^2 \sin\theta \tag{9.14}$$

see Tutorial 6. Hence, in spherical polar coordinates, triple integrals have the form

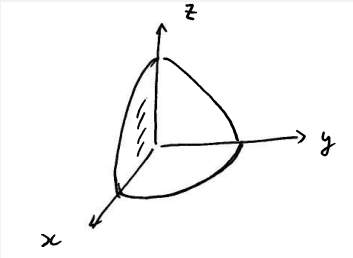
$$\boxed{\iiint_V f dx dy dz = \iiint_{V'} f r^2 \sin\theta dr d\theta d\phi\,.} \tag{9.15}$$

Example 9.8

Evaluate $\iiint_V x dV$, where V is the part of the solid ball

$$x^2 + y^2 + z^2 \leq a^2$$

lying in the first octant.



Solution. In spherical polar coordinates, V is described by the θ range $0 \leq \theta \leq \frac{\pi}{2}$ and the ϕ range $0 \leq \phi \leq \frac{\pi}{2}$, while the r range is $0 \leq r \leq a$. We thus obtain (where J indicates the Jacobian of the coordinate change):

$$\begin{aligned}
\iiint_V x dV &= \int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{2}} \int_0^a \underbrace{r\sin\theta\cos\phi}_{x} \cdot \underbrace{r^2\sin\theta}_{J} dr d\theta d\phi \\
&= \left(\int_0^{\frac{\pi}{2}} \cos\phi d\phi\right) \cdot \left(\int_0^{\frac{\pi}{2}} \sin^2\theta d\theta\right) \cdot \left(\int_0^a r^2 dr\right) \\
&= [\sin\phi]_0^{\frac{\pi}{2}} \cdot \left(\tfrac{1}{2}\int_0^{\frac{\pi}{2}} (1 - \cos 2\theta) d\theta\right) \cdot \frac{a^4}{4} \\
&= \frac{\pi a^4}{16}\,.
\end{aligned}$$

■

9.1.4 Latitude and longitude

Locations on the surface of the earth are often specified using latitude and longitude. These are, more or less, the angles in spherical polar coordinates, with some minor differences.

The origin is chosen to lie on the center of the earth, and the z-axis is chosen to lie along the axis of rotation, with the positive direction going through the north pole. The x-axis is then chosen so that the xz-plane cuts through a particular point in Greenwich in London (these axes were chosen when Britain still had an empire and "ruled the waves", so needed

to be able to navigate; there is actually a brass line marking the spot where the xz-plane emerges from the ground in a park in Greenwich).

Latitude corresponds to the angle θ, except that it is measured as angles north and south of the equator, with latitudes north of the equator taken as positive, and south taken as negative. Values of latitudes are always within the range $-\pi/2$ to $\pi/2$, with $-\pi/2$ corresponding to the south pole and $\pi/2$ corresponding to the north pole (in spherical polar coordinates, these points would correspond to $\theta = \pi$ and $\theta = 0$, respectively).

Longitude corresponds to the angle ϕ, except that it is measured as angles east and west of the xz-plane, with a positive sign to the east and a negative sign to the west. Values of longitude are always within the range $-\pi$ to π.